DEA-C01

AWS Certified Data Engineer - Associate

Exam A

QUESTION 1
A data engineer is configuring an AWS Glue job to read data from an Amazon S3 bucket. The data engineer has set up the necessary AWS Glue connection details and an associated IAM role. However, when the data engineer attempts to run the AWS Glue job, the data engineer receives an error message that indicates that there are problems with the Amazon S3 VPC gateway endpoint.

The data engineer must resolve the error and connect the AWS Glue job to the S3 bucket.

Which solution will meet this requirement?

A. Update the AWS Glue security group to allow inbound traffic from the Amazon S3 VPC gateway endpoint.
B. Configure an S3 bucket policy to explicitly grant the AWS Glue job permissions to access the S3 bucket.
C. Review the AWS Glue job code to ensure that the AWS Glue connection details include a fully qualified domain name.
D. Verify that the VPC's route table includes inbound and outbound routes for the Amazon S3 VPC gateway endpoint.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 2
A retail company has a customer data hub in an Amazon S3 bucket. Employees from many countries use the data hub to support company-wide analytics. A governance team must ensure that the company's data analysts can access data only for customers who are within the same country as the analysts.

Which solution will meet these requirements with the LEAST operational effort?

A. Create a separate table for each country's customer data. Provide access to each analyst based on the country that the analyst serves.
B. Register the S3 bucket as a data lake location in AWS Lake Formation. Use the Lake Formation row-level security features to enforce the company's access policies.
C. Move the data to AWS Regions that are close to the countries where the customers are. Provide access to each analyst based on the country that the analyst serves.
D. Load the data into Amazon Redshift. Create a view for each country. Create separate IAM roles for each country to provide access to data from each country. Assign the appropriate roles to the analysts.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 3
A media company wants to improve a system that recommends media content to customer based on user behavior and preferences. To improve the recommendation system, the company needs to incorporate insights from third-party datasets into the company's existing analytics platform.

The company wants to minimize the effort and time required to incorporate third-party datasets.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use API calls to access and integrate third-party datasets from AWS Data Exchange.
B. Use API calls to access and integrate third-party datasets from AWS DataSync.
C. Use Amazon Kinesis Data Streams to access and integrate third-party datasets from AWS

CodeCommit repositories.
D. Use Amazon Kinesis Data Streams to access and integrate third-party datasets from Amazon Elastic Container Registry (Amazon ECR).

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 4
A financial company wants to implement a data mesh. The data mesh must support centralized data governance, data analysis, and data access control. The company has decided to use AWS Glue for data catalogs and extract, transform, and load (ETL) operations.

Which combination of AWS services will implement a data mesh? (Choose two.)

A. Use Amazon Aurora for data storage. Use an Amazon Redshift provisioned cluster for data analysis.
B. Use Amazon S3 for data storage. Use Amazon Athena for data analysis.
C. Use AWS Glue DataBrew for centralized data governance and access control.
D. Use Amazon RDS for data storage. Use Amazon EMR for data analysis.
E. Use AWS Lake Formation for centralized data governance and access control.

Correct Answer: BE
Section: (none)

Explanation/Reference:

QUESTION 5
A data engineer maintains custom Python scripts that perform a data formatting process that many AWS Lambda functions use. When the data engineer needs to modify the Python scripts, the data engineer must manually update all the Lambda functions.

The data engineer requires a less manual way to update the Lambda functions.

Which solution will meet this requirement?

A. Store a pointer to the custom Python scripts in the execution context object in a shared Amazon S3 bucket.
B. Package the custom Python scripts into Lambda layers. Apply the Lambda layers to the Lambda functions.
C. Store a pointer to the custom Python scripts in environment variables in a shared Amazon S3 bucket.
D. Assign the same alias to each Lambda function. Call reach Lambda function by specifying the function's alias.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 6
A company created an extract, transform, and load (ETL) data pipeline in AWS Glue. A data engineer must crawl a table that is in Microsoft SQL Server. The data engineer needs to extract, transform, and load the output of the crawl to an Amazon S3 bucket. The data engineer also must orchestrate the data pipeline.

Which AWS service or feature will meet these requirements MOST cost-effectively?

A. AWS Step Functions

B. AWS Glue workflows
C. AWS Glue Studio
D. Amazon Managed Workflows for Apache Airflow (Amazon MWAA)

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 7
A financial services company stores financial data in Amazon Redshift. A data engineer wants to run real-time queries on the financial data to support a web-based trading application. The data engineer wants to run the queries from within the trading application.

Which solution will meet these requirements with the LEAST operational overhead?

A. Establish WebSocket connections to Amazon Redshift.
B. Use the Amazon Redshift Data API.
C. Set up Java Database Connectivity (JDBC) connections to Amazon Redshift.
D. Store frequently accessed data in Amazon S3. Use Amazon S3 Select to run the queries.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 8
A company uses Amazon Athena for one-time queries against data that is in Amazon S3. The company has several use cases. The company must implement permission controls to separate query processes and access to query history among users, teams, and applications that are in the same AWS account.

Which solution will meet these requirements?

A. Create an S3 bucket for each use case. Create an S3 bucket policy that grants permissions to appropriate individual IAM users. Apply the S3 bucket policy to the S3 bucket.
B. Create an Athena workgroup for each use case. Apply tags to the workgroup. Create an IAM policy that uses the tags to apply appropriate permissions to the workgroup.
C. Create an IAM role for each use case. Assign appropriate permissions to the role for each use case. Associate the role with Athena.
D. Create an AWS Glue Data Catalog resource policy that grants permissions to appropriate individual IAM users for each use case. Apply the resource policy to the specific tables that Athena uses.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 9
A data engineer needs to schedule a workflow that runs a set of AWS Glue jobs every day. The data engineer does not require the Glue jobs to run or finish at a specific time.

Which solution will run the Glue jobs in the MOST cost-effective way?

A. Choose the FLEX execution class in the Glue job properties.
B. Use the Spot Instance type in Glue job properties.
C. Choose the STANDARD execution class in the Glue job properties.

D. Choose the latest version in the GlueVersion field in the Glue job properties.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 10
A data engineer needs to create an AWS Lambda function that converts the format of data from .csv to Apache Parquet. The Lambda function must run only if a user uploads a .csv file to an Amazon S3 bucket.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an S3 event notification that has an event type of s3:ObjectCreated:*. Use a filter rule to generate notifications only when the suffix includes .csv. Set the Amazon Resource Name (ARN) of the Lambda function as the destination for the event notification.
B. Create an S3 event notification that has an event type of s3:ObjectTagging:* for objects that have a tag set to .csv. Set the Amazon Resource Name (ARN) of the Lambda function as the destination for the event notification.
C. Create an S3 event notification that has an event type of s3:*. Use a filter rule to generate notifications only when the suffix includes .csv. Set the Amazon Resource Name (ARN) of the Lambda function as the destination for the event notification.
D. Create an S3 event notification that has an event type of s3:ObjectCreated:*. Use a filter rule to generate notifications only when the suffix includes .csv. Set an Amazon Simple Notification Service (Amazon SNS) topic as the destination for the event notification. Subscribe the Lambda function to the SNS topic.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 11
A data engineer needs Amazon Athena queries to finish faster. The data engineer notices that all the files the Athena queries use are currently stored in uncompressed .csv format. The data engineer also notices that users perform most queries by selecting a specific column.

Which solution will MOST speed up the Athena query performance?

A. Change the data format from .csv to JSON format. Apply Snappy compression.
B. Compress the .csv files by using Snappy compression.
C. Change the data format from .csv to Apache Parquet. Apply Snappy compression.
D. Compress the .csv files by using gzip compression.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 12
A manufacturing company collects sensor data from its factory floor to monitor and enhance operational efficiency. The company uses Amazon Kinesis Data Streams to publish the data that the sensors collect to a data stream. Then Amazon Kinesis Data Firehose writes the data to an Amazon S3 bucket.

The company needs to display a real-time view of operational efficiency on a large screen in the manufacturing facility.

Which solution will meet these requirements with the LOWEST latency?

A. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to process the sensor data. Use a connector for Apache Flink to write data to an Amazon Timestream database. Use the Timestream database as a source to create a Grafana dashboard.
B. Configure the S3 bucket to send a notification to an AWS Lambda function when any new object is created. Use the Lambda function to publish the data to Amazon Aurora. Use Aurora as a source to create an Amazon QuickSight dashboard.
C. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to process the sensor data. Create a new Data Firehose delivery stream to publish data directly to an Amazon Timestream database. Use the Timestream database as a source to create an Amazon QuickSight dashboard.
D. Use AWS Glue bookmarks to read sensor data from the S3 bucket in real time. Publish the data to an Amazon Timestream database. Use the Timestream database as a source to create a Grafana dashboard.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 13
A company stores daily records of the financial performance of investment portfolios in .csv format in an Amazon S3 bucket. A data engineer uses AWS Glue crawlers to crawl the S3 data.

The data engineer must make the S3 data accessible daily in the AWS Glue Data Catalog.

Which solution will meet these requirements?

A. Create an IAM role that includes the AmazonS3FullAccess policy. Associate the role with the crawler. Specify the S3 bucket path of the source data as the crawler's data store. Create a daily schedule to run the crawler. Configure the output destination to a new path in the existing S3 bucket.
B. Create an IAM role that includes the AWSGlueServiceRole policy. Associate the role with the crawler. Specify the S3 bucket path of the source data as the crawler's data store. Create a daily schedule to run the crawler. Specify a database name for the output.
C. Create an IAM role that includes the AmazonS3FullAccess policy. Associate the role with the crawler. Specify the S3 bucket path of the source data as the crawler's data store. Allocate data processing units (DPUs) to run the crawler every day. Specify a database name for the output.
D. Create an IAM role that includes the AWSGlueServiceRole policy. Associate the role with the crawler. Specify the S3 bucket path of the source data as the crawler's data store. Allocate data processing units (DPUs) to run the crawler every day. Configure the output destination to a new path in the existing S3 bucket.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 14
A company loads transaction data for each day into Amazon Redshift tables at the end of each day. The company wants to have the ability to track which tables have been loaded and which tables still need to be loaded.

A data engineer wants to store the load statuses of Redshift tables in an Amazon DynamoDB table. The data engineer creates an AWS Lambda function to publish the details of the load statuses to DynamoDB.

How should the data engineer invoke the Lambda function to write load statuses to the DynamoDB table?

A. Use a second Lambda function to invoke the first Lambda function based on Amazon EventBridge events.

B. Use the Amazon Redshift Data API to publish an event to Amazon EventBridge. Configure an EventBridge rule to invoke the Lambda function.
C. Use the Amazon Redshift Data API to publish a message to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the SQS queue to invoke the Lambda function.
D. Use a second Lambda function to invoke the first Lambda function based on AWS CloudTrail events.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 15
A data engineer needs to securely transfer 5 TB of data from an on-premises data center to an Amazon S3 bucket. Approximately 5% of the data changes every day. Updates to the data need to be regularly proliferated to the S3 bucket. The data includes files that are in multiple formats. The data engineer needs to automate the transfer process and must schedule the process to run periodically.

Which AWS service should the data engineer use to transfer the data in the MOST operationally efficient way?

A. AWS DataSync
B. AWS Glue
C. AWS Direct Connect
D. Amazon S3 Transfer Acceleration

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 16
A company uses an on-premises Microsoft SQL Server database to store financial transaction data. The company migrates the transaction data from the on-premises database to AWS at the end of each month. The company has noticed that the cost to migrate data from the on-premises database to an Amazon RDS for SQL Server database has increased recently.

The company requires a cost-effective solution to migrate the data to AWS. The solution must cause minimal downtown for the applications that access the database.

Which AWS service should the company use to meet these requirements?

A. AWS Lambda
B. AWS Database Migration Service (AWS DMS)
C. AWS Direct Connect
D. AWS DataSync

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 17
A data engineer is building a data pipeline on AWS by using AWS Glue extract, transform, and load (ETL) jobs. The data engineer needs to process data from Amazon RDS and MongoDB, perform transformations, and load the transformed data into Amazon Redshift for analytics. The data updates must occur every hour.

Which combination of tasks will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Configure AWS Glue triggers to run the ETL jobs every hour.
B. Use AWS Glue DataBrew to clean and prepare the data for analytics.
C. Use AWS Lambda functions to schedule and run the ETL jobs every hour.
D. Use AWS Glue connections to establish connectivity between the data sources and Amazon Redshift.
E. Use the Redshift Data API to load transformed data into Amazon Redshift.

Correct Answer: AD
Section: (none)

Explanation/Reference:

QUESTION 18
A company uses an Amazon Redshift cluster that runs on RA3 nodes. The company wants to scale read and write capacity to meet demand. A data engineer needs to identify a solution that will turn on concurrency scaling.

Which solution will meet this requirement?

A. Turn on concurrency scaling in workload management (WLM) for Redshift Serverless workgroups.
B. Turn on concurrency scaling at the workload management (WLM) queue level in the Redshift cluster.
C. Turn on concurrency scaling in the settings during the creation of any new Redshift cluster.
D. Turn on concurrency scaling for the daily usage quota for the Redshift cluster.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 19
A data engineer must orchestrate a series of Amazon Athena queries that will run every day. Each query can run for more than 15 minutes.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

A. Use an AWS Lambda function and the Athena Boto3 client start_query_execution API call to invoke the Athena queries programmatically.
B. Create an AWS Step Functions workflow and add two states. Add the first state before the Lambda function. Configure the second state as a Wait state to periodically check whether the Athena query has finished using the Athena Boto3 get_query_execution API call. Configure the workflow to invoke the next query when the current query has finished running.
C. Use an AWS Glue Python shell job and the Athena Boto3 client start_query_execution API call to invoke the Athena queries programmatically.
D. Use an AWS Glue Python shell script to run a sleep timer that checks every 5 minutes to determine whether the current Athena query has finished running successfully. Configure the Python shell script to invoke the next query when the current query has finished running.
E. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to orchestrate the Athena queries in AWS Batch.

Correct Answer: AB
Section: (none)

Explanation/Reference:

QUESTION 20

A company is migrating on-premises workloads to AWS. The company wants to reduce overall operational overhead. The company also wants to explore serverless options.

The company's current workloads use Apache Pig, Apache Oozie, Apache Spark, Apache Hbase, and Apache Flink. The on-premises workloads process petabytes of data in seconds. The company must maintain similar or better performance after the migration to AWS.

Which extract, transform, and load (ETL) service will meet these requirements?

A. AWS Glue
B. Amazon EMR
C. AWS Lambda
D. Amazon Redshift

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 21
A data engineer must use AWS services to ingest a dataset into an Amazon S3 data lake. The data engineer profiles the dataset and discovers that the dataset contains personally identifiable information (PII). The data engineer must implement a solution to profile the dataset and obfuscate the PII.

Which solution will meet this requirement with the LEAST operational effort?

A. Use an Amazon Kinesis Data Firehose delivery stream to process the dataset. Create an AWS Lambda transform function to identify the PII. Use an AWS SDK to obfuscate the PII. Set the S3 data lake as the target for the delivery stream.
B. Use the Detect PII transform in AWS Glue Studio to identify the PII. Obfuscate the PII. Use an AWS Step Functions state machine to orchestrate a data pipeline to ingest the data into the S3 data lake.
C. Use the Detect PII transform in AWS Glue Studio to identify the PII. Create a rule in AWS Glue Data Quality to obfuscate the PII. Use an AWS Step Functions state machine to orchestrate a data pipeline to ingest the data into the S3 data lake.
D. Ingest the dataset into Amazon DynamoDB. Create an AWS Lambda function to identify and obfuscate the PII in the DynamoDB table and to transform the data. Use the same Lambda function to ingest the data into the S3 data lake.

Correct Answer: ~~C~~
Section: (none)

Explanation/Reference:

QUESTION 22
A company maintains multiple extract, transform, and load (ETL) workflows that ingest data from the company's operational databases into an Amazon S3 based data lake. The ETL workflows use AWS Glue and Amazon EMR to process data.

The company wants to improve the existing architecture to provide automated orchestration and to require minimal manual effort.

Which solution will meet these requirements with the LEAST operational overhead?

A. AWS Glue workflows
B. AWS Step Functions tasks
C. AWS Lambda functions
D. Amazon Managed Workflows for Apache Airflow (Amazon MWAA) workflows

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 23
A company currently stores all of its data in Amazon S3 by using the S3 Standard storage class.

A data engineer examined data access patterns to identify trends. During the first 6 months, most data files are accessed several times each day. Between 6 months and 2 years, most data files are accessed once or twice each month. After 2 years, data files are accessed only once or twice each year.

The data engineer needs to use an S3 Lifecycle policy to develop new data storage rules. The new storage solution must continue to provide high availability.

Which solution will meet these requirements in the MOST cost-effective way?

A. Transition objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 6 months. Transfer objects to S3 Glacier Flexible Retrieval after 2 years.
B. Transition objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 6 months. Transfer objects to S3 Glacier Flexible Retrieval after 2 years.
C. Transition objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 6 months. Transfer objects to S3 Glacier Deep Archive after 2 years.
D. Transition objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 6 months. Transfer objects to S3 Glacier Deep Archive after 2 years.

Correct Answer: C
Section: (none)

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Explanation/Reference:

QUESTION 24
A company maintains an Amazon Redshift provisioned cluster that the company uses for extract, transform, and load (ETL) operations to support critical analysis tasks. A sales team within the company maintains a Redshift cluster that the sales team uses for business intelligence (BI) tasks.

The sales team recently requested access to the data that is in the ETL Redshift cluster so the team can perform weekly summary analysis tasks. The sales team needs to join data from the ETL cluster with data that is in the sales team's BI cluster.

The company needs a solution that will share the ETL cluster data with the sales team without interrupting the critical analysis tasks. The solution must minimize usage of the computing resources of the ETL cluster.

Which solution will meet these requirements?

A. Set up the sales team BI cluster as a consumer of the ETL cluster by using Redshift data sharing.
B. Create materialized views based on the sales team's requirements. Grant the sales team direct access to the ETL cluster.
C. Create database views based on the sales team's requirements. Grant the sales team direct access to the ETL cluster.
D. Unload a copy of the data from the ETL cluster to an Amazon S3 bucket every week. Create an Amazon Redshift Spectrum table based on the content of the ETL cluster.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 25
A data engineer needs to join data from multiple sources to perform a one-time analysis job. The data is stored in Amazon DynamoDB, Amazon RDS, Amazon Redshift, and Amazon S3.

Which solution will meet this requirement MOST cost-effectively?

A. Use an Amazon EMR provisioned cluster to read from all sources. Use Apache Spark to join the data and perform the analysis.
B. Copy the data from DynamoDB, Amazon RDS, and Amazon Redshift into Amazon S3. Run Amazon Athena queries directly on the S3 files.
C. Use Amazon Athena Federated Query to join the data from all data sources.
D. Use Redshift Spectrum to query data from DynamoDB, Amazon RDS, and Amazon S3 directly from Redshift.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 26
A company is planning to use a provisioned Amazon EMR cluster that runs Apache Spark jobs to perform big data analysis. The company requires high reliability. A big data team must follow best practices for running cost-optimized and long-running workloads on Amazon EMR. The team must find a solution that will maintain the company's current level of performance.

Which combination of resources will meet these requirements MOST cost-effectively? (Choose two.)

A. Use Hadoop Distributed File System (HDFS) as a persistent data store.
B. Use Amazon S3 as a persistent data store.
C. Use x86-based instances for core nodes and task nodes.
D. Use Graviton instances for core nodes and task nodes.
E. Use Spot Instances for all primary nodes.

Correct Answer: BD
Section: (none)

Explanation/Reference:

QUESTION 27
A company wants to implement real-time analytics capabilities. The company wants to use Amazon Kinesis Data Streams and Amazon Redshift to ingest and process streaming data at the rate of several gigabytes per second. The company wants to derive near real-time insights by using existing business intelligence (BI) and analytics tools.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Kinesis Data Streams to stage data in Amazon S3. Use the COPY command to load data from Amazon S3 directly into Amazon Redshift to make the data immediately available for real-time analysis.
B. Access the data from Kinesis Data Streams by using SQL queries. Create materialized views directly on top of the stream. Refresh the materialized views regularly to query the most recent stream data.
C. Create an external schema in Amazon Redshift to map the data from Kinesis Data Streams to an Amazon Redshift object. Create a materialized view to read data from the stream. Set the materialized view to auto refresh.
D. Connect Kinesis Data Streams to Amazon Kinesis Data Firehose. Use Kinesis Data Firehose to stage the data in Amazon S3. Use the COPY command to load the data from Amazon S3 to a table in Amazon Redshift.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 28
A company uses an Amazon QuickSight dashboard to monitor usage of one of the company's applications. The company uses AWS Glue jobs to process data for the dashboard. The company stores the data in a single Amazon S3 bucket. The company adds new data every day.

A data engineer discovers that dashboard queries are becoming slower over time. The data engineer determines that the root cause of the slowing queries is long-running AWS Glue jobs.

Which actions should the data engineer take to improve the performance of the AWS Glue jobs? (Choose two.)

A. Partition the data that is in the S3 bucket. Organize the data by year, month, and day.
B. Increase the AWS Glue instance size by scaling up the worker type.
C. Convert the AWS Glue schema to the DynamicFrame schema class.
D. Adjust AWS Glue job scheduling frequency so the jobs run half as many times each day.
E. Modify the IAM role that grants access to AWS glue to grant access to all S3 features.

Correct Answer: AB
Section: (none)

Explanation/Reference:

QUESTION 29
A data engineer needs to use AWS Step Functions to design an orchestration workflow. The workflow must parallel process a large collection of data files and apply a specific transformation to each file.

Which Step Functions state should the data engineer use to meet these requirements?

A. Parallel state
B. Choice state
C. Map state
D. Wait state

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 30
A company is migrating a legacy application to an Amazon S3 based data lake. A data engineer reviewed data that is associated with the legacy application. The data engineer found that the legacy data contained some duplicate information.

The data engineer must identify and remove duplicate information from the legacy application data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Write a custom extract, transform, and load (ETL) job in Python. Use the DataFrame.drop_duplicates() function by importing the Pandas library to perform data deduplication.
B. Write an AWS Glue extract, transform, and load (ETL) job. Use the FindMatches machine learning (ML) transform to transform the data to perform data deduplication.
C. Write a custom extract, transform, and load (ETL) job in Python. Import the Python dedupe library. Use

the dedupe library to perform data deduplication.

D. Write an AWS Glue extract, transform, and load (ETL) job. Import the Python dedupe library. Use the dedupe library to perform data deduplication.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 31
A company is building an analytics solution. The solution uses Amazon S3 for data lake storage and Amazon Redshift for a data warehouse. The company wants to use Amazon Redshift Spectrum to query the data that is in Amazon S3.

Which actions will provide the FASTEST queries? (Choose two.)

A. Use gzip compression to compress individual files to sizes that are between 1 GB and 5 GB.
B. Use a columnar storage file format.
C. Partition the data based on the most common query predicates.
D. Split the data into files that are less than 10 KB.
E. Use file formats that are not splittable.

Correct Answer: BC
Section: (none)

Explanation/Reference:

QUESTION 32
A company uses Amazon RDS to store transactional data. The company runs an RDS DB instance in a private subnet. A developer wrote an AWS Lambda function with default settings to insert, update, or delete data in the DB instance.

The developer needs to give the Lambda function the ability to connect to the DB instance privately without using the public internet.

Which combination of steps will meet this requirement with the LEAST operational overhead? (Choose two.)

A. Turn on the public access setting for the DB instance.
B. Update the security group of the DB instance to allow only Lambda function invocations on the database port.
C. Configure the Lambda function to run in the same subnet that the DB instance uses.
D. Attach the same security group to the Lambda function and the DB instance. Include a self-referencing rule that allows access through the database port.
E. Update the network ACL of the private subnet to include a self-referencing rule that allows access through the database port.

Correct Answer: CD
Section: (none)

Explanation/Reference:

QUESTION 33
A company has a frontend ReactJS website that uses Amazon API Gateway to invoke REST APIs. The APIs perform the functionality of the website. A data engineer needs to write a Python script that can be occasionally invoked through API Gateway. The code must return results to API Gateway.

Which solution will meet these requirements with the LEAST operational overhead?

A. Deploy a custom Python script on an Amazon Elastic Container Service (Amazon ECS) cluster.
B. Create an AWS Lambda Python function with provisioned concurrency.
C. Deploy a custom Python script that can integrate with API Gateway on Amazon Elastic Kubernetes Service (Amazon EKS).
D. Create an AWS Lambda function. Ensure that the function is warm by scheduling an Amazon EventBridge rule to invoke the Lambda function every 5 minutes by using mock events.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 34
A company has a production AWS account that runs company workloads. The company's security team created a security AWS account to store and analyze security logs from the production AWS account. The security logs in the production AWS account are stored in Amazon CloudWatch Logs.

The company needs to use Amazon Kinesis Data Streams to deliver the security logs to the security AWS account.

Which solution will meet these requirements?

A. Create a destination data stream in the production AWS account. In the security AWS account, create an IAM role that has cross-account permissions to Kinesis Data Streams in the production AWS account.
B. Create a destination data stream in the security AWS account. Create an IAM role and a trust policy to grant CloudWatch Logs the permission to put data into the stream. Create a subscription filter in the security AWS account.
C. Create a destination data stream in the production AWS account. In the production AWS account, create an IAM role that has cross-account permissions to Kinesis Data Streams in the security AWS account.
D. Create a destination data stream in the security AWS account. Create an IAM role and a trust policy to grant CloudWatch Logs the permission to put data into the stream. Create a subscription filter in the production AWS account.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 35
A company uses Amazon S3 to store semi-structured data in a transactional data lake. Some of the data files are small, but other data files are tens of terabytes.

A data engineer must perform a change data capture (CDC) operation to identify changed data from the data source. The data source sends a full snapshot as a JSON file every day and ingests the changed data into the data lake.

Which solution will capture the changed data MOST cost-effectively?

A. Create an AWS Lambda function to identify the changes between the previous data and the current data. Configure the Lambda function to ingest the changes into the data lake.
B. Ingest the data into Amazon RDS for MySQL. Use AWS Database Migration Service (AWS DMS) to write the changed data to the data lake.
C. Use an open source data lake format to merge the data source with the S3 data lake to insert the new data and update the existing data.

D. Ingest the data into an Amazon Aurora MySQL DB instance that runs Aurora Serverless. Use AWS Database Migration Service (AWS DMS) to write the changed data to the data lake.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 36
A data engineer runs Amazon Athena queries on data that is in an Amazon S3 bucket. The Athena queries use AWS Glue Data Catalog as a metadata table.

The data engineer notices that the Athena query plans are experiencing a performance bottleneck. The data engineer determines that the cause of the performance bottleneck is the large number of partitions that are in the S3 bucket. The data engineer must resolve the performance bottleneck and reduce Athena query planning time.

Which solutions will meet these requirements? (Choose two.)

A. Create an AWS Glue partition index. Enable partition filtering.
B. Bucket the data based on a column that the data have in common in a WHERE clause of the user query.
C. Use Athena partition projection based on the S3 bucket prefix.
D. Transform the data that is in the S3 bucket to Apache Parquet format.
E. Use the Amazon EMR S3DistCP utility to combine smaller objects in the S3 bucket into larger objects.

Correct Answer: AC
Section: (none)

Explanation/Reference:

QUESTION 37
A data engineer must manage the ingestion of real-time streaming data into AWS. The data engineer wants to perform real-time analytics on the incoming streaming data by using time-based aggregations over a window of up to 30 minutes. The data engineer needs a solution that is highly fault tolerant.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use an AWS Lambda function that includes both the business and the analytics logic to perform time-based aggregations over a window of up to 30 minutes for the data in Amazon Kinesis Data Streams.
B. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to analyze the data that might occasionally contain duplicates by using multiple types of aggregations.
C. Use an AWS Lambda function that includes both the business and the analytics logic to perform aggregations for a tumbling window of up to 30 minutes, based on the event timestamp.
D. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to analyze the data by using multiple types of aggregations to perform time-based analytics over a window of up to 30 minutes.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 38
A company is planning to upgrade its Amazon Elastic Block Store (Amazon EBS) General Purpose SSD storage from gp2 to gp3. The company wants to prevent any interruptions in its Amazon EC2 instances that will cause data loss during the migration to the upgraded storage.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create snapshots of the gp2 volumes. Create new gp3 volumes from the snapshots. Attach the new gp3 volumes to the EC2 instances.
B. Create new gp3 volumes. Gradually transfer the data to the new gp3 volumes. When the transfer is complete, mount the new gp3 volumes to the EC2 instances to replace the gp2 volumes.
C. Change the volume type of the existing gp2 volumes to gp3. Enter new values for volume size, IOPS, and throughput.
D. Use AWS DataSync to create new gp3 volumes. Transfer the data from the original gp2 volumes to the new gp3 volumes.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 39
A company is migrating its database servers from Amazon EC2 instances that run Microsoft SQL Server to Amazon RDS for Microsoft SQL Server DB instances. The company's analytics team must export large data elements every day until the migration is complete. The data elements are the result of SQL joins across multiple tables. The data must be in Apache Parquet format. The analytics team must store the data in Amazon S3.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Create a view in the EC2 instance-based SQL Server databases that contains the required data elements. Create an AWS Glue job that selects the data directly from the view and transfers the data in Parquet format to an S3 bucket. Schedule the AWS Glue job to run every day.
B. Schedule SQL Server Agent to run a daily SQL query that selects the desired data elements from the EC2 instance-based SQL Server databases. Configure the query to direct the output .csv objects to an S3 bucket. Create an S3 event that invokes an AWS Lambda function to transform the output format from .csv to Parquet.
C. Use a SQL query to create a view in the EC2 instance-based SQL Server databases that contains the required data elements. Create and run an AWS Glue crawler to read the view. Create an AWS Glue job that retrieves the data and transfers the data in Parquet format to an S3 bucket. Schedule the AWS Glue job to run every day.
D. Create an AWS Lambda function that queries the EC2 instance-based databases by using Java Database Connectivity (JDBC). Configure the Lambda function to retrieve the required data, transform the data into Parquet format, and transfer the data into an S3 bucket. Use Amazon EventBridge to schedule the Lambda function to run every day.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 40
A data engineering team is using an Amazon Redshift data warehouse for operational reporting. The team wants to prevent performance issues that might result from long- running queries. A data engineer must choose a system table in Amazon Redshift to record anomalies when a query optimizer identifies conditions that might indicate performance issues.

Which table views should the data engineer use to meet this requirement?

A. STL_USAGE_CONTROL
B. STL_ALERT_EVENT_LOG
C. STL_QUERY_METRICS
D. STL_PLAN_INFO

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 41
A data engineer must ingest a source of structured data that is in .csv format into an Amazon S3 data lake. The .csv files contain 15 columns. Data analysts need to run Amazon Athena queries on one or two columns of the dataset. The data analysts rarely query the entire file.

Which solution will meet these requirements MOST cost-effectively?

A. Use an AWS Glue PySpark job to ingest the source data into the data lake in .csv format.
B. Create an AWS Glue extract, transform, and load (ETL) job to read from the .csv structured data source. Configure the job to ingest the data into the data lake in JSON format.
C. Use an AWS Glue PySpark job to ingest the source data into the data lake in Apache Avro format.
D. Create an AWS Glue extract, transform, and load (ETL) job to read from the .csv structured data source. Configure the job to write the data into the data lake in Apache Parquet format.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 42
A company has five offices in different AWS Regions. Each office has its own human resources (HR) department that uses a unique IAM role. The company stores employee records in a data lake that is based on Amazon S3 storage.

A data engineering team needs to limit access to the records. Each HR department should be able to access records for only employees who are within the HR department's Region.

Which combination of steps should the data engineering team take to meet this requirement with the LEAST operational overhead? (Choose two.)

A. Use data filters for each Region to register the S3 paths as data locations.
B. Register the S3 path as an AWS Lake Formation location.
C. Modify the IAM roles of the HR departments to add a data filter for each department's Region.
D. Enable fine-grained access control in AWS Lake Formation. Add a data filter for each Region.
E. Create a separate S3 bucket for each Region. Configure an IAM policy to allow S3 access. Restrict access based on Region.

Correct Answer: BD
Section: (none)

Explanation/Reference:

QUESTION 43
A company uses AWS Step Functions to orchestrate a data pipeline. The pipeline consists of Amazon EMR jobs that ingest data from data sources and store the data in an Amazon S3 bucket. The pipeline also includes EMR jobs that load the data to Amazon Redshift.

The company's cloud infrastructure team manually built a Step Functions state machine. The cloud infrastructure team launched an EMR cluster into a VPC to support the EMR jobs. However, the deployed Step Functions state machine is not able to run the EMR jobs.

Which combination of steps should the company take to identify the reason the Step Functions state machine is not able to run the EMR jobs? (Choose two.)

A. Use AWS CloudFormation to automate the Step Functions state machine deployment. Create a step to pause the state machine during the EMR jobs that fail. Configure the step to wait for a human user to send approval through an email message. Include details of the EMR task in the email message for further analysis.
B. Verify that the Step Functions state machine code has all IAM permissions that are necessary to create and run the EMR jobs. Verify that the Step Functions state machine code also includes IAM permissions to access the Amazon S3 buckets that the EMR jobs use. Use Access Analyzer for S3 to check the S3 access properties.
C. Check for entries in Amazon CloudWatch for the newly created EMR cluster. Change the AWS Step Functions state machine code to use Amazon EMR on EKS. Change the IAM access policies and the security group configuration for the Step Functions state machine code to reflect inclusion of Amazon Elastic Kubernetes Service (Amazon EKS).
D. Query the flow logs for the VPC. Determine whether the traffic that originates from the EMR cluster can successfully reach the data providers. Determine whether any security group that might be attached to the Amazon EMR cluster allows connections to the data source servers on the informed ports.
E. Check the retry scenarios that the company configured for the EMR jobs. Increase the number of seconds in the interval between each EMR task. Validate that each fallback state has the appropriate catch for each decision state. Configure an Amazon Simple Notification Service (Amazon SNS) topic to store the error messages.

Correct Answer: BD
Section: (none)

Explanation/Reference:

QUESTION 44
A company is developing an application that runs on Amazon EC2 instances. Currently, the data that the application generates is temporary. However, the company needs to persist the data, even if the EC2 instances are terminated.

A data engineer must launch new EC2 instances from an Amazon Machine Image (AMI) and configure the instances to preserve the data.

Which solution will meet this requirement?

A. Launch new EC2 instances by using an AMI that is backed by an EC2 instance store volume that contains the application data. Apply the default settings to the EC2 instances.
B. Launch new EC2 instances by using an AMI that is backed by a root Amazon Elastic Block Store (Amazon EBS) volume that contains the application data. Apply the default settings to the EC2 instances.
C. Launch new EC2 instances by using an AMI that is backed by an EC2 instance store volume. Attach an Amazon Elastic Block Store (Amazon EBS) volume to contain the application data. Apply the default settings to the EC2 instances.
D. Launch new EC2 instances by using an AMI that is backed by an Amazon Elastic Block Store (Amazon EBS) volume. Attach an additional EC2 instance store volume to contain the application data. Apply the default settings to the EC2 instances.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 45
A company uses Amazon Athena to run SQL queries for extract, transform, and load (ETL) tasks by using Create Table As Select (CTAS). The company must use Apache Spark instead of SQL to generate analytics.

Which solution will give the company the ability to use Spark to access Athena?

A. Athena query settings
B. Athena workgroup
C. Athena data source
D. Athena query editor

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 46
A company needs to partition the Amazon S3 storage that the company uses for a data lake. The partitioning will use a path of the S3 object keys in the following format: s3://bucket/prefix/year=2023/month=01/day=01.

A data engineer must ensure that the AWS Glue Data Catalog synchronizes with the S3 storage when the company adds new partitions to the bucket.

Which solution will meet these requirements with the LEAST latency?

A. Schedule an AWS Glue crawler to run every morning.
B. Manually run the AWS Glue CreatePartition API twice each day.
C. Use code that writes data to Amazon S3 to invoke the Boto3 AWS Glue create_partition API call.
D. Run the MSCK REPAIR TABLE command from the AWS Glue console.

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Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 47
A media company uses software as a service (SaaS) applications to gather data by using third-party tools. The company needs to store the data in an Amazon S3 bucket. The company will use Amazon Redshift to perform analytics based on the data.

Which AWS service or feature will meet these requirements with the LEAST operational overhead?

A. Amazon Managed Streaming for Apache Kafka (Amazon MSK)
B. Amazon AppFlow
C. AWS Glue Data Catalog
D. Amazon Kinesis

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 48
A data engineer is using Amazon Athena to analyze sales data that is in Amazon S3. The data engineer writes a query to retrieve sales amounts for 2023 for several products from a table named sales_data. However, the query does not return results for all of the products that are in the sales_data table. The data engineer needs to troubleshoot the query to resolve the issue.

The data engineer's original query is as follows:

```
SELECT product_name, sum(sales_amount)
FROM sales_data
WHERE year = 2023
GROUP BY product_name
```

How should the data engineer modify the Athena query to meet these requirements?

A. Replace sum(sales_amount) with count(*) for the aggregation.
B. Change WHERE year = 2023 to WHERE extract(year FROM sales_data) = 2023.
C. Add HAVING sum(sales_amount) > 0 after the GROUP BY clause.
D. Remove the GROUP BY clause.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 49
A data engineer has a one-time task to read data from objects that are in Apache Parquet format in an Amazon S3 bucket. The data engineer needs to query only one column of the data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure an AWS Lambda function to load data from the S3 bucket into a pandas dataframe. Write a SQL SELECT statement on the dataframe to query the required column.
B. Use S3 Select to write a SQL SELECT statement to retrieve the required column from the S3 objects.
C. Prepare an AWS Glue DataBrew project to consume the S3 objects and to query the required column.
D. Run an AWS Glue crawler on the S3 objects. Use a SQL SELECT statement in Amazon Athena to query the required column.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 50
A company uses Amazon Redshift for its data warehouse. The company must automate refresh schedules for Amazon Redshift materialized views.

Which solution will meet this requirement with the LEAST effort?

A. Use Apache Airflow to refresh the materialized views.
B. Use an AWS Lambda user-defined function (UDF) within Amazon Redshift to refresh the materialized views.
C. Use the query editor v2 in Amazon Redshift to refresh the materialized views.
D. Use an AWS Glue workflow to refresh the materialized views.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 51
A data engineer must orchestrate a data pipeline that consists of one AWS Lambda function and one AWS Glue job. The solution must integrate with AWS services.

Which solution will meet these requirements with the LEAST management overhead?

A. Use an AWS Step Functions workflow that includes a state machine. Configure the state machine to run the Lambda function and then the AWS Glue job.
B. Use an Apache Airflow workflow that is deployed on an Amazon EC2 instance. Define a directed acyclic graph (DAG) in which the first task is to call the Lambda function and the second task is to call the AWS Glue job.
C. Use an AWS Glue workflow to run the Lambda function and then the AWS Glue job.
D. Use an Apache Airflow workflow that is deployed on Amazon Elastic Kubernetes Service (Amazon EKS). Define a directed acyclic graph (DAG) in which the first task is to call the Lambda function and the second task is to call the AWS Glue job.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 52
A company needs to set up a data catalog and metadata management for data sources that run in the AWS Cloud. The company will use the data catalog to maintain the metadata of all the objects that are in a set of data stores. The data stores include structured sources such as Amazon RDS and Amazon Redshift. The data stores also include semistructured sources such as JSON files and .xml files that are stored in Amazon S3.

The company needs a solution that will update the data catalog on a regular basis. The solution also must detect changes to the source metadata.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Aurora as the data catalog. Create AWS Lambda functions that will connect to the data catalog. Configure the Lambda functions to gather the metadata information from multiple sources and to update the Aurora data catalog. Schedule the Lambda functions to run periodically.
B. Use the AWS Glue Data Catalog as the central metadata repository. Use AWS Glue crawlers to connect to multiple data stores and to update the Data Catalog with metadata changes. Schedule the crawlers to run periodically to update the metadata catalog.
C. Use Amazon DynamoDB as the data catalog. Create AWS Lambda functions that will connect to the data catalog. Configure the Lambda functions to gather the metadata information from multiple sources and to update the DynamoDB data catalog. Schedule the Lambda functions to run periodically.
D. Use the AWS Glue Data Catalog as the central metadata repository. Extract the schema for Amazon RDS and Amazon Redshift sources, and build the Data Catalog. Use AWS Glue crawlers for data that is in Amazon S3 to infer the schema and to automatically update the Data Catalog.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 53
A company stores data from an application in an Amazon DynamoDB table that operates in provisioned capacity mode. The workloads of the application have predictable throughput load on a regular schedule. Every Monday, there is an immediate increase in activity early in the morning. The application has very low usage during weekends.

The company must ensure that the application performs consistently during peak usage times.

Which solution will meet these requirements in the MOST cost-effective way?

A. Increase the provisioned capacity to the maximum capacity that is currently present during peak load times.
B. Divide the table into two tables. Provision each table with half of the provisioned capacity of the original

table. Spread queries evenly across both tables.
C. Use AWS Application Auto Scaling to schedule higher provisioned capacity for peak usage times. Schedule lower capacity during off-peak times.
D. Change the capacity mode from provisioned to on-demand. Configure the table to scale up and scale down based on the load on the table.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 54
A company is planning to migrate on-premises Apache Hadoop clusters to Amazon EMR. The company also needs to migrate a data catalog into a persistent storage solution.

The company currently stores the data catalog in an on-premises Apache Hive metastore on the Hadoop clusters. The company requires a serverless solution to migrate the data catalog.

Which solution will meet these requirements MOST cost-effectively?

A. Use AWS Database Migration Service (AWS DMS) to migrate the Hive metastore into Amazon S3. Configure AWS Glue Data Catalog to scan Amazon S3 to produce the data catalog.
B. Configure a Hive metastore in Amazon EMR. Migrate the existing on-premises Hive metastore into Amazon EMR. Use AWS Glue Data Catalog to store the company's data catalog as an external data catalog.
C. Configure an external Hive metastore in Amazon EMR. Migrate the existing on-premises Hive metastore into Amazon EMR. Use Amazon Aurora MySQL to store the company's data catalog.
D. Configure a new Hive metastore in Amazon EMR. Migrate the existing on-premises Hive metastore into Amazon EMR. Use the new metastore as the company's data catalog.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 55
A company uses an Amazon Redshift provisioned cluster as its database. The Redshift cluster has five reserved ra3.4xlarge nodes and uses key distribution.

A data engineer notices that one of the nodes frequently has a CPU load over 90%. SQL Queries that run on the node are queued. The other four nodes usually have a CPU load under 15% during daily operations.

The data engineer wants to maintain the current number of compute nodes. The data engineer also wants to balance the load more evenly across all five compute nodes.

Which solution will meet these requirements?

A. Change the sort key to be the data column that is most often used in a WHERE clause of the SQL SELECT statement.
B. Change the distribution key to the table column that has the largest dimension.
C. Upgrade the reserved node from ra3.4xlarge to ra3.16xlarge.
D. Change the primary key to be the data column that is most often used in a WHERE clause of the SQL SELECT statement.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 56
A security company stores IoT data that is in JSON format in an Amazon S3 bucket. The data structure can change when the company upgrades the IoT devices. The company wants to create a data catalog that includes the IoT data. The company's analytics department will use the data catalog to index the data.

Which solution will meet these requirements MOST cost-effectively?

A. Create an AWS Glue Data Catalog. Configure an AWS Glue Schema Registry. Create a new AWS Glue workload to orchestrate the ingestion of the data that the analytics department will use into Amazon Redshift Serverless.
B. Create an Amazon Redshift provisioned cluster. Create an Amazon Redshift Spectrum database for the analytics department to explore the data that is in Amazon S3. Create Redshift stored procedures to load the data into Amazon Redshift.
C. Create an Amazon Athena workgroup. Explore the data that is in Amazon S3 by using Apache Spark through Athena. Provide the Athena workgroup schema and tables to the analytics department.
D. Create an AWS Glue Data Catalog. Configure an AWS Glue Schema Registry. Create AWS Lambda user defined functions (UDFs) by using the Amazon Redshift Data API. Create an AWS Step Functions job to orchestrate the ingestion of the data that the analytics department will use into Amazon Redshift Serverless.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 57
A company stores details about transactions in an Amazon S3 bucket. The company wants to log all writes to the S3 bucket into another S3 bucket that is in the same AWS Region.

Which solution will meet this requirement with the LEAST operational effort?

A. Configure an S3 Event Notifications rule for all activities on the transactions S3 bucket to invoke an AWS Lambda function. Program the Lambda function to write the event to Amazon Kinesis Data Firehose. Configure Kinesis Data Firehose to write the event to the logs S3 bucket.
B. Create a trail of management events in AWS CloudTraiL. Configure the trail to receive data from the transactions S3 bucket. Specify an empty prefix and write-only events. Specify the logs S3 bucket as the destination bucket.
C. Configure an S3 Event Notifications rule for all activities on the transactions S3 bucket to invoke an AWS Lambda function. Program the Lambda function to write the events to the logs S3 bucket.
D. Create a trail of data events in AWS CloudTraiL. Configure the trail to receive data from the transactions S3 bucket. Specify an empty prefix and write-only events. Specify the logs S3 bucket as the destination bucket.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 58
A data engineer needs to maintain a central metadata repository that users access through Amazon EMR and Amazon Athena queries. The repository needs to provide the schema and properties of many tables. Some of the metadata is stored in Apache Hive. The data engineer needs to import the metadata from Hive into the central metadata repository.

Which solution will meet these requirements with the LEAST development effort?

A. Use Amazon EMR and Apache Ranger.
B. Use a Hive metastore on an EMR cluster.
C. Use the AWS Glue Data Catalog.
D. Use a metastore on an Amazon RDS for MySQL DB instance.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 59
A company needs to build a data lake in AWS. The company must provide row-level data access and column-level data access to specific teams. The teams will access the data by using Amazon Athena, Amazon Redshift Spectrum, and Apache Hive from Amazon EMR.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon S3 for data lake storage. Use S3 access policies to restrict data access by rows and columns. Provide data access through Amazon S3.
B. Use Amazon S3 for data lake storage. Use Apache Ranger through Amazon EMR to restrict data access by rows and columns. Provide data access by using Apache Pig.
C. Use Amazon Redshift for data lake storage. Use Redshift security policies to restrict data access by rows and columns. Provide data access by using Apache Spark and Amazon Athena federated queries.
D. Use Amazon S3 for data lake storage. Use AWS Lake Formation to restrict data access by rows and columns. Provide data access through AWS Lake Formation.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 60
An airline company is collecting metrics about flight activities for analytics. The company is conducting a proof of concept (POC) test to show how analytics can provide insights that the company can use to increase on-time departures.

The POC test uses objects in Amazon S3 that contain the metrics in .csv format. The POC test uses Amazon Athena to query the data. The data is partitioned in the S3 bucket by date.

As the amount of data increases, the company wants to optimize the storage solution to improve query performance.

Which combination of solutions will meet these requirements? (Choose two.)

A. Add a randomized string to the beginning of the keys in Amazon S3 to get more throughput across partitions.
B. Use an S3 bucket that is in the same account that uses Athena to query the data.
C. Use an S3 bucket that is in the same AWS Region where the company runs Athena queries.
D. Preprocess the .csv data to JSON format by fetching only the document keys that the query requires.
E. Preprocess the .csv data to Apache Parquet format by fetching only the data blocks that are needed for predicates.

Correct Answer: CE
Section: (none)

Explanation/Reference:

QUESTION 61
A company uses Amazon RDS for MySQL as the database for a critical application. The database workload is mostly writes, with a small number of reads.

A data engineer notices that the CPU utilization of the DB instance is very high. The high CPU utilization is slowing down the application. The data engineer must reduce the CPU utilization of the DB Instance.

Which actions should the data engineer take to meet this requirement? (Choose two.)

A. Use the Performance Insights feature of Amazon RDS to identify queries that have high CPU utilization. Optimize the problematic queries.
B. Modify the database schema to include additional tables and indexes.
C. Reboot the RDS DB instance once each week.
D. Upgrade to a larger instance size.
E. Implement caching to reduce the database query load.

Correct Answer: AD
Section: (none)

Explanation/Reference:

QUESTION 62
A company has used an Amazon Redshift table that is named Orders for 6 months. The company performs weekly updates and deletes on the table. The table has an interleaved sort key on a column that contains AWS Regions.

The company wants to reclaim disk space so that the company will not run out of storage space. The company also wants to analyze the sort key column.

Which Amazon Redshift command will meet these requirements?

A. VACUUM FULL Orders
B. VACUUM DELETE ONLY Orders
C. VACUUM REINDEX Orders
D. VACUUM SORT ONLY Orders

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 63
A manufacturing company wants to collect data from sensors. A data engineer needs to implement a solution that ingests sensor data in near real time.

The solution must store the data to a persistent data store. The solution must store the data in nested JSON format. The company must have the ability to query from the data store with a latency of less than 10 milliseconds.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use a self-hosted Apache Kafka cluster to capture the sensor data. Store the data in Amazon S3 for querying.
B. Use AWS Lambda to process the sensor data. Store the data in Amazon S3 for querying.
C. Use Amazon Kinesis Data Streams to capture the sensor data. Store the data in Amazon DynamoDB for querying.
D. Use Amazon Simple Queue Service (Amazon SQS) to buffer incoming sensor data. Use AWS Glue to

store the data in Amazon RDS for querying.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 64
A company stores data in a data lake that is in Amazon S3. Some data that the company stores in the data lake contains personally identifiable information (PII). Multiple user groups need to access the raw data. The company must ensure that user groups can access only the PII that they require.

Which solution will meet these requirements with the LEAST effort?

A. Use Amazon Athena to query the data. Set up AWS Lake Formation and create data filters to establish levels of access for the company's IAM roles. Assign each user to the IAM role that matches the user's PII access requirements.
B. Use Amazon QuickSight to access the data. Use column-level security features in QuickSight to limit the PII that users can retrieve from Amazon S3 by using Amazon Athena. Define QuickSight access levels based on the PII access requirements of the users.
C. Build a custom query builder UI that will run Athena queries in the background to access the data. Create user groups in Amazon Cognito. Assign access levels to the user groups based on the PII access requirements of the users.
D. Create IAM roles that have different levels of granular access. Assign the IAM roles to IAM user groups. Use an identity-based policy to assign access levels to user groups at the column level.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 65
A data engineer must build an extract, transform, and load (ETL) pipeline to process and load data from 10 source systems into 10 tables that are in an Amazon Redshift database. All the source systems generate .csv, JSON, or Apache Parquet files every 15 minutes. The source systems all deliver files into one Amazon S3 bucket. The file sizes range from 10 MB to 20 GB. The ETL pipeline must function correctly despite changes to the data schema.

Which data pipeline solutions will meet these requirements? (Choose two.)

A. Use an Amazon EventBridge rule to run an AWS Glue job every 15 minutes. Configure the AWS Glue job to process and load the data into the Amazon Redshift tables.
B. Use an Amazon EventBridge rule to invoke an AWS Glue workflow job every 15 minutes. Configure the AWS Glue workflow to have an on-demand trigger that runs an AWS Glue crawler and then runs an AWS Glue job when the crawler finishes running successfully. Configure the AWS Glue job to process and load the data into the Amazon Redshift tables.
C. Configure an AWS Lambda function to invoke an AWS Glue crawler when a file is loaded into the S3 bucket. Configure an AWS Glue job to process and load the data into the Amazon Redshift tables. Create a second Lambda function to run the AWS Glue job. Create an Amazon EventBridge rule to invoke the second Lambda function when the AWS Glue crawler finishes running successfully.
D. Configure an AWS Lambda function to invoke an AWS Glue workflow when a file is loaded into the S3 bucket. Configure the AWS Glue workflow to have an on-demand trigger that runs an AWS Glue crawler and then runs an AWS Glue job when the crawler finishes running successfully. Configure the AWS Glue job to process and load the data into the Amazon Redshift tables.
E. Configure an AWS Lambda function to invoke an AWS Glue job when a file is loaded into the S3 bucket. Configure the AWS Glue job to read the files from the S3 bucket into an Apache Spark DataFrame. Configure the AWS Glue job to also put smaller partitions of the DataFrame into an Amazon Kinesis Data Firehose delivery stream. Configure the delivery stream to load data into the Amazon Redshift tables.

Correct Answer: BD
Section: (none)

Explanation/Reference:

QUESTION 66
A financial company wants to use Amazon Athena to run on-demand SQL queries on a petabyte-scale dataset to support a business intelligence (BI) application. An AWS Glue job that runs during non-business hours updates the dataset once every day. The BI application has a standard data refresh frequency of 1 hour to comply with company policies.

A data engineer wants to cost optimize the company's use of Amazon Athena without adding any additional infrastructure costs.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure an Amazon S3 Lifecycle policy to move data to the S3 Glacier Deep Archive storage class after 1 day.
B. Use the query result reuse feature of Amazon Athena for the SQL queries.
C. Add an Amazon ElastiCache cluster between the BI application and Athena.
D. Change the format of the files that are in the dataset to Apache Parquet.

Correct Answer: B
Section: (none)

Explanation/Reference:

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QUESTION 67
A company's data engineer needs to optimize the performance of table SQL queries. The company stores data in an Amazon Redshift cluster. The data engineer cannot increase the size of the cluster because of budget constraints.

The company stores the data in multiple tables and loads the data by using the EVEN distribution style. Some tables are hundreds of gigabytes in size. Other tables are less than 10 MB in size.

Which solution will meet these requirements?

A. Keep using the EVEN distribution style for all tables. Specify primary and foreign keys for all tables.
B. Use the ALL distribution style for large tables. Specify primary and foreign keys for all tables.
C. Use the ALL distribution style for rarely updated small tables. Specify primary and foreign keys for all tables.
D. Specify a combination of distribution, sort, and partition keys for all tables.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 68
A company receives .csv files that contain physical address data. The data is in columns that have the following names: Door_No, Street_Name, City, and Zip_Code. The company wants to create a single column to store these values in the following format:

```
{
    "Door_No": "24",
    "Street_Name": "AAA street",
    "City": "BBB",
    "Zip_Code": "111111"
}
```

Which solution will meet this requirement with the LEAST coding effort?

A. Use AWS Glue DataBrew to read the files. Use the NEST_TO_ARRAY transformation to create the new column.
B. Use AWS Glue DataBrew to read the files. Use the NEST_TO_MAP transformation to create the new column.
C. Use AWS Glue DataBrew to read the files. Use the PIVOT transformation to create the new column.
D. Write a Lambda function in Python to read the files. Use the Python data dictionary type to create the new column.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 69
A company receives call logs as Amazon S3 objects that contain sensitive customer information. The company must protect the S3 objects by using encryption. The company must also use encryption keys that only specific employees can access.

Which solution will meet these requirements with the LEAST effort?

A. Use an AWS CloudHSM cluster to store the encryption keys. Configure the process that writes to Amazon S3 to make calls to CloudHSM to encrypt and decrypt the objects. Deploy an IAM policy that restricts access to the CloudHSM cluster.
B. Use server-side encryption with customer-provided keys (SSE-C) to encrypt the objects that contain customer information. Restrict access to the keys that encrypt the objects.
C. Use server-side encryption with AWS KMS keys (SSE-KMS) to encrypt the objects that contain customer information. Configure an IAM policy that restricts access to the KMS keys that encrypt the objects.
D. Use server-side encryption with Amazon S3 managed keys (SSE-S3) to encrypt the objects that contain customer information. Configure an IAM policy that restricts access to the Amazon S3 managed keys that encrypt the objects.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 70
A company stores petabytes of data in thousands of Amazon S3 buckets in the S3 Standard storage class. The data supports analytics workloads that have unpredictable and variable data access patterns.

The company does not access some data for months. However, the company must be able to retrieve all data within milliseconds. The company needs to optimize S3 storage costs.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use S3 Storage Lens standard metrics to determine when to move objects to more cost-optimized storage classes. Create S3 Lifecycle policies for the S3 buckets to move objects to cost-optimized storage classes. Continue to refine the S3 Lifecycle policies in the future to optimize storage costs.
B. Use S3 Storage Lens activity metrics to identify S3 buckets that the company accesses infrequently. Configure S3 Lifecycle rules to move objects from S3 Standard to the S3 Standard-Infrequent Access (S3 Standard-IA) and S3 Glacier storage classes based on the age of the data.
C. Use S3 Intelligent-Tiering. Activate the Deep Archive Access tier.
D. Use S3 Intelligent-Tiering. Use the default access tier.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 71
During a security review, a company identified a vulnerability in an AWS Glue job. The company discovered that credentials to access an Amazon Redshift cluster were hard coded in the job script.

A data engineer must remediate the security vulnerability in the AWS Glue job. The solution must securely store the credentials.

Which combination of steps should the data engineer take to meet these requirements? (Choose two.)

A. Store the credentials in the AWS Glue job parameters.
B. Store the credentials in a configuration file that is in an Amazon S3 bucket.
C. Access the credentials from a configuration file that is in an Amazon S3 bucket by using the AWS Glue job.
D. Store the credentials in AWS Secrets Manager.
E. Grant the AWS Glue job IAM role access to the stored credentials.

Correct Answer: DE
Section: (none)

Explanation/Reference:

QUESTION 72
A data engineer uses Amazon Redshift to run resource-intensive analytics processes once every month. Every month, the data engineer creates a new Redshift provisioned cluster. The data engineer deletes the Redshift provisioned cluster after the analytics processes are complete every month. Before the data engineer deletes the cluster each month, the data engineer unloads backup data from the cluster to an Amazon S3 bucket.

The data engineer needs a solution to run the monthly analytics processes that does not require the data engineer to manage the infrastructure manually.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Step Functions to pause the Redshift cluster when the analytics processes are complete and to resume the cluster to run new processes every month.
B. Use Amazon Redshift Serverless to automatically process the analytics workload.
C. Use the AWS CLI to automatically process the analytics workload.
D. Use AWS CloudFormation templates to automatically process the analytics workload.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 73
A company receives a daily file that contains customer data in .xls format. The company stores the file in Amazon S3. The daily file is approximately 2 GB in size.

A data engineer concatenates the column in the file that contains customer first names and the column that contains customer last names. The data engineer needs to determine the number of distinct customers in the file.

Which solution will meet this requirement with the LEAST operational effort?

A. Create and run an Apache Spark job in an AWS Glue notebook. Configure the job to read the S3 file and calculate the number of distinct customers.
B. Create an AWS Glue crawler to create an AWS Glue Data Catalog of the S3 file. Run SQL queries from Amazon Athena to calculate the number of distinct customers.
C. Create and run an Apache Spark job in Amazon EMR Serverless to calculate the number of distinct customers.
D. Use AWS Glue DataBrew to create a recipe that uses the COUNT_DISTINCT aggregate function to calculate the number of distinct customers.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 74
A healthcare company uses Amazon Kinesis Data Streams to stream real-time health data from wearable devices, hospital equipment, and patient records.

A data engineer needs to find a solution to process the streaming data. The data engineer needs to store the data in an Amazon Redshift Serverless warehouse. The solution must support near real-time analytics of the streaming data and the previous day's data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Load data into Amazon Kinesis Data Firehose. Load the data into Amazon Redshift.
B. Use the streaming ingestion feature of Amazon Redshift.
C. Load the data into Amazon S3. Use the COPY command to load the data into Amazon Redshift.
D. Use the Amazon Aurora zero-ETL integration with Amazon Redshift.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 75
A data engineer needs to use an Amazon QuickSight dashboard that is based on Amazon Athena queries on data that is stored in an Amazon S3 bucket. When the data engineer connects to the QuickSight dashboard, the data engineer receives an error message that indicates insufficient permissions.

Which factors could cause to the permissions-related errors? (Choose two.)

A. There is no connection between QuickSight and Athena.
B. The Athena tables are not cataloged.
C. QuickSight does not have access to the S3 bucket.
D. QuickSight does not have access to decrypt S3 data.

E. There is no IAM role assigned to QuickSight.

Correct Answer: CD
Section: (none)

Explanation/Reference:

QUESTION 76
A company stores datasets in JSON format and .csv format in an Amazon S3 bucket. The company has Amazon RDS for Microsoft SQL Server databases, Amazon DynamoDB tables that are in provisioned capacity mode, and an Amazon Redshift cluster. A data engineering team must develop a solution that will give data scientists the ability to query all data sources by using syntax similar to SQL.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Glue to crawl the data sources. Store metadata in the AWS Glue Data Catalog. Use Amazon Athena to query the data. Use SQL for structured data sources. Use PartiQL for data that is stored in JSON format.
B. Use AWS Glue to crawl the data sources. Store metadata in the AWS Glue Data Catalog. Use Redshift Spectrum to query the data. Use SQL for structured data sources. Use PartiQL for data that is stored in JSON format.
C. Use AWS Glue to crawl the data sources. Store metadata in the AWS Glue Data Catalog. Use AWS Glue jobs to transform data that is in JSON format to Apache Parquet or .csv format. Store the transformed data in an S3 bucket. Use Amazon Athena to query the original and transformed data from the S3 bucket.
D. Use AWS Lake Formation to create a data lake. Use Lake Formation jobs to transform the data from all data sources to Apache Parquet format. Store the transformed data in an S3 bucket. Use Amazon Athena or Redshift Spectrum to query the data.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 77
A data engineer is configuring Amazon SageMaker Studio to use AWS Glue interactive sessions to prepare data for machine learning (ML) models.

The data engineer receives an access denied error when the data engineer tries to prepare the data by using SageMaker Studio.

Which change should the engineer make to gain access to SageMaker Studio?

A. Add the AWSGlueServiceRole managed policy to the data engineer's IAM user.
B. Add a policy to the data engineer's IAM user that includes the sts:AssumeRole action for the AWS Glue and SageMaker service principals in the trust policy.
C. Add the AmazonSageMakerFullAccess managed policy to the data engineer's IAM user.
D. Add a policy to the data engineer's IAM user that allows the sts:AddAssociation action for the AWS Glue and SageMaker service principals in the trust policy.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 78
A company extracts approximately 1 TB of data every day from data sources such as SAP HANA, Microsoft SQL Server, MongoDB, Apache Kafka, and Amazon DynamoDB. Some of the data sources have

undefined data schemas or data schemas that change.

A data engineer must implement a solution that can detect the schema for these data sources. The solution must extract, transform, and load the data to an Amazon S3 bucket. The company has a service level agreement (SLA) to load the data into the S3 bucket within 15 minutes of data creation.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon EMR to detect the schema and to extract, transform, and load the data into the S3 bucket. Create a pipeline in Apache Spark.
B. Use AWS Glue to detect the schema and to extract, transform, and load the data into the S3 bucket. Create a pipeline in Apache Spark.
C. Create a PySpark program in AWS Lambda to extract, transform, and load the data into the S3 bucket.
D. Create a stored procedure in Amazon Redshift to detect the schema and to extract, transform, and load the data into a Redshift Spectrum table. Access the table from Amazon S3.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 79
A company has multiple applications that use datasets that are stored in an Amazon S3 bucket. The company has an ecommerce application that generates a dataset that contains personally identifiable information (PII). The company has an internal analytics application that does not require access to the PII.

To comply with regulations, the company must not share PII unnecessarily. A data engineer needs to implement a solution that with redact PII dynamically, based on the needs of each application that accesses the dataset.

Which solution will meet the requirements with the LEAST operational overhead?

A. Create an S3 bucket policy to limit the access each application has. Create multiple copies of the dataset. Give each dataset copy the appropriate level of redaction for the needs of the application that accesses the copy.
B. Create an S3 Object Lambda endpoint. Use the S3 Object Lambda endpoint to read data from the S3 bucket. Implement redaction logic within an S3 Object Lambda function to dynamically redact PII based on the needs of each application that accesses the data.
C. Use AWS Glue to transform the data for each application. Create multiple copies of the dataset. Give each dataset copy the appropriate level of redaction for the needs of the application that accesses the copy.
D. Create an API Gateway endpoint that has custom authorizers. Use the API Gateway endpoint to read data from the S3 bucket. Initiate a REST API call to dynamically redact PII based on the needs of each application that accesses the data.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 80
A data engineer needs to build an extract, transform, and load (ETL) job. The ETL job will process daily incoming .csv files that users upload to an Amazon S3 bucket. The size of each S3 object is less than 100 MB.

Which solution will meet these requirements MOST cost-effectively?

A. Write a custom Python application. Host the application on an Amazon Elastic Kubernetes Service (Amazon EKS) cluster.

B. Write a PySpark ETL script. Host the script on an Amazon EMR cluster.
C. Write an AWS Glue PySpark job. Use Apache Spark to transform the data.
D. Write an AWS Glue Python shell job. Use pandas to transform the data.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 81
A data engineer creates an AWS Glue Data Catalog table by using an AWS Glue crawler that is named Orders. The data engineer wants to add the following new partitions:

s3://transactions/orders/order_date=2023-01-01
s3://transactions/orders/order_date=2023-01-02

The data engineer must edit the metadata to include the new partitions in the table without scanning all the folders and files in the location of the table.

Which data definition language (DDL) statement should the data engineer use in Amazon Athena?

A. ALTER TABLE Orders ADD PARTITION(order_date='2023-01-01') LOCATION 's3://transactions/orders/order_date=2023-01-01';
ALTER TABLE Orders ADD PARTITION(order_date='2023-01-02') LOCATION 's3://transactions/orders/order_date=2023-01-02';
B. MSCK REPAIR TABLE Orders;
C. REPAIR TABLE Orders;
D. ALTER TABLE Orders MODIFY PARTITION(order_date='2023-01-01') LOCATION 's3://transactions/orders/2023-01-01';
ALTER TABLE Orders MODIFY PARTITION(order_date='2023-01-02') LOCATION 's3://transactions/orders/2023-01-02';

Correct Answer: A
Section: (none)

Explanation/Reference:
Reference:

QUESTION 82
A company stores 10 to 15 TB of uncompressed .csv files in Amazon S3. The company is evaluating Amazon Athena as a one-time query engine.

The company wants to transform the data to optimize query runtime and storage costs.

Which file format and compression solution will meet these requirements for Athena queries?

A. .csv format compressed with zip
B. JSON format compressed with bzip2
C. Apache Parquet format compressed with Snappy
D. Apache Avro format compressed with LZO

Correct Answer: C
Section: (none)

Explanation/Reference:
Reference:

QUESTION 83
A company uses Apache Airflow to orchestrate the company's current on-premises data pipelines. The

company runs SQL data quality check tasks as part of the pipelines. The company wants to migrate the pipelines to AWS and to use AWS managed services.

Which solution will meet these requirements with the LEAST amount of refactoring?

A. Setup AWS Outposts in the AWS Region that is nearest to the location where the company uses Airflow. Migrate the servers into Outposts hosted Amazon EC2 instances. Update the pipelines to interact with the Outposts hosted EC2 instances instead of the on-premises pipelines.
B. Create a custom Amazon Machine Image (AMI) that contains the Airflow application and the code that the company needs to migrate. Use the custom AMI to deploy Amazon EC2 instances. Update the network connections to interact with the newly deployed EC2 instances.
C. Migrate the existing Airflow orchestration configuration into Amazon Managed Workflows for Apache Airflow (Amazon MWAA). Create the data quality checks during the ingestion to validate the data quality by using SQL tasks in Airflow.
D. Convert the pipelines to AWS Step Functions workflows. Recreate the data quality checks in SQL as Python based AWS Lambda functions.

Correct Answer: C
Section: (none)

Explanation/Reference:
Reference:

QUESTION 84
A company uses Amazon EMR as an extract, transform, and load (ETL) pipeline to transform data that comes from multiple sources. A data engineer must orchestrate the pipeline to maximize performance.

Which AWS service will meet this requirement MOST cost effectively?

A. Amazon EventBridge
B. Amazon Managed Workflows for Apache Airflow (Amazon MWAA)
C. AWS Step Functions
D. AWS Glue Workflows

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 85
An online retail company stores Application Load Balancer (ALB) access logs in an Amazon S3 bucket. The company wants to use Amazon Athena to query the logs to analyze traffic patterns.

A data engineer creates an unpartitioned table in Athena. As the amount of the data gradually increases, the response time for queries also increases. The data engineer wants to improve the query performance in Athena.

Which solution will meet these requirements with the LEAST operational effort?

A. Create an AWS Glue job that determines the schema of all ALB access logs and writes the partition metadata to AWS Glue Data Catalog.
B. Create an AWS Glue crawler that includes a classifier that determines the schema of all ALB access logs and writes the partition metadata to AWS Glue Data Catalog.
C. Create an AWS Lambda function to transform all ALB access logs. Save the results to Amazon S3 in Apache Parquet format. Partition the metadata. Use Athena to query the transformed data.
D. Use Apache Hive to create bucketed tables. Use an AWS Lambda function to transform all ALB access logs.

Correct Answer: B

Section: (none)

Explanation/Reference:
Reference:

QUESTION 86
A company has a business intelligence platform on AWS. The company uses an AWS Storage Gateway Amazon S3 File Gateway to transfer files from the company's on-premises environment to an Amazon S3 bucket.

A data engineer needs to setup a process that will automatically launch an AWS Glue workflow to run a series of AWS Glue jobs when each file transfer finishes successfully.

Which solution will meet these requirements with the LEAST operational overhead?

A. Determine when the file transfers usually finish based on previous successful file transfers. Set up an Amazon EventBridge scheduled event to initiate the AWS Glue jobs at that time of day.
B. Set up an Amazon EventBridge event that initiates the AWS Glue workflow after every successful S3 File Gateway file transfer event.
C. Set up an on-demand AWS Glue workflow so that the data engineer can start the AWS Glue workflow when each file transfer is complete.
D. Set up an AWS Lambda function that will invoke the AWS Glue Workflow. Set up an event for the creation of an S3 object as a trigger for the Lambda function.

Correct Answer: B
Section: (none)

Explanation/Reference:

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QUESTION 87
A retail company uses Amazon Aurora PostgreSQL to process and store live transactional data. The company uses an Amazon Redshift cluster for a data warehouse.

An extract, transform, and load (ETL) job runs every morning to update the Redshift cluster with new data from the PostgreSQL database. The company has grown rapidly and needs to cost optimize the Redshift cluster.

A data engineer needs to create a solution to archive historical data. The data engineer must be able to run analytics queries that effectively combine data from live transactional data in PostgreSQL, current data in Redshift, and archived historical data. The solution must keep only the most recent 15 months of data in Amazon Redshift to reduce costs.

Which combination of steps will meet these requirements? (Choose two.)

A. Configure the Amazon Redshift Federated Query feature to query live transactional data that is in the PostgreSQL database.
B. Configure Amazon Redshift Spectrum to query live transactional data that is in the PostgreSQL database.
C. Schedule a monthly job to copy data that is older than 15 months to Amazon S3 by using the UNLOAD command. Delete the old data from the Redshift cluster. Configure Amazon Redshift Spectrum to access historical data in Amazon S3.
D. Schedule a monthly job to copy data that is older than 15 months to Amazon S3 Glacier Flexible Retrieval by using the UNLOAD command. Delete the old data from the Redshift cluster. Configure Redshift Spectrum to access historical data from S3 Glacier Flexible Retrieval.
E. Create a materialized view in Amazon Redshift that combines live, current, and historical data from different sources.

Correct Answer: CD
Section: (none)

Explanation/Reference:

QUESTION 88
A manufacturing company has many IoT devices in facilities around the world. The company uses Amazon Kinesis Data Streams to collect data from the devices. The data includes device ID, capture date, measurement type, measurement value, and facility ID. The company uses facility ID as the partition key.

The company's operations team recently observed many WriteThroughputExceeded exceptions. The operations team found that some shards were heavily used but other shards were generally idle.

How should the company resolve the issues that the operations team observed?

A. Change the partition key from facility ID to a randomly generated key.
B. Increase the number of shards.
C. Archive the data on the producer's side.
D. Change the partition key from facility ID to capture date.

Correct Answer: A
Section: (none)

Explanation/Reference:
Reference:

QUESTION 89
A data engineer wants to improve the performance of SQL queries in Amazon Athena that run against a sales data table.

The data engineer wants to understand the execution plan of a specific SQL statement. The data engineer also wants to see the computational cost of each operation in a SQL query.

Which statement does the data engineer need to run to meet these requirements?

A. `EXPLAIN SELECT * FROM sales;`
B. `EXPLAIN ANALYZE FROM sales;`
C. `EXPLAIN ANALYZE SELECT * FROM sales;`
D. `EXPLAIN FROM sales;`

Correct Answer: C
Section: (none)

Explanation/Reference:
Reference:

QUESTION 90
A company plans to provision a log delivery stream within a VPC. The company configured the VPC flow logs to publish to Amazon CloudWatch Logs. The company needs to send the flow logs to Splunk in near real time for further analysis.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure an Amazon Kinesis Data Streams data stream to use Splunk as the destination. Create a CloudWatch Logs subscription filter to send log events to the data stream.
B. Create an Amazon Kinesis Data Firehose delivery stream to use Splunk as the destination. Create a CloudWatch Logs subscription filter to send log events to the delivery stream.
C. Create an Amazon Kinesis Data Firehose delivery stream to use Splunk as the destination. Create an AWS Lambda function to send the flow logs from CloudWatch Logs to the delivery stream.
D. Configure an Amazon Kinesis Data Streams data stream to use Splunk as the destination. Create an AWS Lambda function to send the flow logs from CloudWatch Logs to the data stream.

Correct Answer: B
Section: (none)

Explanation/Reference:
Reference:

QUESTION 91
A company has a data lake on AWS. The data lake ingests sources of data from business units. The company uses Amazon Athena for queries. The storage layer is Amazon S3 with an AWS Glue Data Catalog as a metadata repository.

The company wants to make the data available to data scientists and business analysts. However, the company first needs to manage fine-grained, column-level data access for Athena based on the user roles and responsibilities.

Which solution will meet these requirements?

A. Set up AWS Lake Formation. Define security policy-based rules for the users and applications by IAM role in Lake Formation.
B. Define an IAM resource-based policy for AWS Glue tables. Attach the same policy to IAM user groups.
C. Define an IAM identity-based policy for AWS Glue tables. Attach the same policy to IAM roles. Associate the IAM roles with IAM groups that contain the users.
D. Create a resource share in AWS Resource Access Manager (AWS RAM) to grant access to IAM users.

Correct Answer: A
Section: (none)

Explanation/Reference:
Reference:

QUESTION 92
A company has developed several AWS Glue extract, transform, and load (ETL) jobs to validate and transform data from Amazon S3. The ETL jobs load the data into Amazon RDS for MySQL in batches once every day. The ETL jobs use a DynamicFrame to read the S3 data.

The ETL jobs currently process all the data that is in the S3 bucket. However, the company wants the jobs to process only the daily incremental data.

Which solution will meet this requirement with the LEAST coding effort?

A. Create an ETL job that reads the S3 file status and logs the status in Amazon DynamoDB.
B. Enable job bookmarks for the ETL jobs to update the state after a run to keep track of previously processed data.
C. Enable job metrics for the ETL jobs to help keep track of processed objects in Amazon CloudWatch.
D. Configure the ETL jobs to delete processed objects from Amazon S3 after each run.

Correct Answer: B
Section: (none)

Explanation/Reference:
Reference:

QUESTION 93
An online retail company has an application that runs on Amazon EC2 instances that are in a VPC. The company wants to collect flow logs for the VPC and analyze network traffic.

Which solution will meet these requirements MOST cost-effectively?

A. Publish flow logs to Amazon CloudWatch Logs. Use Amazon Athena for analytics.
B. Publish flow logs to Amazon CloudWatch Logs. Use an Amazon OpenSearch Service cluster for analytics.

C. Publish flow logs to Amazon S3 in text format. Use Amazon Athena for analytics.
D. Publish flow logs to Amazon S3 in Apache Parquet format. Use Amazon Athena for analytics.

Correct Answer: D
Section: (none)

Explanation/Reference:
Reference:

QUESTION 94
A retail company stores transactions, store locations, and customer information tables in four reserved ra3.4xlarge Amazon Redshift cluster nodes. All three tables use even table distribution.

The company updates the store location table only once or twice every few years.

A data engineer notices that Redshift queues are slowing down because the whole store location table is constantly being broadcast to all four compute nodes for most queries. The data engineer wants to speed up the query performance by minimizing the broadcasting of the store location table.

Which solution will meet these requirements in the MOST cost-effective way?

A. Change the distribution style of the store location table from EVEN distribution to ALL distribution.
B. Change the distribution style of the store location table to KEY distribution based on the column that has the highest dimension.
C. Add a join column named store_id into the sort key for all the tables.
D. Upgrade the Redshift reserved node to a larger instance size in the same instance family.

Correct Answer: A
Section: (none)

Explanation/Reference:
Reference:

QUESTION 95
A company has a data warehouse that contains a table that is named Sales. The company stores the table in Amazon Redshift. The table includes a column that is named city_name. The company wants to query the table to find all rows that have a city_name that starts with "San" or "El".

Which SQL query will meet this requirement?

A. Select * from Sales where city_name ~ ‘$(San|El)*’;
B. Select * from Sales where city_name ~ ‘^(San|El)*’;
C. Select * from Sales where city_name ~’$(San&El)*’;
D. Select * from Sales where city_name ~ ‘^(San&El)*’;

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 96
A company needs to send customer call data from its on-premises PostgreSQL database to AWS to generate near real-time insights. The solution must capture and load updates from operational data stores that run in the PostgreSQL database. The data changes continuously.

A data engineer configures an AWS Database Migration Service (AWS DMS) ongoing replication task. The task reads changes in near real time from the PostgreSQL source database transaction logs for each table. The task then sends the data to an Amazon Redshift cluster for processing.

The data engineer discovers latency issues during the change data capture (CDC) of the task. The data engineer thinks that the PostgreSQL source database is causing the high latency.

Which solution will confirm that the PostgreSQL database is the source of the high latency?

A. Use Amazon CloudWatch to monitor the DMS task. Examine the CDCIncomingChanges metric to identify delays in the CDC from the source database.
B. Verify that logical replication of the source database is configured in the postgresql.conf configuration file.
C. Enable Amazon CloudWatch Logs for the DMS endpoint of the source database. Check for error messages.
D. Use Amazon CloudWatch to monitor the DMS task. Examine the CDCLatencySource metric to identify delays in the CDC from the source database.

Correct Answer: D
Section: (none)

Explanation/Reference:
Reference:

QUESTION 97
A lab uses IoT sensors to monitor humidity, temperature, and pressure for a project. The sensors send 100 KB of data every 10 seconds. A downstream process will read the data from an Amazon S3 bucket every 30 seconds.

Which solution will deliver the data to the S3 bucket with the LEAST latency?

A. Use Amazon Kinesis Data Streams and Amazon Kinesis Data Firehose to deliver the data to the S3 bucket. Use the default buffer interval for Kinesis Data Firehose.
B. Use Amazon Kinesis Data Streams to deliver the data to the S3 bucket. Configure the stream to use 5 provisioned shards.
C. Use Amazon Kinesis Data Streams and call the Kinesis Client Library to deliver the data to the S3 bucket. Use a 5 second buffer interval from an application.
D. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) and Amazon Kinesis Data Firehose to deliver the data to the S3 bucket. Use a 5 second buffer interval for Kinesis Data Firehose.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 98
A company wants to use machine learning (ML) to perform analytics on data that is in an Amazon S3 data lake. The company has two data transformation requirements that will give consumers within the company the ability to create reports.

The company must perform daily transformations on 300 GB of data that is in a variety format that must arrive in Amazon S3 at a scheduled time. The company must perform one-time transformations of terabytes of archived data that is in the S3 data lake. The company uses Amazon Managed Workflows for Apache Airflow (Amazon MWAA) Directed Acyclic Graphs (DAGs) to orchestrate processing.

Which combination of tasks should the company schedule in the Amazon MWAA DAGs to meet these requirements MOST cost-effectively? (Choose two.)

A. For daily incoming data, use AWS Glue crawlers to scan and identify the schema.
B. For daily incoming data, use Amazon Athena to scan and identify the schema.
C. For daily incoming data, use Amazon Redshift to perform transformations.
D. For daily and archived data, use Amazon EMR to perform data transformations.
E. For archived data, use Amazon SageMaker to perform data transformations.

Correct Answer: AD
Section: (none)

Explanation/Reference:
Reference:

QUESTION 99
A retail company uses AWS Glue for extract, transform, and load (ETL) operations on a dataset that contains information about customer orders. The company wants to implement specific validation rules to ensure data accuracy and consistency.

Which solution will meet these requirements?

A. Use AWS Glue job bookmarks to track the data for accuracy and consistency.
B. Create custom AWS Glue Data Quality rulesets to define specific data quality checks.
C. Use the built-in AWS Glue Data Quality transforms for standard data quality validations.
D. Use AWS Glue Data Catalog to maintain a centralized data schema and metadata repository.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 100
An insurance company stores transaction data that the company compressed with gzip.

The company needs to query the transaction data for occasional audits.

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Which solution will meet this requirement in the MOST cost-effective way?

A. Store the data in Amazon Glacier Flexible Retrieval. Use Amazon S3 Glacier Select to query the data.
B. Store the data in Amazon S3. Use Amazon S3 Select to query the data.
C. Store the data in Amazon S3. Use Amazon Athena to query the data.
D. Store the data in Amazon Glacier Instant Retrieval. Use Amazon Athena to query the data.

Correct Answer: B
Section: (none)

Explanation/Reference:
Reference:

QUESTION 101
A data engineer finished testing an Amazon Redshift stored procedure that processes and inserts data into a table that is not mission critical. The engineer wants to automatically run the stored procedure on a daily basis.

Which solution will meet this requirement in the MOST cost-effective way?

A. Create an AWS Lambda function to schedule a cron job to run the stored procedure.
B. Schedule and run the stored procedure by using the Amazon Redshift Data API in an Amazon EC2 Spot Instance.
C. Use query editor v2 to run the stored procedure on a schedule.
D. Schedule an AWS Glue Python shell job to run the stored procedure.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 102
A marketing company collects clickstream data. The company sends the clickstream data to Amazon Kinesis Data Firehose and stores the clickstream data in Amazon S3. The company wants to build a series of dashboards that hundreds of users from multiple departments will use.

The company will use Amazon QuickSight to develop the dashboards. The company wants a solution that can scale and provide daily updates about clickstream activity.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

A. Use Amazon Redshift to store and query the clickstream data.
B. Use Amazon Athena to query the clickstream data
C. Use Amazon S3 analytics to query the clickstream data.
D. Access the query data through a QuickSight direct SQL query.
E. Access the query data through QuickSight SPICE (Super-fast, Parallel, In-memory Calculation Engine). Configure a daily refresh for the dataset.

Correct Answer: BE
Section: (none)

Explanation/Reference:
Reference:

QUESTION 103
A data engineer is building a data orchestration workflow. The data engineer plans to use a hybrid model that includes some on-premises resources and some resources that are in the cloud. The data engineer wants to prioritize portability and open source resources.

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Which service should the data engineer use in both the on-premises environment and the cloud-based environment?

A. AWS Data Exchange
B. Amazon Simple Workflow Service (Amazon SWF)
C. Amazon Managed Workflows for Apache Airflow (Amazon MWAA)
D. AWS Glue

Correct Answer: C
Section: (none)

Explanation/Reference:
Reference:

QUESTION 104
A gaming company uses a NoSQL database to store customer information. The company is planning to migrate to AWS.

The company needs a fully managed AWS solution that will handle high online transaction processing (OLTP) workload, provide single-digit millisecond performance, and provide high availability around the world.

Which solution will meet these requirements with the LEAST operational overhead?

A. Amazon Keyspaces (for Apache Cassandra)
B. Amazon DocumentDB (with MongoDB compatibility)
C. Amazon DynamoDB
D. Amazon Timestream

Correct Answer: C

Section: (none)

Explanation/Reference:
Reference:

QUESTION 105
A data engineer creates an AWS Lambda function that an Amazon EventBridge event will invoke. When the data engineer tries to invoke the Lambda function by using an EventBridge event, an AccessDeniedException message appears.

How should the data engineer resolve the exception?

A. Ensure that the trust policy of the Lambda function execution role allows EventBridge to assume the execution role.
B. Ensure that both the IAM role that EventBridge uses and the Lambda function's resource-based policy have the necessary permissions.
C. Ensure that the subnet where the Lambda function is deployed is configured to be a private subnet.
D. Ensure that EventBridge schemas are valid and that the event mapping configuration is correct.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 106
A company uses a data lake that is based on an Amazon S3 bucket. To comply with regulations, the company must apply two layers of server-side encryption to files that are uploaded to the S3 bucket. The company wants to use an AWS Lambda function to apply the necessary encryption.

Which solution will meet these requirements?

A. Use both server-side encryption with AWS KMS keys (SSE-KMS) and the Amazon S3 Encryption Client.
B. Use dual-layer server-side encryption with AWS KMS keys (DSSE-KMS).
C. Use server-side encryption with customer-provided keys (SSE-C) before files are uploaded.
D. Use server-side encryption with AWS KMS keys (SSE-KMS).

Correct Answer: B
Section: (none)

Explanation/Reference:
Reference:

QUESTION 107
A data engineer notices that Amazon Athena queries are held in a queue before the queries run.

How can the data engineer prevent the queries from queueing?

A. Increase the query result limit.
B. Configure provisioned capacity for an existing workgroup.
C. Use federated queries.
D. Allow users who run the Athena queries to an existing workgroup.

Correct Answer: B
Section: (none)

Explanation/Reference:
Reference: https://docs.aws.amazon.com/athena/latest/ug/performance-tuning.html

QUESTION 108
A data engineer needs to debug an AWS Glue job that reads from Amazon S3 and writes to Amazon Redshift. The data engineer enabled the bookmark feature for the AWS Glue job.
The data engineer has set the maximum concurrency for the AWS Glue job to 1.

The AWS Glue job is successfully writing the output to Amazon Redshift. However, the Amazon S3 files that were loaded during previous runs of the AWS Glue job are being reprocessed by subsequent runs.

What is the likely reason the AWS Glue job is reprocessing the files?

A. The AWS Glue job does not have the s3:GetObjectAcl permission that is required for bookmarks to work correctly.
B. The maximum concurrency for the AWS Glue job is set to 1.
C. The data engineer incorrectly specified an older version of AWS Glue for the Glue job.
D. The AWS Glue job does not have a required commit statement.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 109
An ecommerce company wants to use AWS to migrate data pipelines from an on-premises environment into the AWS Cloud. The company currently uses a third-party tool in the on-premises environment to orchestrate data ingestion processes.

The company wants a migration solution that does not require the company to manage servers. The solution must be able to orchestrate Python and Bash scripts. The solution must not require the company to refactor any code.

Which solution will meet these requirements with the LEAST operational overhead?

A. AWS Lambda
B. Amazon Managed Workflows for Apache Airflow (Amazon MVVAA)
C. AWS Step Functions
D. AWS Glue

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 110
A retail company stores data from a product lifecycle management (PLM) application in an on-premises MySQL database. The PLM application frequently updates the database when transactions occur.

The company wants to gather insights from the PLM application in near real time. The company wants to integrate the insights with other business datasets and to analyze the combined dataset by using an Amazon Redshift data warehouse.

The company has already established an AWS Direct Connect connection between the on-premises infrastructure and AWS.

Which solution will meet these requirements with the LEAST development effort?

A. Run a scheduled AWS Glue extract, transform, and load (ETL) job to get the MySQL database updates by using a Java Database Connectivity (JDBC) connection. Set Amazon Redshift as the destination for the ETL job.
B. Run a full load plus CDC task in AWS Database Migration Service (AWS DMS) to continuously

replicate the MySQL database changes. Set Amazon Redshift as the destination for the task.
C. Use the Amazon AppFlow SDK to build a custom connector for the MySQL database to continuously replicate the database changes. Set Amazon Redshift as the destination for the connector.
D. Run scheduled AWS DataSync tasks to synchronize data from the MySQL database. Set Amazon Redshift as the destination for the tasks.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 111
A marketing company uses Amazon S3 to store clickstream data. The company queries the data at the end of each day by using a SQL JOIN clause on S3 objects that are stored in separate buckets.

The company creates key performance indicators (KPIs) based on the objects. The company needs a serverless solution that will give users the ability to query data by partitioning the data. The solution must maintain the atomicity, consistency, isolation, and durability (ACID) properties of the data.

Which solution will meet these requirements MOST cost-effectively?

A. Amazon S3 Select
B. Amazon Redshift Spectrum
C. Amazon Athena
D. Amazon EMR

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 112
A company wants to migrate data from an Amazon RDS for PostgreSQL DB instance in the eu-east-1 Region of an AWS account named Account_A. The company will migrate the data to an Amazon Redshift cluster in the eu-west-1 Region of an AWS account named Account_B.

Which solution will give AWS Database Migration Service (AWS DMS) the ability to replicate data between two data stores?

A. Set up an AWS DMS replication instance in Account_B in eu-west-1.
B. Set up an AWS DMS replication instance in Account_B in eu-east-1.
C. Set up an AWS DMS replication instance in a new AWS account in eu-west-1.
D. Set up an AWS DMS replication instance in Account_A in eu-east-1.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 113
A company uses Amazon S3 as a data lake. The company sets up a data warehouse by using a multi-node Amazon Redshift cluster. The company organizes the data files in the data lake based on the data source of each data file.

The company loads all the data files into one table in the Redshift cluster by using a separate COPY command for each data file location. This approach takes a long time to load all the data files into the table. The company must increase the speed of the data ingestion. The company does not want to increase the cost of the process.

Which solution will meet these requirements?

A. Use a provisioned Amazon EMR cluster to copy all the data files into one folder. Use a COPY command to load the data into Amazon Redshift.
B. Load all the data files in parallel into Amazon Aurora. Run an AWS Glue job to load the data into Amazon Redshift.
C. Use an AWS Give job to copy all the data files into one folder. Use a COPY command to load the data into Amazon Redshift.
D. Create a manifest file that contains the data file locations. Use a COPY command to load the data into Amazon Redshift.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 114
A company plans to use Amazon Kinesis Data Firehose to store data in Amazon S3. The source data consists of 2 MB .csv files. The company must convert the .csv files to JSON format. The company must store the files in Apache Parquet format.

Which solution will meet these requirements with the LEAST development effort?

A. Use Kinesis Data Firehose to convert the .csv files to JSON. Use an AWS Lambda function to store the files in Parquet format.
B. Use Kinesis Data Firehose to convert the .csv files to JSON and to store the files in Parquet format.
C. Use Kinesis Data Firehose to invoke an AWS Lambda function that transforms the .csv files to JSON and stores the files in Parquet format.
D. Use Kinesis Data Firehose to invoke an AWS Lambda function that transforms the .csv files to JSON. Use Kinesis Data Firehose to store the files in Parquet format.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 115
A company is using an AWS Transfer Family server to migrate data from an on-premises environment to AWS. Company policy mandates the use of TLS 1.2 or above to encrypt the data in transit.

Which solution will meet these requirements?

A. Generate new SSH keys for the Transfer Family server. Make the old keys and the new keys available for use.
B. Update the security group rules for the on-premises network to allow only connections that use TLS 1.2 or above.
C. Update the security policy of the Transfer Family server to specify a minimum protocol version of TLS 1.2
D. Install an SSL certificate on the Transfer Family server to encrypt data transfers by using TLS 1.2.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 116
A company wants to migrate an application and an on-premises Apache Kafka server to AWS. The application processes incremental updates that an on-premises Oracle database sends to the Kafka server. The company wants to use the replatform migration strategy instead of the refactor strategy.

Which solution will meet these requirements with the LEAST management overhead?

A. Amazon Kinesis Data Streams
B. Amazon Managed Streaming for Apache Kafka (Amazon MSK) provisioned cluster
C. Amazon Kinesis Data Firehose
D. Amazon Managed Streaming for Apache Kafka (Amazon MSK) Serverless

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 117
A data engineer is building an automated extract, transform, and load (ETL) ingestion pipeline by using AWS Glue. The pipeline ingests compressed files that are in an Amazon S3 bucket. The ingestion pipeline must support incremental data processing.

Which AWS Glue feature should the data engineer use to meet this requirement?

A. Workflows
B. Triggers
C. Job bookmarks
D. Classifiers

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Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 118
A banking company uses an application to collect large volumes of transactional data. The company uses Amazon Kinesis Data Streams for real-time analytics. The company's application uses the PutRecord action to send data to Kinesis Data Streams.

A data engineer has observed network outages during certain times of day. The data engineer wants to configure exactly-once delivery for the entire processing pipeline.

Which solution will meet this requirement?

A. Design the application so it can remove duplicates during processing by embedding a unique ID in each record at the source.
B. Update the checkpoint configuration of the Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) data collection application to avoid duplicate processing of events.
C. Design the data source so events are not ingested into Kinesis Data Streams multiple times.
D. Stop using Kinesis Data Streams. Use Amazon EMR instead. Use Apache Flink and Apache Spark Streaming in Amazon EMR.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 119
A company stores logs in an Amazon S3 bucket. When a data engineer attempts to access several log files, the data engineer discovers that some files have been unintentionally deleted.

The data engineer needs a solution that will prevent unintentional file deletion in the future.

Which solution will meet this requirement with the LEAST operational overhead?

A. Manually back up the S3 bucket on a regular basis.
B. Enable S3 Versioning for the S3 bucket.
C. Configure replication for the S3 bucket.
D. Use an Amazon S3 Glacier storage class to archive the data that is in the S3 bucket.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 120
A telecommunications company collects network usage data throughout each day at a rate of several thousand data points each second. The company runs an application to process the usage data in real time. The company aggregates and stores the data in an Amazon Aurora DB instance.

Sudden drops in network usage usually indicate a network outage. The company must be able to identify sudden drops in network usage so the company can take immediate remedial actions.

Which solution will meet this requirement with the LEAST latency?

A. Create an AWS Lambda function to query Aurora for drops in network usage. Use Amazon EventBridge to automatically invoke the Lambda function every minute.
B. Modify the processing application to publish the data to an Amazon Kinesis data stream. Create an Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) application to detect drops in network usage.
C. Replace the Aurora database with an Amazon DynamoDB table. Create an AWS Lambda function to query the DynamoDB table for drops in network usage every minute. Use DynamoDB Accelerator (DAX) between the processing application and DynamoDB table.
D. Create an AWS Lambda function within the Database Activity Streams feature of Aurora to detect drops in network usage.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 121
A data engineer is processing and analyzing multiple terabytes of raw data that is in Amazon S3. The data engineer needs to clean and prepare the data. Then the data engineer needs to load the data into Amazon Redshift for analytics.

The data engineer needs a solution that will give data analysts the ability to perform complex queries. The solution must eliminate the need to perform complex extract, transform, and load (ETL) processes or to manage infrastructure.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon EMR to prepare the data. Use AWS Step Functions to load the data into Amazon Redshift. Use Amazon QuickSight to run queries.
B. Use AWS Glue DataBrew to prepare the data. Use AWS Glue to load the data into Amazon Redshift.

Use Amazon Redshift to run queries.

C. Use AWS Lambda to prepare the data. Use Amazon Kinesis Data Firehose to load the data into Amazon Redshift. Use Amazon Athena to run queries.
D. Use AWS Glue to prepare the data. Use AWS Database Migration Service (AVVS DMS) to load the data into Amazon Redshift. Use Amazon Redshift Spectrum to run queries.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 122
A company uses an AWS Lambda function to transfer files from a legacy SFTP environment to Amazon S3 buckets. The Lambda function is VPC enabled to ensure that all communications between the Lambda function and other AVS services that are in the same VPC environment will occur over a secure network.

The Lambda function is able to connect to the SFTP environment successfully. However, when the Lambda function attempts to upload files to the S3 buckets, the Lambda function returns timeout errors. A data engineer must resolve the timeout issues in a secure way.

Which solution will meet these requirements in the MOST cost-effective way?

A. Create a NAT gateway in the public subnet of the VPC. Route network traffic to the NAT gateway.
B. Create a VPC gateway endpoint for Amazon S3. Route network traffic to the VPC gateway endpoint.
C. Create a VPC interface endpoint for Amazon S3. Route network traffic to the VPC interface endpoint.
D. Use a VPC internet gateway to connect to the internet. Route network traffic to the VPC internet gateway.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 123
A company reads data from customer databases that run on Amazon RDS. The databases contain many inconsistent fields. For example, a customer record field that iPnamed place_id in one database is named location_id in another database. The company needs to link customer records across different databases, even when customer record fields do not match.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create a provisioned Amazon EMR cluster to process and analyze data in the databases. Connect to the Apache Zeppelin notebook. Use the FindMatches transform to find duplicate records in the data.
B. Create an AWS Glue crawler to craw the databases. Use the FindMatches transform to find duplicate records in the data. Evaluate and tune the transform by evaluating the performance and results.
C. Create an AWS Glue crawler to craw the databases. Use Amazon SageMaker to construct Apache Spark ML pipelines to find duplicate records in the data.
D. Create a provisioned Amazon EMR cluster to process and analyze data in the databases. Connect to the Apache Zeppelin notebook. Use an Apache Spark ML model to find duplicate records in the data. Evaluate and tune the model by evaluating the performance and results.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 124

A finance company receives data from third-party data providers and stores the data as objects in an Amazon S3 bucket.

The company ran an AWS Glue crawler on the objects to create a data catalog. The AWS Glue crawler created multiple tables. However, the company expected that the crawler would create only one table.

The company needs a solution that will ensure the AVS Glue crawler creates only one table.

Which combination of solutions will meet this requirement? (Choose two.)

A. Ensure that the object format, compression type, and schema are the same for each object.
B. Ensure that the object format and schema are the same for each object. Do not enforce consistency for the compression type of each object.
C. Ensure that the schema is the same for each object. Do not enforce consistency for the file format and compression type of each object.
D. Ensure that the structure of the prefix for each S3 object name is consistent.
E. Ensure that all S3 object names follow a similar pattern.

Correct Answer: AD
Section: (none)

Explanation/Reference:

QUESTION 125
An application consumes messages from an Amazon Simple Queue Service (Amazon SQS) queue. The application experiences occasional downtime. As a result of the downtime, messages within the queue expire and are deleted after 1 day. The message deletions cause data loss for the application.

Which solutions will minimize data loss for the application? (Choose two.)

A. Increase the message retention period
B. Increase the visibility timeout.
C. Attach a dead-letter queue (DLQ) to the SQS queue.
D. Use a delay queue to delay message delivery
E. Reduce message processing time.

Correct Answer: AC
Section: (none)

Explanation/Reference:

QUESTION 126
A company is creating near real-time dashboards to visualize time series data. The company ingests data into Amazon Managed Streaming for Apache Kafka (Amazon MSK). A customized data pipeline consumes the data. The pipeline then writes data to Amazon Keyspaces (for Apache Cassandra), Amazon OpenSearch Service, and Apache Avro objects in Amazon S3.

Which solution will make the data available for the data visualizations with the LEAST latency?

A. Create OpenSearch Dashboards by using the data from OpenSearch Service.
B. Use Amazon Athena with an Apache Hive metastore to query the Avro objects in Amazon S3. Use Amazon Managed Grafana to connect to Athena and to create the dashboards.
C. Use Amazon Athena to query the data from the Avro objects in Amazon S3. Configure Amazon Keyspaces as the data catalog. Connect Amazon QuickSight to Athena to create the dashboards.
D. Use AWS Glue to catalog the data. Use S3 Select to query the Avro objects in Amazon S3. Connect Amazon QuickSight to the S3 bucket to create the dashboards.

Correct Answer: A

Section: (none)

Explanation/Reference:

QUESTION 127
A media company wants to use Amazon OpenSearch Service to analyze rea-time data about popular musical artists and songs. The company expects to ingest millions of new data events every day. The new data events will arrive through an Amazon Kinesis data stream. The company must transform the data and then ingest the data into the OpenSearch Service domain.

Which method should the company use to ingest the data with the LEAST operational overhead?

A. Use Amazon Kinesis Data Firehose and an AWS Lambda function to transform the data and deliver the transformed data to OpenSearch Service.
B. Use a Logstash pipeline that has prebuilt filters to transform the data and deliver the transformed data to OpenSearch Service.
C. Use an AWS Lambda function to call the Amazon Kinesis Agent to transform the data and deliver the transformed data OpenSearch Service.
D. Use the Kinesis Client Library (KCL) to transform the data and deliver the transformed data to OpenSearch Service.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 128
A company stores customer data tables that include customer addresses in an AWS Lake Formation data lake. To comply with new regulations, the company must ensure that users cannot access data for customers who are in Canada.

The company needs a solution that will prevent user access to rows for customers who are in Canada.

Which solution will meet this requirement with the LEAST operational effort?

A. Set a row-level filter to prevent user access to a row where the country is Canada.
B. Create an IAM role that restricts user access to an address where the country is Canada.
C. Set a column-level filter to prevent user access to a row where the country is Canada.
D. Apply a tag to all rows where Canada is the country. Prevent user access where the tag is equal to “Canada”.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 129
A company has implemented a lake house architecture in Amazon Redshift. The company needs to give users the ability to authenticate into Redshift query editor by using a third-party identity provider (IdP).

A data engineer must set up the authentication mechanism.

What is the first step the data engineer should take to meet this requirement?

A. Register the third-party IdP as an identity provider in the configuration settings of the Redshift cluster.
B. Register the third-party IdP as an identity provider from within Amazon Redshift.
C. Register the third-party IdP as an identity provider for AVS Secrets Manager. Configure Amazon

Redshift to use Secrets Manager to manage user credentials.
D. Register the third-party IdP as an identity provider for AWS Certificate Manager (ACM). Configure Amazon Redshift to use ACM to manage user credentials.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 130
A company currently uses a provisioned Amazon EMR cluster that includes general purpose Amazon EC2 instances. The EMR cluster uses EMR managed scaling between one to five task nodes for the company's long-running Apache Spark extract, transform, and load (ETL) job. The company runs the ETL job every day.

When the company runs the ETL job, the EMR cluster quickly scales up to five nodes. The EMR cluster often reaches maximum CPU usage, but the memory usage remains under 30%.

The company wants to modify the EMR cluster configuration to reduce the EMR costs to run the daily ETL job.

Which solution will meet these requirements MOST cost-effectively?

A. Increase the maximum number of task nodes for EMR managed scaling to 10.
B. Change the task node type from general purpose EC2 instances to memory optimized EC2 instances.
C. Switch the task node type from general purpose EC2 instances to compute optimized EC2 instances.
D. Reduce the scaling cooldown period for the provisioned EMR cluster.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 131
A company uploads .csv files to an Amazon S3 bucket. The company's data platform team has set up an AWS Glue crawler to perform data discovery and to create the tables and schemas.

An AWS Glue job writes processed data from the tables to an Amazon Redshift database. The AWS Glue job handles column mapping and creates the Amazon Redshift tables in the Redshift database appropriately.

If the company reruns the AWS Glue job for any reason, duplicate records are introduced into the Amazon Redshift tables. The company needs a solution that will update the Redshift tables without duplicates.

Which solution will meet these requirements?

A. Modify the AWS Glue job to copy the rows into a staging Redshift table. Add SQL commands to update the existing rows with new values from the staging Redshift table.
B. Modify the AWS Glue job to load the previously inserted data into a MySQL database. Perform an upsert operation in the MySQL database. Copy the results to the Amazon Redshift tables.
C. Use Apache Spark's DataFrame dropDuplicates() API to eliminate duplicates. Write the data to the Redshift tables.
D. Use the AWS Glue ResolveChoice built-in transform to select the value of the column from the most recent record.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 132
A company is using Amazon Redshift to build a data warehouse solution. The company is loading hundreds of files into a fact table that is in a Redshift cluster.

The company wants the data warehouse solution to achieve the greatest possible throughput. The solution must use cluster resources optimally when the company loads data into the fact table.

Which solution will meet these requirements?

A. Use multiple COPY commands to load the data into the Redshift cluster.
B. Use S3DistCp to load multiple files into Hadoop Distributed File System (HDFS). Use an HDFS connector to ingest the data into the Redshift cluster.
C. Use a number of INSERT statements equal to the number of Redshift cluster nodes. Load the data in parallel into each node.
D. Use a single COPY command to load the data into the Redshift cluster.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 133
A company ingests data from multiple data sources and stores the data in an Amazon S3 bucket. An AWS Glue extract, transform, and load (ETL) job transforms the data and writes the transformed data to an Amazon S3 based data lake. The company uses Amazon Athena to query the data that is in the data lake.

The company needs to identify matching records even when the records do not have a common unique identifier.

Which solution will meet this requirement?

A. Use Amazon Macie pattern matching as part of the ETL job.
B. Train and use the AWS Glue PySpark Filter class in the ETL job.
C. Partition tables and use the ETL job to partition the data on a unique identifier.
D. Train and use the AWS Lake Formation FindMatches transform in the ETL job.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 134
A data engineer is using an AWS Glue crawler to catalog data that is in an Amazon S3 bucket. The S3 bucket contains both .csv and json files. The data engineer configured the crawler to exclude the .json files from the catalog.

When the data engineer runs queries in Amazon Athena, the queries also process the excluded .json files. The data engineer wants to resolve this issue. The data engineer needs a solution that will not affect access requirements for the .csv files in the source S3 bucket.

Which solution will meet this requirement with the SHORTEST query times?

A. Adjust the AWS Glue crawler settings to ensure that the AWS Glue crawler also excludes .json files.
B. Use the Athena console to ensure the Athena queries also exclude the .json files.
C. Relocate the .json files to a different path within the S3 bucket.
D. Use S3 bucket policies to block access to the .json files.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 135
A data engineer set up an AWS Lambda function to read an object that is stored in an Amazon S3 bucket. The object is encrypted by an AWS KMS key.

The data engineer configured the Lambda function's execution role to access the S3 bucket. However, the Lambda function encountered an error and failed to retrieve the content of the object.

What is the likely cause of the error?

A. The data engineer misconfigured the permissions of the S3 bucket. The Lambda function could not access the object.
B. The Lambda function is using an outdated SDK version, which caused the read failure.
C. The S3 bucket is located in a different AWS Region than the Region where the data engineer works. Latency issues caused the Lambda function to encounter an error.
D. The Lambda function's execution role does not have the necessary permissions to access the KMS key that can decrypt the S3 object.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 136
A data engineer has implemented data quality rules in 1,000 AWS Glue Data Catalog tables. Because of a recent change in business requirements, the data engineer must edit the data quality rules.

How should the data engineer meet this requirement with the LEAST operational overhead?

A. Create a pipeline in AWS Glue ETL to edit the rules for each of the 1,000 Data Catalog tables. Use an AWS Lambda function to call the corresponding AWS Glue job for each Data Catalog table.
B. Create an AWS Lambda function that makes an API call to AWS Glue Data Quality to make the edits.
C. Create an Amazon EMR cluster. Run a pipeline on Amazon EMR that edits the rules for each Data Catalog table. Use an AWS Lambda function to run the EMR pipeline.
D. Use the AWS Management Console to edit the rules within the Data Catalog.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 137
Two developers are working on separate application releases. The developers have created feature branches named Branch A and Branch B by using a GitHub repository's master branch as the source.

The developer for Branch A deployed code to the production system. The code for Branch B will merge into a master branch in the following week's scheduled application release.

Which command should the developer for Branch B run before the developer raises a pull request to the master branch?

A. git diff branchB master
git commit -m <message>

B. git pull master
C. git rebase master
D. git fetch -b master

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 138
A company stores employee data in Amazon Redshift. A table names Employee uses columns named Region ID, Department ID, and Role ID as a compound sort key.

Which queries will MOST increase the speed of query by using a compound sort key of the table? (Choose two.)

A. Select *from Employee where Region ID='North America';
B. Select *from Employee where Region ID='North America' and Department ID=20;
C. Select *from Employee where Department ID=20 and Region ID='North America';
D. Select *from Employee where Role ID=50;
E. Select *from Employee where Region ID='North America' and Role ID=50;

Correct Answer: BE
Section: (none)

Explanation/Reference:

QUESTION 139
A company receives test results from testing facilities that are located around the world. The company stores the test results in millions of 1 KB JSON files in an Amazon S3 bucket. A data engineer needs to process the files, convert them into Apache Parquet format, and load them into Amazon Redshift tables. The data engineer uses AWS Glue to process the files, AWS Step Functions to orchestrate the processes, and Amazon EventBridge to schedule jobs.

The company recently added more testing facilities. The time required to process files is increasing. The data engineer must reduce the data processing time.

Which solution will MOST reduce the data processing time?

A. Use AWS Lambda to group the raw input files into larger files. Write the larger files back to Amazon S3. Use AWS Glue to process the files. Load the files into the Amazon Redshift tables.
B. Use the AWS Glue dynamic frame file-grouping option to ingest the raw input files. Process the files. Load the files into the Amazon Redshift tables.
C. Use the Amazon Redshift COPY command to move the raw input files from Amazon S3 directly into the Amazon Redshift tables. Process the files in Amazon Redshift.
D. Use Amazon EMR instead of AWS Glue to group the raw input files. Process the files in Amazon EMR. Load the files into the Amazon Redshift tables.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 140
A data engineer uses Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to run data pipelines in an AWS account.

A workflow recently failed to run. The data engineer needs to use Apache Airflow logs to diagnose the failure of the workflow.

Which log type should the data engineer use to diagnose the cause of the failure?

A. YourEnvironmentName-WebServer
B. YourEnvironmentName-Scheduler
C. YourEnvironmentName-DAGProcessing
D. YourEnvironmentName-Task

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 141
A finance company uses Amazon Redshift as a data warehouse. The company stores the data in a shared Amazon S3 bucket. The company uses Amazon Redshift Spectrum to access the data that is stored in the S3 bucket. The data comes from certified third-party data providers. Each third-party data provider has unique connection details.

To comply with regulations, the company must ensure that none of the data is accessible from outside the company's AWS environment.

Which combination of steps should the company take to meet these requirements? (Choose two.)

A. Replace the existing Redshift cluster with a new Redshift cluster that is in a private subnet. Use an interface VPC endpoint to connect to the Redshift cluster. Use a NAT gateway to give Redshift access to the S3 bucket.
B. Create an AWS CloudHSM hardware security module (HSM) for each data provider. Encrypt each data provider's data by using the corresponding HSM for each data provider.
C. Turn on enhanced VPC routing for the Amazon Redshift cluster. Set up an AWS Direct Connect connection and configure a connection between each data provider and the finance company's VPC.
D. Define table constraints for the primary keys and the foreign keys.
E. Use federated queries to access the data from each data provider. Do not upload the data to the S3 bucket. Perform the federated queries through a gateway VPC endpoint.

Correct Answer: AC
Section: (none)

Explanation/Reference:

QUESTION 142
Files from multiple data sources arrive in an Amazon S3 bucket on a regular basis. A data engineer wants to ingest new files into Amazon Redshift in near real time when the new files arrive in the S3 bucket.

Which solution will meet these requirements?

A. Use the query editor v2 to schedule a COPY command to load new files into Amazon Redshift.
B. Use the zero-ETL integration between Amazon Aurora and Amazon Redshift to load new files into Amazon Redshift.
C. Use AWS Glue job bookmarks to extract, transform, and load (ETL) load new files into Amazon Redshift.
D. Use S3 Event Notifications to invoke an AWS Lambda function that loads new files into Amazon Redshift.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 143
A technology company currently uses Amazon Kinesis Data Streams to collect log data in real time. The company wants to use Amazon Redshift for downstream real-time queries and to enrich the log data.

Which solution will ingest data into Amazon Redshift with the LEAST operational overhead?

A. Set up an Amazon Kinesis Data Firehose delivery stream to send data to a Redshift provisioned cluster table.
B. Set up an Amazon Kinesis Data Firehose delivery stream to send data to Amazon S3. Configure a Redshift provisioned cluster to load data every minute.
C. Configure Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to send data directly to a Redshift provisioned cluster table.
D. Use Amazon Redshift streaming ingestion from Kinesis Data Streams and to present data as a materialized view.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 144
A company maintains a data warehouse in an on-premises Oracle database. The company wants to build a data lake on AWS. The company wants to load data warehouse tables into Amazon S3 and synchronize the tables with incremental data that arrives from the data warehouse every day.

Each table has a column that contains monotonically increasing values. The size of each table is less than 50 GB. The data warehouse tables are refreshed every night between 1 AM and 2 AM. A business intelligence team queries the tables between 10 AM and 8 PM every day.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Use an AWS Database Migration Service (AWS DMS) full load plus CDC job to load tables that contain monotonically increasing data columns from the on-premises data warehouse to Amazon S3. Use custom logic in AWS Glue to append the daily incremental data to a full-load copy that is in Amazon S3.
B. Use an AWS Glue Java Database Connectivity (JDBC) connection. Configure a job bookmark for a column that contains monotonically increasing values. Write custom logic to append the daily incremental data to a full-load copy that is in Amazon S3.
C. Use an AWS Database Migration Service (AWS DMS) full load migration to load the data warehouse tables into Amazon S3 every day. Overwrite the previous day's full-load copy every day.
D. Use AWS Glue to load a full copy of the data warehouse tables into Amazon S3 every day. Overwrite the previous day's full-load copy every day.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 145
A company is building a data lake for a new analytics team. The company is using Amazon S3 for storage and Amazon Athena for query analysis. All data that is in Amazon S3 is in Apache Parquet format.

The company is running a new Oracle database as a source system in the company’s data center. The company has 70 tables in the Oracle database. All the tables have primary keys. Data can occasionally change in the source system. The company wants to ingest the tables every day into the data lake.

Which solution will meet this requirement with the LEAST effort?

A. Create an Apache Sqoop job in Amazon EMR to read the data from the Oracle database. Configure the Sqoop job to write the data to Amazon S3 in Parquet format.
B. Create an AWS Glue connection to the Oracle database. Create an AWS Glue bookmark job to ingest the data incrementally and to write the data to Amazon S3 in Parquet format.
C. Create an AWS Database Migration Service (AWS DMS) task for ongoing replication. Set the Oracle database as the source. Set Amazon S3 as the target. Configure the task to write the data in Parquet format.
D. Create an Oracle database in Amazon RDS. Use AWS Database Migration Service (AWS DMS) to migrate the on-premises Oracle database to Amazon RDS. Configure triggers on the tables to invoke AWS Lambda functions to write changed records to Amazon S3 in Parquet format.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 146
A transportation company wants to track vehicle movements by capturing geolocation records. The records are 10 bytes in size. The company receives up to 10.000 records every second. Data transmission delays of a few minutes are acceptable because of unreliable network conditions.

The transportation company wants to use Amazon Kinesis Data Streams to ingest the geolocation data. The company needs a reliable mechanism to send data to Kinesis Data Streams. The company needs to maximize the throughput efficiency of the Kinesis shards.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Kinesis Agent
B. Kinesis Producer Library (KPL)
C. Amazon Kinesis Data Firehose
D. Kinesis SDK

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 147
An investment company needs to manage and extract insights from a volume of semi-structured data that grows continuously.

A data engineer needs to deduplicate the semi-structured data, remove records that are duplicates, and remove common misspellings of duplicates.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use the FindMatches feature of AWS Glue to remove duplicate records.
B. Use non-Windows functions in Amazon Athena to remove duplicate records.
C. Use Amazon Neptune ML and an Apache Gremlin script to remove duplicate records.
D. Use the global tables feature of Amazon DynamoDB to prevent duplicate data.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 148
A company is building an inventory management system and an inventory reordering system to automatically reorder products. Both systems use Amazon Kinesis Data Streams. The inventory management system uses the Amazon Kinesis Producer Library (KPL) to publish data to a stream. The inventory reordering system uses the Amazon Kinesis Client Library (KCL) to consume data from the stream. The company configures the stream to scale up and down as needed.

Before the company deploys the systems to production, the company discovers that the inventory reordering system received duplicated data.

Which factors could have caused the reordering system to receive duplicated data? (Choose two.)

A. The producer experienced network-related timeouts.
B. The stream's value for the IteratorAgeMilliseconds metric was too high.
C. There was a change in the number of shards, record processors, or both.
D. The AggregationEnabled configuration property was set to true.
E. The max_records configuration property was set to a number that was too high.

Correct Answer: AC
Section: (none)

Explanation/Reference:

QUESTION 149
An ecommerce company operates a complex order fulfilment process that spans several operational systems hosted in AWS. Each of the operational systems has a Java Database Connectivity (JDBC)-compliant relational database where the latest processing state is captured.

The company needs to give an operations team the ability to track orders on an hourly basis across the entire fulfillment process.

Which solution will meet these requirements with the LEAST development overhead?

A. Use AWS Glue to build ingestion pipelines from the operational systems into Amazon Redshift Build dashboards in Amazon QuickSight that track the orders.
B. Use AWS Glue to build ingestion pipelines from the operational systems into Amazon DynamoDB. Build dashboards in Amazon QuickSight that track the orders.
C. Use AWS Database Migration Service (AWS DMS) to capture changed records in the operational systems. Publish the changes to an Amazon DynamoDB table in a different AWS region from the source database. Build Grafana dashboards that track the orders.
D. Use AWS Database Migration Service (AWS DMS) to capture changed records in the operational systems. Publish the changes to an Amazon DynamoDB table in a different AWS region from the source database. Build Amazon QuickSight dashboards that track the orders.

Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 150
A data engineer needs to use Amazon Neptune to develop graph applications.

Which programming languages should the engineer use to develop the graph applications? (Choose two.)

A. Gremlin
B. SQL

C. ANSI SQL
D. SPARQL
E. Spark SQL

Correct Answer: AD
Section: (none)

Explanation/Reference:

QUESTION 151
A mobile gaming company wants to capture data from its gaming app. The company wants to make the data available to three internal consumers of the data. The data records are approximately 20 KB in size.

The company wants to achieve optimal throughput from each device that runs the gaming app. Additionally, the company wants to develop an application to process data streams. The stream-processing application must have dedicated throughput for each internal consumer.

Which solution will meet these requirements?

A. Configure the mobile app to call the PutRecords API operation to send data to Amazon Kinesis Data Streams. Use the enhanced fan-out feature with a stream for each internal consumer.
B. Configure the mobile app to call the PutRecordBatch API operation to send data to Amazon Kinesis Data Firehose. Submit an AWS Support case to turn on dedicated throughput for the company's AWS account. Allow each internal consumer to access the stream.
C. Configure the mobile app to use the Amazon Kinesis Producer Library (KPL) to send data to Amazon Kinesis Data Firehose. Use the enhanced fan-out feature with a stream for each internal consumer.
D. Configure the mobile app to call the PutRecords API operation to send data to Amazon Kinesis Data Streams. Host the stream-processing application for each internal consumer on Amazon EC2 instances. Configure auto scaling for the EC2 instances.

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 152
A data engineer maintains a materialized view that is based on an Amazon Redshift database. The view has a column named load_date that stores the date when each row was loaded.

The data engineer needs to reclaim database storage space by deleting all the rows from the materialized view.

Which command will reclaim the MOST database storage space?

A. `DELETE FROM materialized_view_name where 1=1`
B. `TRUNCATE materialized_view_name`
C. `VACUUM table_name where load_date<=current_date`
materializedview
D. `DELETE FROM materialized_view_name where load_date<=current_date`

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 153
A retail company uses an Amazon Redshift data warehouse and an Amazon S3 bucket. The company ingests retail order data into the S3 bucket every day.

The company stores all order data at a single path within the S3 bucket. The data has more than 100 columns. The company ingests the order data from a third-party application that generates more than 30 files in CSV format every day. Each CSV file is between 50 and 70 MB in size.

The company uses Amazon Redshift Spectrum to run queries that select sets of columns. Users aggregate metrics based on daily orders. Recently, users have reported that the performance of the queries has degraded. A data engineer must resolve the performance issues for the queries.

Which combination of steps will meet this requirement with LEAST developmental effort? (Choose two.)

A. Configure the third-party application to create the files in a columnar format.
B. Develop an AWS Glue ETL job to convert the multiple daily CSV files to one file for each day.
C. Partition the order data in the S3 bucket based on order date.
D. Configure the third-party application to create the files in JSON format.
E. Load the JSON data into the Amazon Redshift table in a SUPER type column.

Correct Answer: AC
Section: (none)

Explanation/Reference:
Explanation:
A columnar format like Parquet or ORC optimizes query performance, especially when using services like Redshift Spectrum. Redshift Spectrum allows querying data directly in S3, and columnar formats help reduce the amount of data scanned during queries because only the needed columns are read. This reduces I/O and speeds up the query performance without needing to load all the columns of the dataset. This change is highly beneficial, especially when querying large datasets with many columns like in this scenario.

Partitioning the data in Amazon S3 helps Redshift Spectrum prune unnecessary data, improving query performance. Partitioning by a frequently filtered column like order date allows Redshift Spectrum to scan only relevant partitions, reducing the amount of data that needs to be processed. This leads to faster query times.

While combining files might reduce the number of files and improve performance slightly, it doesn't address the core issue of optimizing the data format and partitioning, which have a much bigger impact on performance.

JSON is not an efficient format for large-scale analytics and tends to have worse performance compared to columnar formats. Columnar formats like Parquet or ORC are preferable in this case.

The SUPER type is useful for semi-structured data in Redshift but isn't directly related to improving query performance in this scenario, where columnar formats and partitioning would provide more benefit.

QUESTION 154
A company stores customer records in Amazon S3. The company must not delete or modify the customer record data for 7 years after each record is created. The root user also must not have the ability to delete or modify the data.

A data engineer wants to use S3 Object Lock to secure the data.

Which solution will meet these requirements?

A. Enable governance mode on the S3 bucket. Use a default retention period of 7 years.
B. Enable compliance mode on the S3 bucket. Use a default retention period of 7 years.
C. Place a legal hold on individual objects in the S3 bucket. Set the retention period to 7 years.
D. Set the retention period for individual objects in the S3 bucket to 7 years.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Compliance Mode is the stricter mode of S3 Object Lock. When enabled, it ensures that no one, not even the root user, can modify or delete the object until the retention period expires. In this case, to meet the requirement of preventing any deletion or modification of the data for 7 years, compliance mode is the correct choice.

Governance Mode allows some users with special permissions (such as the root user) to override the lock and delete or modify the data, which does not meet the strict requirement that even the root user should not have this ability.

Legal Hold is used to prevent deletion of objects without specifying a time-based retention period, but it can be removed by users with sufficient permissions, and the root user might still be able to delete the data.

Retention Period for Individual Objects is possible, but enabling compliance mode provides a more robust solution by applying the same 7-year retention policy bucket-wide, ensuring that no modifications or deletions can occur during that period.

QUESTION 155
A data engineer needs to create a new empty table in Amazon Athena that has the same schema as an existing table named old_table.

Which SQL statement should the data engineer use to meet this requirement?

A. `CREATE TABLE new_table AS SELECT * FROM old_tables;`
B. `INSERT INTO new_table SELECT * FROM old_table;`
C. `CREATE TABLE new_table (LIKE old_table);`
D. `CREATE TABLE new_table AS (SELECT * FROM old_table) WITH NO DATA;`

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Correct Answer: D
Section: (none)

Explanation/Reference:

QUESTION 156
A data engineer needs to create an Amazon Athena table based on a subset of data from an existing Athena table named cities_world. The cities_world table contains cities that are located around the world. The data engineer must create a new table named cities_us to contain only the cities from cities_world that are located in the US.

Which SQL statement should the data engineer use to meet this requirement?

A. `INSERT INTO cities_usa (city,state) SELECT city, state FROM cities_world WHERE country=’usa’;`
B. `MOVE city, state FROM cities_world TO cities_usa WHERE country=’usa’;`
C. `INSERT INTO cities_usa SELECT city, state FROM cities_world WHERE country=’usa’;`
D. `UPDATE cities_usa SET (city, state) = (SELECT city, state FROM cities_world WHERE country=’usa’);`

Correct Answer: A
Section: (none)

Explanation/Reference:

QUESTION 157
A company implements a data mesh that has a central governance account. The company needs to catalog all data in the governance account. The governance account uses AWS Lake Formation to centrally share data and grant access permissions.

The company has created a new data product that includes a group of Amazon Redshift Serverless tables. A data engineer needs to share the data product with a marketing team. The marketing team must have access to only a subset of columns. The data engineer needs to share the same data product with a compliance team. The compliance team must have access to a different subset of columns than the marketing team needs access to.

Which combination of steps should the data engineer take to meet these requirements? (Choose two.)

A. Create views of the tables that need to be shared. Include only the required columns.
B. Create an Amazon Redshift data share that includes the tables that need to be shared.
C. Create an Amazon Redshift managed VPC endpoint in the marketing team’s account. Grant the marketing team access to the views.
D. Share the Amazon Redshift data share to the Lake Formation catalog in the governance account.
E. Share the Amazon Redshift data share to the Amazon Redshift Serverless workgroup in the marketing team's account.

Correct Answer: BD
Section: (none)

Explanation/Reference:

QUESTION 158
A company has a data lake in Amazon S3. The company uses AWS Glue to catalog data and AWS Glue Studio to implement data extract, transform, and load (ETL) pipelines.

The company needs to ensure that data quality issues are checked every time the pipelines run. A data engineer must enhance the existing pipelines to evaluate data quality rules based on predefined thresholds.

Which solution will meet these requirements with the LEAST implementation effort?

A. Add a new transform that is defined by a SQL query to each Glue ETL job. Use the SQL query to implement a ruleset that includes the data quality rules that need to be evaluated.
B. Add a new Evaluate Data Quality transform to each Glue ETL job. Use Data Quality Definition Language (DQDL) to implement a ruleset that includes the data quality rules that need to be evaluated.
C. Add a new custom transform to each Glue ETL job. Use the PyDeequ library to implement a ruleset that includes the data quality rules that need to be evaluated.
D. Add a new custom transform to each Glue ETL job. Use the Great Expectations library to implement a ruleset that includes the data quality rules that need to be evaluated.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Evaluate Data Quality Transform in AWS Glue is a built-in feature that allows the application of Data Quality Definition Language (DQDL) to define and apply data quality rules directly within AWS Glue ETL jobs. This option offers the least implementation effort because it leverages a native AWS Glue feature, allowing data engineers to easily add and enforce data quality checks without needing custom code or external libraries.

While SQL can implement data quality rules, this approach requires manually writing and managing SQL queries within each job, leading to higher implementation effort compared to using built-in Glue functionality.

PyDeequ is an external library that requires custom coding and additional integration effort. Although it is effective, it demands more implementation effort than using AWS Glue’s built-in features.

Great Expectations is another external data quality library. While it is robust, it also involves custom coding and higher maintenance effort compared to AWS Glue's native data quality evaluation tools.

QUESTION 159
A company has an application that uses a microservice architecture. The company hosts the application on an Amazon Elastic Kubernetes Services (Amazon EKS) cluster.

The company wants to set up a robust monitoring system for the application. The company needs to analyze the logs from the EKS cluster and the application. The company needs to correlate the cluster's logs with the application's traces to identify points of failure in the whole application request flow.

Which combination of steps will meet these requirements with the LEAST development effort? (Choose two.)

A. Use FluentBit to collect logs. Use OpenTelemetry to collect traces.
B. Use Amazon CloudWatch to collect logs. Use Amazon Kinesis to collect traces.
C. Use Amazon CloudWatch to collect logs. Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) to collect traces.
D. Use Amazon OpenSearch to correlate the logs and traces.
E. Use AWS Glue to correlate the logs and traces.

Correct Answer: AD
Section: (none)

Explanation/Reference:
Explanation:
FluentBit is a lightweight, efficient tool to collect, process, and forward logs from the EKS cluster. It integrates well with AWS services like CloudWatch or OpenSearch for storing and analyzing logs. OpenTelemetry is an open-source standard for collecting distributed traces, making it ideal for monitoring and analyzing application performance in a microservice architecture. Using these tools requires minimal development effort since they are widely adopted and have strong integrations with AWS.

Amazon OpenSearch (formerly Elasticsearch) is a fully managed service that makes it easy to store, search, and analyze log data, including traces from distributed systems. It is commonly used to correlate logs and traces because it provides robust querying capabilities and visualization features (via Kibana or OpenSearch Dashboards). This solution offers the necessary tools for log-trace correlation with minimal custom development effort.

While CloudWatch is suitable for log collection, Amazon Kinesis is more suited for real-time data streaming rather than collecting traces for correlation. Kinesis also requires more configuration and development effort compared to OpenTelemetry for tracing.

Similar to Kinesis, Amazon MSK is primarily used for data streaming rather than trace collection. It requires additional setup and custom development for correlation compared to the more straightforward integration provided by FluentBit and OpenTelemetry.

AWS Glue is primarily used for ETL (extract, transform, load) operations in a data lake. It is not designed for real-time log and trace correlation and would require significant development effort compared to the built-in capabilities of OpenSearch.

QUESTION 160
A company has a gaming application that stores data in Amazon DynamoDB tables. A data engineer needs to ingest the game data into an Amazon OpenSearch Service cluster. Data updates must occur in near real time.

Which solution will meet these requirements?

A. Use AWS Step Functions to periodically export data from the Amazon DynamoDB tables to an Amazon S3 bucket. Use an AWS Lambda function to load the data into Amazon OpenSearch Service.
B. Configure an AWS Glue job to have a source of Amazon DynamoDB and a destination of Amazon OpenSearch Service to transfer data in near real time.
C. Use Amazon DynamoDB Streams to capture table changes. Use an AWS Lambda function to process

and update the data in Amazon OpenSearch Service.

D. Use a custom OpenSearch plugin to sync data from the Amazon DynamoDB tables.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
DynamoDB Streams can capture changes to items in DynamoDB tables (such as inserts, updates, and deletes) in near real-time. An AWS Lambda function can be triggered by these streams, and the function can process the changes and update the data in Amazon OpenSearch Service. This solution meets the requirement for near real-time updates with minimal latency.

The “Use AWS Step Functions to periodically export data from the Amazon DynamoDB tables to an Amazon S3 bucket. Use an AWS Lambda function to load the data into Amazon OpenSearch Service.” solution involves exporting data periodically, which does not meet the requirement for near real-time updates. Periodic exports could introduce delays.

AWS Glue is generally used for batch processing, and it is not designed for near real-time data ingestion. It would not be an ideal solution for real-time or low-latency requirements.

The “Use a custom OpenSearch plugin to sync data from the Amazon DynamoDB tables.” option suggests building a custom solution, which would involve significant development effort. Moreover, OpenSearch does not natively support such plugins, making this solution less practical compared to using DynamoDB Streams and Lambda, which are designed to work together.

QUESTION 161
A company uses Amazon Redshift as its data warehouse service. A data engineer needs to design a physical data model.

The data engineer encounters a de-normalized table that is growing in size. The table does not have a suitable column to use as the distribution key.

Which distribution style should the data engineer use to meet these requirements with the LEAST maintenance overhead?

A. ALL distribution
B. EVEN distribution
C. AUTO distribution
D. KEY distribution

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AUTO distribution allows Amazon Redshift to automatically manage the distribution style based on the size of the table. With this option, Redshift chooses the most optimal distribution method, and as the table grows, it will adjust the distribution style to maintain performance. This minimizes maintenance overhead since the system adapts without the need for manual tuning.

ALL distribution copies the entire table to all nodes. It can be useful for small, static tables but is inefficient for large or growing tables as it increases storage and I/O costs significantly.

EVEN distribution distributes rows evenly across all nodes. While it reduces data skew, it may not be optimal for larger tables where certain queries would benefit from co-locating data to minimize cross-node traffic.

KEY distribution places data based on the values in a specific column (distribution key). Since the question states that no suitable column exists for a distribution key, this method would not be appropriate.

QUESTION 162

A retail company is expanding its operations globally. The company needs to use Amazon QuickSight to accurately calculate currency exchange rates for financial reports. The company has an existing dashboard that includes a visual that is based on an analysis of a dataset that contains global currency values and exchange rates.

A data engineer needs to ensure that exchange rates are calculated with a precision of four decimal places. The calculations must be precomputed. The data engineer must materialize results in QuickSight super-fast, parallel, in-memory calculation engine (SPICE).

Which solution will meet these requirements?

A. Define and create the calculated field in the dataset.
B. Define and create the calculated field in the analysis.
C. Define and create the calculated field in the visual.
D. Define and create the calculated field in the dashboard.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Defining and creating the calculated field in the dataset ensures that the exchange rate calculations are precomputed before being stored in SPICE (QuickSight's super-fast, parallel, in-memory calculation engine). This allows the calculation to be materialized and reused across different analyses and visuals, ensuring that the precision is maintained and the performance is optimized.

Creating the calculated field in the analysis would compute the result at the analysis level, not within the dataset. This approach doesn't guarantee precomputation within SPICE and may increase the calculation overhead for each analysis.

Defining the field in the visual means the calculation is performed dynamically every time the visual is rendered, which does not leverage SPICE's precomputation benefits and can result in slower performance.

Creating the calculated field at the dashboard level is similar to doing it at the visual level, meaning the calculations are not precomputed and stored in SPICE, leading to potential performance issues and less reuse of the calculation across different analyses or visuals.

QUESTION 163
A company has three subsidiaries. Each subsidiary uses a different data warehousing solution. The first subsidiary hosts its data warehouse in Amazon Redshift. The second subsidiary uses Teradata Vantage on AWS. The third subsidiary uses Google BigQuery.

The company wants to aggregate all the data into a central Amazon S3 data lake. The company wants to use Apache Iceberg as the table format.

A data engineer needs to build a new pipeline to connect to all the data sources, run transformations by using each source engine, join the data, and write the data to Iceberg.

Which solution will meet these requirements with the LEAST operational effort?

A. Use native Amazon Redshift, Teradata, and BigQuery connectors to build the pipeline in AWS Glue. Use native AWS Glue transforms to join the data. Run a Merge operation on the data lake Iceberg table.
B. Use the Amazon Athena federated query connectors for Amazon Redshift, Teradata, and BigQuery to build the pipeline in Athena. Write a SQL query to read from all the data sources, join the data, and run a Merge operation on the data lake Iceberg table.
C. Use the native Amazon Redshift connector, the Java Database Connectivity (JDBC) connector for Teradata, and the open source Apache Spark BigQuery connector to build the pipeline in Amazon EMR. Write code in PySpark to join the data. Run a Merge operation on the data lake Iceberg table.
D. Use the native Amazon Redshift, Teradata, and BigQuery connectors in Amazon Appflow to write data to Amazon S3 and AWS Glue Data Catalog. Use Amazon Athena to join the data. Run a Merge

operation on the data lake Iceberg table.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon Athena federated query allows querying data from multiple data sources, including Amazon Redshift, Teradata, and Google BigQuery, using their federated query connectors. This solution offers a serverless approach, reducing the operational overhead of managing infrastructure while allowing SQL-based transformations across all data sources. Once the data is read and joined, Athena can write the results back to Amazon S3 in the Iceberg table format with a Merge operation.

This approach minimizes the operational effort as Athena manages the complexity of connecting to different databases through its connectors, and you can perform the necessary transformations and data joins using familiar SQL.

While AWS Glue is a powerful ETL tool, it requires more operational effort to manage complex transformations across multiple systems, and managing native transforms across different engines (Redshift, Teradata, BigQuery) in Glue can introduce additional complexity.

Amazon EMR with PySpark can handle the task, but it requires more operational effort to manage and maintain the EMR cluster. Writing and maintaining PySpark code can also be more complex compared to using SQL in Athena.

Appflow is primarily designed for simple data movement between SaaS applications and AWS services, but it does not provide the complex transformation and joining capabilities needed for this scenario. Using Athena after Appflow for joins adds unnecessary complexity compared to directly using federated queries in Athena.

QUESTION 164
A company is building a data stream processing application. The application runs in an Amazon Elastic Kubernetes Service (Amazon EKS) cluster. The application stores processed data in an Amazon DynamoDB table.

The company needs the application containers in the EKS cluster to have secure access to the DynamoDB table. The company does not want to embed AWS credentials in the containers.

Which solution will meet these requirements?

A. Store the AWS credentials in an Amazon S3 bucket. Grant the EKS containers access to the S3 bucket to retrieve the credentials.
B. Attach an IAM role to the EKS worker nodes, Grant the IAM role access to DynamoDB. Use the IAM role to set up IAM roles service accounts (IRSA) functionality.
C. Create an IAM user that has an access key to access the DynamoDB table. Use environment variables in the EKS containers to store the IAM user access key data.
D. Create an IAM user that has an access key to access the DynamoDB table. Use Kubernetes secrets that are mounted in a volume of the EKS cluster nodes to store the user access key data.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
IAM roles for service accounts (IRSA) allow assigning fine-grained permissions to EKS pods based on Kubernetes service accounts. This approach ensures that AWS credentials are not hardcoded or stored in environment variables, which enhances security. Instead, the pods running in the EKS cluster assume the IAM role with permissions to access Amazon DynamoDB securely.

This solution meets the requirement of providing secure access without embedding AWS credentials in the containers, while also allowing for more granular access control at the pod level rather than relying on the node-level IAM role.

Storing AWS credentials in S3 is insecure and not a recommended practice. This approach does not meet the security requirement of avoiding embedded AWS credentials.

Storing AWS credentials in environment variables is insecure and not recommended. This exposes credentials within the container, increasing the risk of credential leakage.

Although Kubernetes secrets can provide some protection, storing long-term AWS credentials (access key and secret key) in secrets still presents a security risk. The preferred method is to use temporary credentials via IAM roles (IRSA), which rotate automatically.

QUESTION 165
A data engineer needs to onboard a new data producer into AWS. The data producer needs to migrate data products to AWS.

The data producer maintains many data pipelines that support a business application. Each pipeline must have service accounts and their corresponding credentials. The data engineer must establish a secure connection from the data producer's on-premises data center to AWS. The data engineer must not use the public internet to transfer data from an on-premises data center to AWS.

Which solution will meet these requirements?

A. Instruct the new data producer to create Amazon Machine Images (AMIs) on Amazon Elastic Container Service (Amazon ECS) to store the code base of the application. Create security groups in a public subnet that allow connections only to the on-premises data center.
B. Create an AWS Direct Connect connection to the on-premises data center. Store the service account credentials in AWS Secrets manager.
C. Create a security group in a public subnet. Configure the security group to allow only connections from the CIDR blocks that correspond to the data producer. Create Amazon S3 buckets than contain presigned URLS that have one-day expiration dates.
D. Create an AWS Direct Connect connection to the on-premises data center. Store the application keys in AWS Secrets Manager. Create Amazon S3 buckets that contain presigned URLS that have one-day expiration dates.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Direct Connect provides a dedicated, private network connection between the on-premises data center and AWS, ensuring that data is not transferred over the public internet. This fulfills the requirement of secure, private connectivity.

AWS Secrets Manager securely stores, rotates, and manages access to sensitive information such as service account credentials, reducing the need to embed credentials in code or configuration files.

QUESTION 166
A data engineer configured an AWS Glue Data Catalog for data that is stored in Amazon S3 buckets. The data engineer needs to configure the Data Catalog to receive incremental updates.

The data engineer sets up event notifications for the S3 bucket and creates an Amazon Simple Queue Service (Amazon SQS) queue to receive the S3 events.

Which combination of steps should the data engineer take to meet these requirements with LEAST operational overhead? (Choose two.)

A. Create an S3 event-based AWS Glue crawler to consume events from the SQS queue.
B. Define a time-based schedule to run the AWS Glue crawler, and perform incremental updates to the Data Catalog.
C. Use an AWS Lambda function to directly update the Data Catalog based on S3 events that the SQS queue receives.

D. Manually initiate the AWS Glue crawler to perform updates to the Data Catalog when there is a change in the S3 bucket.
E. Use AWS Step Functions to orchestrate the process of updating the Data Catalog based on S3 events that the SQS queue receives.

Correct Answer: AC
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue crawlers can be configured to work with S3 event notifications and consume events from SQS queues. This allows the crawler to automatically detect changes in the S3 bucket and update the Data Catalog incrementally, minimizing operational overhead by automating the updates.

You can use AWS Lambda to process the S3 events as they are received in the SQS queue. The Lambda function can be configured to automatically update the Glue Data Catalog, ensuring that changes are reflected in near real-time without manual intervention, which also reduces operational overhead.

A time-based schedule does not meet the requirement for receiving incremental updates in real time. Scheduling adds unnecessary delay and operational complexity compared to event-driven updates.

Manually triggering the crawler would introduce high operational overhead and is not aligned with the requirement to reduce the manual process for updating the Data Catalog.

While Step Functions could be used for orchestration, this adds complexity compared to using an event-driven Lambda function, which provides a simpler and more direct solution for processing the S3 events.

QUESTION 167
A company uses AWS Glue Data Catalog to index data that is uploaded to an Amazon S3 bucket every day. The company uses a daily batch processes in an extract, transform, and load (ETL) pipeline to upload data from external sources into the S3 bucket.

The company runs a daily report on the S3 data. Some days, the company runs the report before all the daily data has been uploaded to the S3 bucket. A data engineer must be able to send a message that identifies any incomplete data to an existing Amazon Simple Notification Service (Amazon SNS) topic.

Which solution will meet this requirement with the LEAST operational overhead?

A. Create data quality checks for the source datasets that the daily reports use. Create a new AWS managed Apache Airflow cluster. Run the data quality checks by using Airflow tasks that run data quality queries on the columns data type and the presence of null values. Configure Airflow Directed Acyclic Graphs (DAGs) to send an email notification that informs the data engineer about the incomplete datasets to the SNS topic.
B. Create data quality checks on the source datasets that the daily reports use. Create a new Amazon EMR cluster. Use Apache Spark SQL to create Apache Spark jobs in the EMR cluster that run data quality queries on the columns data type and the presence of null values. Orchestrate the ETL pipeline by using an AWS Step Functions workflow. Configure the workflow to send an email notification that informs the data engineer about the incomplete datasets to the SNS topic.
C. Create data quality checks on the source datasets that the daily reports use. Create data quality actions by using AWS Glue workflows to confirm the completeness and consistency of the datasets. Configure the data quality actions to create an event in Amazon EventBridge if a dataset is incomplete. Configure EventBridge to send the event that informs the data engineer about the incomplete datasets to the Amazon SNS topic.
D. Create AWS Lambda functions that run data quality queries on the columns data type and the presence of null values. Orchestrate the ETL pipeline by using an AWS Step Functions workflow that runs the Lambda functions. Configure the Step Functions workflow to send an email notification that informs the data engineer about the incomplete datasets to the SNS topic.

Correct Answer: C
Section: (none)

Explanation/Reference:

Explanation:
AWS Glue workflows and data quality actions allow for seamless integration with AWS Glue to ensure that data quality checks are automatically performed after ETL jobs. This solution leverages the existing AWS Glue infrastructure and EventBridge to trigger notifications without needing complex orchestration tools like Apache Airflow or EMR, resulting in lower operational overhead.

Amazon EventBridge can easily be configured to trigger a notification to Amazon SNS when a dataset is incomplete, providing a highly automated and low-maintenance solution.

Airflow requires managing a separate cluster and configuring Directed Acyclic Graphs (DAGs), which introduces more operational complexity and overhead than AWS Glue workflows.

Using Amazon EMR introduces unnecessary complexity and higher operational overhead, as it requires provisioning and managing a cluster for Spark jobs, which is more complex than using AWS Glue and EventBridge.

While Lambda can be a lightweight option, orchestrating Lambda functions for data quality checks with Step Functions adds unnecessary complexity when AWS Glue workflows already offer built-in capabilities for data quality actions with lower overhead.

QUESTION 168
A company stores customer data that contains personally identifiable information (PII) in an Amazon Redshift cluster. The company's marketing, claims, and analytics teams need to be able to access the customer data.

The marketing team should have access to obfuscated claim information but should have full access to customer contact information. The claims team should have access to customer information for each claim that the team processes. The analytics team should have access only to obfuscated PII data.

Which solution will enforce these data access requirements with the LEAST administrative overhead?

A. Create a separate Redshift cluster for each team. Load only the required data for each team. Restrict access to clusters based on the teams.
B. Create views that include required fields for each of the data requirements. Grant the teams access only to the view that each team requires.
C. Create a separate Amazon Redshift database role for each team. Define masking policies that apply for each team separately. Attach appropriate masking policies to each team role.
D. Move the customer data to an Amazon S3 bucket. Use AWS Lake Formation to create a data lake. Use fine-grained security capabilities to grant each team appropriate permissions to access the data.

Correct Answer: C
Section: (none)

Explanation/Reference:

QUESTION 169
A financial company recently added more features to its mobile app. The new features required the company to create a new topic in an existing Amazon Managed Streaming for Apache Kafka (Amazon MSK) cluster.

A few days after the company added the new topic, Amazon CloudWatch raised an alarm on the RootDiskUsed metric for the MSK cluster.

How should the company address the CloudWatch alarm?

A. Expand the storage of the MSK broker. Configure the MSK cluster storage to expand automatically.
B. Expand the storage of the Apache ZooKeeper nodes.
C. Update the MSK broker instance to a larger instance type. Restart the MSK cluster.
D. Specify the Target Volume-in-GiB parameter for the existing topic.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
The RootDiskUsed metric in Amazon MSK indicates that the storage space on the broker's disk is filling up. To resolve this, the company can expand the storage for the MSK brokers to prevent them from running out of disk space.

Amazon MSK supports automatic storage scaling, which can be configured to avoid similar issues in the future. By enabling auto-scaling, MSK will automatically expand the storage when needed, ensuring that the brokers have enough space to handle new topics or additional data.

ZooKeeper is used for managing the state of Kafka brokers, but the RootDiskUsed metric pertains to the Kafka brokers, not the ZooKeeper nodes. Expanding ZooKeeper storage will not resolve the issue.

While increasing the instance size may improve performance, it does not address the issue of running out of disk space on the broker's root volume, which is causing the CloudWatch alarm.

Kafka topics do not have a direct volume specification. Storage for Kafka is managed at the broker level, not at the individual topic level. Therefore, this option does not apply to the issue of disk usage.

QUESTION 170
A data engineer needs to build an enterprise data catalog based on the company's Amazon S3 buckets and Amazon RDS databases. The data catalog must include storage format metadata for the data in the catalog.

Which solution will meet these requirements with the LEAST effort?

A. Use an AWS Glue crawler to scan the S3 buckets and RDS databases and build a data catalog. Use data stewards to inspect the data and update the data catalog with the data format.
B. Use an AWS Glue crawler to build a data catalog. Use AWS Glue crawler classifiers to recognize the format of data and store the format in the catalog.
C. Use Amazon Macie to build a data catalog and to identify sensitive data elements. Collect the data format information from Macie.
D. Use scripts to scan data elements and to assign data classifications based on the format of the data.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue crawlers can automatically scan data in Amazon S3 buckets and Amazon RDS databases to build a data catalog. Glue crawlers also have classifiers that can automatically detect the format of the data (such as CSV, JSON, Parquet, etc.) and store this information as metadata in the Data Catalog. This solution automates the process of cataloging and format recognition, meeting the requirement with the least effort.

The "Use an AWS Glue crawler to scan the S3 buckets and RDS databases and build a data catalog. Use data stewards to inspect the data and update the data catalog with the data format." option requires manual inspection and updating of the data catalog by data stewards, which adds significant effort and is unnecessary since Glue crawlers can automatically detect the format.

Amazon Macie is primarily used for identifying sensitive data (e.g., PII), not for building a comprehensive data catalog or identifying data formats. It doesn't meet the requirement of cataloging storage format metadata.

Writing custom scripts to scan and classify data based on format is much more labor-intensive compared to using an automated Glue crawler, which handles this task with much less effort.

QUESTION 171
A company analyzes data in a data lake every quarter to perform inventory assessments. A data engineer

uses AWS Glue DataBrew to detect any personally identifiable formation (PII) about customers within the data. The company's privacy policy considers some custom categories of information to be PII. However, the categories are not included in standard DataBrew data quality rules.

The data engineer needs to modify the current process to scan for the custom PII categories across multiple datasets within the data lake.

Which solution will meet these requirements with the LEAST operational overhead?

A. Manually review the data for custom PII categories.
B. Implement custom data quality rules in DataBrew. Apply the custom rules across datasets.
C. Develop custom Python scripts to detect the custom PII categories. Call the scripts from DataBrew.
D. Implement regex patterns to extract PII information from fields during extract transform, and load (ETL) operations into the data lake.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue DataBrew allows users to create custom data quality rules, including rules for detecting non-standard PII categories. By implementing custom rules, the data engineer can automate the detection of PII (including the custom categories) across multiple datasets without needing to manually review or develop custom scripts. This provides a solution with the least operational overhead, as it leverages the existing DataBrew framework and automation capabilities.

Manually reviewing the data is highly labor-intensive, error-prone, and not scalable, especially across multiple datasets.

While possible, writing custom Python scripts adds operational overhead, as it requires developing, maintaining, and integrating scripts with DataBrew. This increases complexity compared to implementing custom rules directly within DataBrew.

While regex can be useful, incorporating it directly into ETL processes adds operational complexity and doesn't leverage the capabilities of DataBrew, which is already designed for data cleaning and quality validation.

QUESTION 172
A company receives a data file from a partner each day in an Amazon S3 bucket. The company uses a daily AWS Glue extract, transform, and load (ETL) pipeline to clean and transform each data file. The output of the ETL pipeline is written to a CSV file named Daily.csv in a second S3 bucket.

Occasionally, the daily data file is empty or is missing values for required fields. When the file is missing data, the company can use the previous day's CSV file.

A data engineer needs to ensure that the previous day's data file is overwritten only if the new daily file is complete and valid.

Which solution will meet these requirements with the LEAST effort?

A. Invoke an AWS Lambda function to check the file for missing data and to fill in missing values in required fields.
B. Configure the AWS Glue ETL pipeline to use AWS Glue Data Quality rules. Develop rules in Data Quality Definition Language (DQDL) to check for missing values in required fields and empty files.
C. Use AWS Glue Studio to change the code in the ETL pipeline to fill in any missing values in the required fields with the most common values for each field.
D. Run a SQL query in Amazon Athena to read the CSV file and drop missing rows. Copy the corrected CSV file to the second S3 bucket.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue Data Quality rules can be integrated directly into the AWS Glue ETL pipeline to automatically check for missing or invalid data in the daily file. You can use Data Quality Definition Language (DQDL) to define rules that verify the completeness and validity of the data before the pipeline writes the output. This solution ensures that the previous day's file is only overwritten if the new data is complete and valid, minimizing manual intervention and operational effort.

While Lambda could be used, it requires additional development and integration outside of the existing Glue ETL pipeline. This adds complexity and operational overhead.

The “Use AWS Glue Studio to change the code in the ETL pipeline to fill in any missing values in the required fields with the most common values for each field” approach assumes that filling missing values with the most common ones is acceptable, but it might not always be the correct solution for all cases. The requirement specifies that the file should be validated and replaced only if it's complete and valid.

Using Athena to drop missing rows doesn’t fully address the problem, as it could result in loss of data and still requires manual intervention to validate the file before copying it. Additionally, it introduces unnecessary steps compared to using built-in AWS Glue Data Quality capabilities.

QUESTION 173
A marketing company uses Amazon S3 to store marketing data. The company uses versioning in some buckets. The company runs several jobs to read and load data into the buckets.

To help cost-optimize its storage, the company wants to gather information about incomplete multipart uploads and outdated versions that are present in the S3 buckets.

Which solution will meet these requirements with the LEAST operational effort?

A. Use AWS CLI to gather the information.
B. Use Amazon S3 Inventory configurations reports to gather the information.
C. Use the Amazon S3 Storage Lens dashboard to gather the information.
D. Use AWS usage reports for Amazon S3 to gather the information.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon S3 Storage Lens provides a comprehensive view of S3 storage usage and activity. It includes metrics such as the number of incomplete multipart uploads and the number of non-current object versions, making it an ideal tool for gathering insights into storage inefficiencies like outdated versions and incomplete uploads. This solution requires minimal operational effort as it is built into S3 and provides a user-friendly dashboard with detailed insights.

Using the AWS CLI to manually list and check multipart uploads and object versions would require scripting and regular execution, leading to higher operational overhead.

S3 Inventory provides reports on the objects in your bucket, but it does not directly provide information on incomplete multipart uploads or outdated versions. You would need to process the data further to gather the specific insights.

AWS usage reports are more focused on billing and usage rather than detailed storage optimization metrics like incomplete multipart uploads and outdated object versions.

QUESTION 174
A gaming company uses Amazon Kinesis Data Streams to collect clickstream data. The company uses Amazon Data Firehose delivery streams to store the data in JSON format in Amazon S3. Data scientists at the company use Amazon Athena to query the most recent data to obtain business insights.

The company wants to reduce Athena costs but does not want to recreate the data pipeline.

Which solution will meet these requirements with the LEAST management effort?

A. Change the Firehose output format to Apache Parquet. Provide a custom S3 object YYYYMMDD prefix expression and specify a large buffer size. For the existing data, create an AWS Glue extract, transform, and load (ETL) job. Configure the ETL job to combine small JSON files, convert the JSON files to large Parquet files, and add the YYYYMMDD prefix. Use the ALTER TABLE ADD PARTITION statement to reflect the partition on the existing Athena table.
B. Create an Apache Spark job that combines JSON files and converts the JSON files to Apache Parquet files. Launch an Amazon EMR ephemeral cluster every day to run the Spark job to create new Parquet files in a different S3 location. Use the ALTER TABLE SET LOCATION statement to reflect the new S3 location on the existing Athena table.
C. Create a Kinesis data stream as a delivery destination for Firehose. Use Amazon Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics) to run Apache Flink on the Kinesis data stream. Use Flink to aggregate the data and save the data to Amazon S3 in Apache Parquet format with a custom S3 object YYYYMMDD prefix. Use the ALTER TABLE ADD PARTITION statement to reflect the partition on the existing Athena table.
D. Integrate an AWS Lambda function with Firehose to convert source records to Apache Parquet and write them to Amazon S3. In parallel, run an AWS Glue extract, transform, and load (ETL) job to combine the JSON files and convert the JSON files to large Parquet files. Create a custom S3 object YYYYMMDD prefix. Use the ALTER TABLE ADD PARTITION statement to reflect the partition on the existing Athena table.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Changing the Firehose output format to Apache Parquet reduces Amazon Athena query costs because Parquet is a columnar storage format, which is much more efficient for queries compared to row-based formats like JSON. Parquet helps reduce the amount of data scanned by Athena, thus lowering costs.

Configuring Firehose to use a large buffer size ensures fewer, larger files, which also improves query performance in Athena.

For the existing JSON data, using an AWS Glue ETL job to convert the files into Parquet and apply the necessary partitioning (e.g., YYYYMMDD) will help optimize future queries without needing to recreate the data pipeline. Using the ALTER TABLE ADD PARTITION statement in Athena will allow the existing table to reflect the new partitions.

This solution minimizes management effort by reconfiguring Amazon Kinesis Data Firehose and using AWS Glue for handling existing data, without requiring significant changes to the existing pipeline.

Running an Apache Spark job on Amazon EMR requires more operational management, as you would need to launch and manage ephemeral EMR clusters daily, which increases complexity.

Amazon Managed Service for Apache Flink introduces additional components (Kinesis Data Streams and Flink), which adds complexity and management overhead compared to using the existing Firehose and simply changing its output format.

Using Lambda for real-time transformation and combining that with a Glue ETL job introduces unnecessary complexity. Firehose already supports direct output to Parquet format, making the Lambda function redundant.

QUESTION 175
A company needs a solution to manage costs for an existing Amazon DynamoDB table. The company also needs to control the size of the table. The solution must not disrupt any ongoing read or write operations. The company wants to use a solution that automatically deletes data from the table after 1 month.

Which solution will meet these requirements with the LEAST ongoing maintenance?

A. Use the DynamoDB TTL feature to automatically expire data based on timestamps.
B. Configure a scheduled Amazon EventBridge rule to invoke an AWS Lambda function to check for data

that is older than 1 month. Configure the Lambda function to delete old data.

C. Configure a stream on the DynamoDB table to invoke an AWS Lambda function. Configure the Lambda function to delete data in the table that is older than 1 month.

D. Use an AWS Lambda function to periodically scan the DynamoDB table for data that is older than 1 month. Configure the Lambda function to delete old data.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
DynamoDB TTL (Time to Live) is a built-in feature that allows you to automatically delete expired items from your table based on a timestamp attribute. Once TTL is enabled, DynamoDB automatically removes the items after they reach the expiration time, without disrupting any ongoing read or write operations. This solution provides the least ongoing maintenance because it's fully managed by DynamoDB and doesn't require manual interventions or additional Lambda functions.

QUESTION 176
A company uses Amazon S3 to store data and Amazon QuickSight to create visualizations,

The company has an S3 bucket in an AWS account named Hub-Account. The S3 bucket is encrypted by an AWS Key Management Service (AWS KMS) key. The company's QuickSight instance is in a separate account named BI-Account.

The company updates the S3 bucket policy to grant access to the QuickSight service role. The company wants to enable cross-account access to allow QuickSight to interact with the S3 bucket.

Which combination of steps will meet this requirement? (Choose two.)

A. Use the existing AWS KMS key to encrypt connections from QuickSight to the S3 bucket.

B. Add the S3 bucket as a resource that the QuickSight service role can access.

C. Use AWS Resource Access Manager (AWS RAM) to share the S3 bucket with the BI-Account account.

D. Add an IAM policy to the QuickSight service role to give QuickSight access to the KMS key that encrypts the S3 bucket.

E. Add the KMS key as a resource that the QuickSight service role can access.

Correct Answer: E
Section: (none)

Explanation/Reference:

QUESTION 177
A car sales company maintains data about cars that are listed for sale in an area. The company receives data about new car listings from vendors who upload the data daily as compressed files into Amazon S3. The compressed files are up to 5 KB in size. The company wants to see the most up-to-date listings as soon as the data is uploaded to Amazon S3.

A data engineer must automate and orchestrate the data processing workflow of the listings to feed a dashboard. The data engineer must also provide the ability to perform one-time queries and analytical reporting. The query solution must be scalable.

Which solution will meet these requirements MOST cost-effectively?

A. Use an Amazon EMR cluster to process incoming data. Use AWS Step Functions to orchestrate workflows. Use Apache Hive for one-time queries and analytical reporting. Use Amazon OpenSearch Service to bulk ingest the data into compute optimized instances. Use OpenSearch Dashboards in OpenSearch Service for the dashboard.

B. Use a provisioned Amazon EMR cluster to process incoming data. Use AWS Step Functions to orchestrate workflows. Use Amazon Athena for one-time queries and analytical reporting. Use Amazon QuickSight for the dashboard.

C. Use AWS Glue to process incoming data. Use AWS Step Functions to orchestrate workflows. Use Amazon Redshift Spectrum for one-time queries and analytical reporting. Use OpenSearch Dashboards in Amazon OpenSearch Service for the dashboard.
D. Use AWS Glue to process incoming data. Use AWS Lambda and S3 Event Notifications to orchestrate workflows. Use Amazon Athena for one-time queries and analytical reporting. Use Amazon QuickSight for the dashboard.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue is a fully managed ETL (Extract, Transform, Load) service that is cost-effective and scalable for processing incoming data. It can handle the compressed data files efficiently.

S3 Event Notifications and AWS Lambda provide a low-cost, serverless orchestration solution that automatically triggers workflows as soon as new files are uploaded to S3. This ensures real-time or near real-time processing of the data without the need to manage long-running services like EMR.

Amazon Athena is a serverless, scalable query service that allows users to run SQL queries on data directly in S3, which is ideal for one-time queries and analytical reporting without provisioning infrastructure.

Amazon QuickSight provides a cost-effective, fully managed service for visualizing the data, making it suitable for the dashboard requirements.

This solution is cost-effective because it minimizes the need for provisioned infrastructure and uses serverless and managed services like Lambda, Glue, Athena, and QuickSight, all of which automatically scale with demand.

QUESTION 178
A company has AWS resources in multiple AWS Regions. The company has an Amazon EFS file system in each Region where the company operates. The company’s data science team operates within only a single Region. The data that the data science team works with must remain within the team's Region.

A data engineer needs to create a single dataset by processing files that are in each of the company's Regional EFS file systems. The data engineer wants to use an AWS Step Functions state machine to orchestrate AWS Lambda functions to process the data.

Which solution will meet these requirements with the LEAST effort?

A. Peer the VPCs that host the EFS file systems in each Region with the VPC that is in the data science team’s Region. Enable EFS file locking. Configure the Lambda functions in the data science team's Region to mount each of the Region specific file systems. Use the Lambda functions to process the data.
B. Configure each of the Regional EFS file systems to replicate data to the data science team's Region. In the data science team’s Region, configure the Lambda functions to mount the replica file systems. Use the Lambda functions to process the data.
C. Deploy the Lambda functions to each Region. Mount the Regional EFS file systems to the Lambda functions. Use the Lambda functions to process the data. Store the output in an Amazon S3 bucket in the data science team’s Region.
D. Use AWS DataSync to transfer files from each of the Regional EFS files systems to the file system that is in the data science team's Region. Configure the Lambda functions in the data science team's Region to mount the file system that is in the same Region. Use the Lambda functions to process the data.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
AWS DataSync is designed for transferring data between AWS storage services across regions, such as

EFS, S3, and other storage systems. By using DataSync, you can efficiently transfer data from each of the Regional EFS file systems to the EFS file system in the data science team's Region. This allows the data to remain within the data science team's Region while being processed, which meets the requirement that data cannot leave the Region.

Once the data is transferred, Lambda functions in the same Region as the data science team can mount the local EFS file system and process the data, minimizing complexity and the risk of network-related latency or failures from cross-region data processing.

This approach ensures minimal operational effort because DataSync automates the transfer process and works seamlessly with EFS, while keeping the data localized for processing by Lambda functions in the same Region.

QUESTION 179
A company hosts its applications on Amazon EC2 instances. The company must use SSL/TLS connections that encrypt data in transit to communicate securely with AWS infrastructure that is managed by a customer.

A data engineer needs to implement a solution to simplify the generation, distribution, and rotation of digital certificates. The solution must automatically renew and deploy SSL/TLS certificates.

Which solution will meet these requirements with the LEAST operational overhead?

A. Store self-managed certificates on the EC2 instances.
B. Use AWS Certificate Manager (ACM).
C. Implement custom automation scripts in AWS Secrets Manager.
D. Use Amazon Elastic Container Service (Amazon ECS) Service Connect.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Certificate Manager (ACM) simplifies the process of generating, distributing, and managing SSL/TLS certificates. ACM allows you to automatically renew and deploy certificates with minimal operational overhead. Once integrated with services like EC2, ACM can handle the entire lifecycle of the certificates, from creation to renewal and deployment, without manual intervention.

Managing certificates manually introduces significant operational overhead. You would need to handle generation, deployment, renewal, and rotation manually, which is error-prone and time-consuming.

While you could store certificates in AWS Secrets Manager, you would still need to implement custom scripts for certificate generation, rotation, and deployment, increasing operational complexity compared to ACM.

Amazon ECS Service Connect is used for simplifying service discovery and communication within ECS. While it can secure traffic between ECS services, it is not designed for managing SSL/TLS certificates on EC2 instances.

QUESTION 180
A company saves customer data to an Amazon S3 bucket. The company uses server-side encryption with AWS KMS keys (SSE-KMS) to encrypt the bucket. The dataset includes personally identifiable information (PII) such as social security numbers and account details.

Data that is tagged as PII must be masked before the company uses customer data for analysis. Some users must have secure access to the PII data during the pre-processing phase. The company needs a low-maintenance solution to mask and secure the PII data throughout the entire engineering pipeline.

Which combination of solutions will meet these requirements? (Choose two.)

A. Use AWS Glue DataBrew to perform extract, transform, and load (ETL) tasks that mask the PII data before analysis.

B. Use Amazon GuardDuty to monitor access patterns for the PII data that is used in the engineering pipeline.
C. Configure an Amazon Macie discovery job for the S3 bucket.
D. Use AWS Identity and Access Management (IAM) to manage permissions and to control access to the PII data.
E. Write custom scripts in an application to mask the PII data and to control access.

Correct Answer: AD
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue DataBrew is a data preparation service that simplifies the process of cleaning and normalizing data, including masking PII data. It allows you to visually transform data and apply masking rules before the data is used for analysis. This solution is low-maintenance and seamlessly integrates with S3 for processing data.

AWS Identity and Access Management (IAM) can be used to manage permissions and control access to the PII data. IAM provides the necessary fine-grained access control for determining which users or roles can view or modify the PII data during the pre-processing and analysis phases.

GuardDuty is primarily a threat detection service. While it can monitor unusual access patterns, it does not provide the necessary tools for masking or securing PII data during the engineering pipeline.

Amazon Macie is useful for discovering and classifying PII in S3 buckets, but it does not handle the actual masking or securing of PII data. It is more for classification and monitoring rather than processing or access control.

Custom scripts can be error-prone, labor-intensive, and require ongoing maintenance. This approach introduces unnecessary complexity compared to the managed solution offered by AWS Glue DataBrew.

QUESTION 181
A data engineer is launching an Amazon EMR cluster. The data that the data engineer needs to load into the new cluster is currently in an Amazon S3 bucket. The data engineer needs to ensure that data is encrypted both at rest and in transit.

The data that is in the S3 bucket is encrypted by an AWS Key Management Service (AWS KMS) key. The data engineer has an Amazon S3 path that has a Privacy Enhanced Mail (PEM) file.

Which solution will meet these requirements?

A. Create an Amazon EMR security configuration. Specify the appropriate AWS KMS key for at-rest encryption for the S3 bucket. Create a second security configuration. Specify the Amazon S3 path of the PEM file for in-transit encryption. Create the EMR cluster, and attach both security configurations to the cluster.
B. Create an Amazon EMR security configuration. Specify the appropriate AWS KMS key for local disk encryption for the S3 bucket. Specify the Amazon S3 path of the PEM file for in-transit encryption. Use the security configuration during EMR cluster creation.
C. Create an Amazon EMR security configuration. Specify the appropriate AWS KMS key for at-rest encryption for the S3 bucket. Specify the Amazon S3 path of the PEM file for in-transit encryption. Use the security configuration during EMR cluster creation.
D. Create an Amazon EMR security configuration. Specify the appropriate AWS KMS key for at-rest encryption for the S3 bucket. Specify the Amazon S3 path of the PEM file for in-transit encryption. Create the EMR cluster, and attach the security configuration to the cluster.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon EMR security configurations allow you to define encryption settings for both at-rest and in-transit

data. In this case, the data in the S3 bucket is already encrypted with an AWS KMS key, so you need to ensure the EMR cluster uses the same key for at-rest encryption when accessing the S3 bucket.

For in-transit encryption, you need to provide a PEM file that contains the SSL/TLS certificate, which ensures that data transferred to and from the cluster is encrypted.

By specifying both the KMS key for at-rest encryption and the PEM file for in-transit encryption in a single security configuration, you ensure that the data is encrypted during storage and transmission.

QUESTION 182
A retail company is using an Amazon Redshift cluster to support real-time inventory management. The company has deployed an ML model on a real-time endpoint in Amazon SageMaker.

The company wants to make real-time inventory recommendations. The company also wants to make predictions about future inventory needs.

Which solutions will meet these requirements? (Choose two.)

A. Use Amazon Redshift ML to generate inventory recommendations.
B. Use SQL to invoke a remote SageMaker endpoint for prediction.
C. Use Amazon Redshift ML to schedule regular data exports for offline model training.
D. Use SageMaker Autopilot to create inventory management dashboards in Amazon Redshift.
E. Use Amazon Redshift as a file storage system to archive old inventory management reports.

Correct Answer: AB
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift ML integrates machine learning (ML) directly into the Redshift environment, allowing you to build and use ML models with SQL commands. By leveraging Redshift ML, the company can make real-time inventory recommendations based on historical and current data directly within Redshift.

Redshift can invoke external services, such as a SageMaker real-time endpoint, using SQL queries. This allows the company to send real-time data from Redshift to SageMaker and receive predictions (e.g., inventory forecasting) in real time, meeting the need for real-time predictions.

QUESTION 183
A company stores CSV files in an Amazon S3 bucket. A data engineer needs to process the data in the CSV files and store the processed data in a new S3 bucket.

The process needs to rename a column, remove specific columns, ignore the second row of each file, create a new column based on the values of the first row of the data, and filter the results by a numeric value of a column.

Which solution will meet these requirements with the LEAST development effort?

A. Use AWS Glue Python jobs to read and transform the CSV files.
B. Use an AWS Glue custom crawler to read and transform the CSV files.
C. Use an AWS Glue workflow to build a set of jobs to crawl and transform the CSV files.
D. Use AWS Glue DataBrew recipes to read and transform the CSV files.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue DataBrew is a visual data preparation tool that allows you to clean, normalize, and transform data without writing code. Using DataBrew recipes, you can easily perform transformations such as renaming columns, removing specific columns, ignoring certain rows, creating new columns, and filtering data based on column values. This solution requires the least development effort because it provides a no-

code/low-code interface for performing these tasks.

While AWS Glue Python jobs can handle these transformations, they would require writing custom code, which involves more development effort compared to using DataBrew.

AWS Glue crawlers are used for cataloging data and are not suitable for performing complex transformations like ignoring rows, renaming columns, or creating new columns.

Using an AWS Glue workflow to build a set of jobs to crawl and transform the CSV files adds unnecessary complexity. You would need to orchestrate multiple jobs and workflows, which requires more setup and development compared to using DataBrew for the transformations.

QUESTION 184
A company uses Amazon Redshift as its data warehouse. Data encoding is applied to the existing tables of the data warehouse. A data engineer discovers that the compression encoding applied to some of the tables is not the best fit for the data.

The data engineer needs to improve the data encoding for the tables that have sub-optimal encoding.

Which solution will meet this requirement?

A. Run the ANALYZE command against the identified tables. Manually update the compression encoding of columns based on the output of the command.
B. Run the ANALYZE COMPRESSION command against the identified tables. Manually update the compression encoding of columns based on the output of the command.
C. Run the VACUUM REINDEX command against the identified tables.
D. Run the VACUUM RECLUSTER command against the identified tables.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
The ANALYZE COMPRESSION command in Amazon Redshift evaluates the existing data in a table and suggests the most optimal compression encoding for each column. After running this command, the data engineer can manually update the table to apply the recommended compression encodings. This approach ensures the best fit for data compression, improving storage efficiency and query performance.

The ANALYZE command collects statistics for query optimization, but it does not provide compression encoding recommendations. It is focused on query performance, not data compression.

The VACUUM REINDEX command does not exist in Amazon Redshift. VACUUM commands are generally used for reclaiming disk space and sorting tables, not for compression optimization.

VACUUM RECLUSTER does not exist in Amazon Redshift. VACUUM operations focus on reorganizing and sorting data but do not address compression encoding.

QUESTION 185
The company stores a large volume of customer records in Amazon S3. To comply with regulations, the company must be able to access new customer records immediately for the first 30 days after the records are created. The company accesses records that are older than 30 days infrequently.

The company needs to cost-optimize its Amazon S3 storage.

Which solution will meet these requirements MOST cost-effectively?

A. Apply a lifecycle policy to transition records to S3 Standard Infrequent-Access (S3 Standard-IA) storage after 30 days.
B. Use S3 Intelligent-Tiering storage.
C. Transition records to S3 Glacier Deep Archive storage after 30 days.
D. Use S3 Standard-Infrequent Access (S3 Standard-IA) storage for all customer records.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
S3 Standard-IA (Infrequent Access) is ideal for data that is accessed less frequently but requires immediate access when needed. Since the company needs immediate access to customer records for the first 30 days and infrequent access afterward, applying an S3 lifecycle policy to automatically transition the data to S3 Standard-IA after 30 days provides a cost-effective solution. This way, records remain easily accessible during the initial 30 days and are moved to a cheaper storage tier when they are accessed less often.

S3 Intelligent-Tiering automatically moves data between frequent and infrequent access tiers based on access patterns, but it incurs additional monitoring and automation charges. This option is typically more expensive compared to S3 Standard-IA if the access pattern is well understood (like in this case, where data is accessed frequently for the first 30 days and infrequently thereafter).

S3 Glacier Deep Archive is a low-cost storage option for archival data that is rarely accessed and requires hours for retrieval. It does not meet the requirement for immediate access to records after 30 days.

Storing all data in S3 Standard-IA is more expensive during the first 30 days when the data is accessed frequently. S3 Standard is more cost-effective for frequently accessed data during that period.

QUESTION 186
A data engineer is using Amazon QuickSight to build a dashboard to report a company's revenue in multiple AWS Regions. The data engineer wants the dashboard to display the total revenue for a Region, regardless of the drill-down levels shown in the visual.

Which solution will meet these requirements?

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A. Create a table calculation.
B. Create a simple calculated field.
C. Create a level-aware calculation - aggregate (LAC-A) function.
D. Create a level-aware calculation - window (LAC-W) function.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Level-Aware Calculations (LAC) in Amazon QuickSight allow users to create calculations that are independent of the levels of aggregation in a visual. Specifically, the LAC-A (Aggregate) function lets you calculate metrics such as the total revenue for a region regardless of the drill-down levels displayed. This means that even if the user drills down into more granular data, the total revenue for the Region will still be displayed in the dashboard, meeting the requirements.

Table calculations are context-dependent, meaning they calculate values based on the visual's current granularity and drill-down levels. This would not meet the requirement to show total revenue regardless of drill-down.

A simple calculated field operates at the current visual's level of detail and would change as you drill down. It would not maintain the total revenue for a region regardless of drill-down.

LAC-W (Window) functions are typically used for running totals, moving averages, or ranking, which are calculated over a range of data but still affected by drill-down levels. This would not ensure the total revenue is displayed regardless of drill-down.

QUESTION 187
A retail company stores customer data in an Amazon S3 bucket. Some of the customer data contains personally identifiable information (PII) about customers. The company must not share PII data with business partners.

A data engineer must determine whether a dataset contains PII before making objects in the dataset available to business partners.

Which solution will meet this requirement with the LEAST manual intervention?

A. Configure the S3 bucket and S3 objects to allow access to Amazon Macie. Use automated sensitive data discovery in Macie.
B. Configure AWS CloudTrail to monitor S3 PUT operations. Inspect the CloudTrail trails to identify operations that save PII.
C. Create an AWS Lambda function to identify PII in S3 objects. Schedule the function to run periodically.
D. Create a table in AWS Glue Data Catalog. Write custom SQL queries to identify PII in the table. Use Amazon Athena to run the queries.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Macie is a fully managed data security service that automatically discovers, classifies, and protects sensitive data such as personally identifiable information (PII) in Amazon S3. Macie's automated sensitive data discovery feature can scan S3 objects for PII without requiring manual intervention, making it the most efficient and automated solution for identifying sensitive data before sharing it with business partners.

CloudTrail is useful for monitoring API calls but is not designed to inspect the contents of S3 objects for PII. This would require manual effort to inspect the data saved in S3, which is inefficient.

While Lambda could be used to scan for PII, it would require custom code to identify sensitive data. This approach increases manual effort for development and maintenance compared to using a fully managed service like Macie.

Writing custom SQL queries to identify PII is a manual process and requires the creation and management of Glue tables and Athena queries. This approach adds complexity and manual intervention compared to the automated capabilities of Macie.

QUESTION 188
A data engineer needs to create an empty copy of an existing table in Amazon Athena to perform data processing tasks. The existing table in Athena contains 1,000 rows.

Which query will meet this requirement?

A.
```
CREATE TABLE new_table
LIKE old_table;
```
B.
```
CREATE TABLE new_table
AS SELECT *
FROM old_table
WITH NO DATA;
```
C.
```
CREATE TABLE new_table
AS SELECT *
FROM old_table;
```
D.
```
CREATE TABLE new_table
as SELECT *
FROM old_cable
WHERE 1=1;
```

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 189
A company has a data lake in Amazon S3. The company collects AWS CloudTrail logs for multiple applications. The company stores the logs in the data lake, catalogs the logs in AWS Glue, and partitions the logs based on the year. The company uses Amazon Athena to analyze the logs.

Recently, customers reported that a query on one of the Athena tables did not return any data. A data engineer must resolve the issue.

Which combination of troubleshooting steps should the data engineer take? (Choose two.)

A. Confirm that Athena is pointing to the correct Amazon S3 location.
B. Increase the query timeout duration.
C. Use the MSCK REPAIR TABLE command.
D. Restart Athena.
E. Delete and recreate the problematic Athena table.

Correct Answer: AC
Section: (none)

Explanation/Reference:
Explanation:
One common issue could be that the table's S3 location is incorrect or has changed. The data engineer should check that the Athena table is correctly referencing the S3 bucket or path where the CloudTrail logs are stored.

If new partitions have been added to the S3 bucket (such as logs for a new year), but these partitions haven't been reflected in the Athena table, running the MSCK REPAIR TABLE command will load new partitions into the table's metadata, ensuring that Athena can query the new data.

A query timeout would not cause data to be missing from the results. This option does not address the issue of missing data in the query results.

Athena is a serverless query service, so there is no option to "restart" Athena. This is not applicable.

While deleting and recreating the table could resolve the issue, it's generally not necessary and is more disruptive. The problem can often be resolved with less intrusive steps like checking the S3 location and repairing partitions.

QUESTION 190
A data engineer wants to orchestrate a set of extract, transform, and load (ETL) jobs that run on AWS. The ETL jobs contain tasks that must run Apache Spark jobs on Amazon EMR, make API calls to Salesforce, and load data into Amazon Redshift.

The ETL jobs need to handle failures and retries automatically. The data engineer needs to use Python to orchestrate the jobs.

Which service will meet these requirements?

A. Amazon Managed Workflows for Apache Airflow (Amazon MWAA)
B. AWS Step Functions
C. AWS Glue
D. Amazon EventBridge

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Managed Workflows for Apache Airflow (Amazon MWAA) is a fully managed service that makes it easy to run Apache Airflow at scale. Airflow is a popular tool for orchestrating complex workflows, including ETL jobs, and can handle various task types such as running Apache Spark jobs on Amazon EMR, making

API calls to external systems like Salesforce, and loading data into Amazon Redshift. Airflow allows you to define workflows using Python and offers built-in mechanisms for handling failures, retries, and retries automatically, making it a great fit for this use case.

While AWS Step Functions can orchestrate workflows and handle retries and errors, it is less suited for Python-based orchestration compared to Airflow. Also, Step Functions is more suited for AWS services and does not offer built-in support for external systems like Salesforce, which would require custom integrations.

AWS Glue is a serverless ETL service, but it primarily focuses on data transformation tasks. Glue does not handle orchestration of complex workflows involving non-Glue jobs, such as calling Salesforce APIs or running Spark jobs on EMR, as effectively as Airflow.

EventBridge is an event-driven service, not an orchestration service. It is designed for event-based workflows but does not offer the workflow orchestration capabilities needed for handling tasks like retries, error handling, and running complex workflows across different systems.

QUESTION 191
A data engineer maintains custom Python scripts that perform a data formatting process that many AWS Lambda functions use. When the data engineer needs to modify the Python scripts, the data engineer must manually update all the Lambda functions.

The data engineer requires a less manual way to update the Lambda functions.

Which solution will meet this requirement?

A. Store the custom Python scripts in a shared Amazon S3 bucket. Store a pointer to the custom scripts in the execution context object.
B. Package the custom Python scripts into Lambda layers. Apply the Lambda layers to the Lambda functions.
C. Store the custom Python scripts in a shared Amazon S3 bucket. Store a pointer to the customer scripts in environment variables.
D. Assign the same alias to each Lambda function. Call each Lambda function by specifying the function's alias.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Lambda layers allow you to package external libraries, custom scripts, or dependencies (like Python scripts) separately from the Lambda function code. By using Lambda layers, you can update the custom Python scripts in one central place (the layer) without manually updating each Lambda function. When the layer is updated, all Lambda functions that use the layer automatically get the latest version of the scripts.

QUESTION 192
A company stores customer data in an Amazon S3 bucket. Multiple teams in the company want to use the customer data for downstream analysis. The company needs to ensure that the teams do not have access to personally identifiable information (PII) about the customers.

Which solution will meet this requirement with LEAST operational overhead?

A. Use Amazon Macie to create and run a sensitive data discovery job to detect and remove PII.
B. Use S3 Object Lambda to access the data, and use Amazon Comprehend to detect and remove PII.
C. Use Amazon Data Firehose and Amazon Comprehend to detect and remove PII.
D. Use an AWS Glue DataBrew job to store the PII data in a second S3 bucket. Perform analysis on the data that remains in the original S3 bucket.

Correct Answer: B
Section: (none)

Explanation/Reference:

QUESTION 193
A company stores its processed data in an S3 bucket. The company has a strict data access policy. The company uses IAM roles to grant teams within the company different levels of access to the S3 bucket.

The company wants to receive notifications when a user violates the data access policy. Each notification must include the username of the user who violated the policy.

Which solution will meet these requirements?

A. Use AWS Config rules to detect violations of the data access policy. Set up compliance alarms.
B. Use Amazon CloudWatch metrics to gather object-level metrics. Set up CloudWatch alarms.
C. Use AWS CloudTrail to track object-level events for the S3 bucket. Forward events to Amazon CloudWatch to set up CloudWatch alarms.
D. Use Amazon S3 server access logs to monitor access to the bucket. Forward the access logs to an Amazon CloudWatch log group. Use metric filters on the log group to set up CloudWatch alarms.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS CloudTrail provides detailed logging of AWS API calls, including object-level events for Amazon S3. You can configure CloudTrail to track data events for the S3 bucket, which will log each access to the objects in the bucket, including the username of the IAM entity (user or role) that accessed the data. By forwarding these CloudTrail events to Amazon CloudWatch, you can set up alarms to trigger notifications when policy violations are detected, and include the violating user's identity.

AWS Config is primarily designed for monitoring resource configurations, not for tracking real-time access events like data access violations. It does not directly provide information about which user accessed the S3 objects.

CloudWatch metrics can monitor S3 storage or request metrics, but it does not provide detailed logging of object-level access, including the username of the violator. You need CloudTrail to get detailed event information.

While S3 server access logs can track access to objects, they lack real-time processing and do not provide the same level of detail as CloudTrail. Additionally, processing access logs with CloudWatch metric filters adds more complexity compared to using CloudTrail, which is designed for this use case.

QUESTION 194
A company needs to load customer data that comes from a third party into an Amazon Redshift data warehouse. The company stores order data and product data in the same data warehouse. The company wants to use the combined dataset to identify potential new customers.

A data engineer notices that one of the fields in the source data includes values that are in JSON format.

How should the data engineer load the JSON data into the data warehouse with the LEAST effort?

A. Use the SUPER data type to store the data in the Amazon Redshift table.
B. Use AWS Glue to flatten the JSON data and ingest it into the Amazon Redshift table.
C. Use Amazon S3 to store the JSON data. Use Amazon Athena to query the data.
D. Use an AWS Lambda function to flatten the JSON data. Store the data in Amazon S3.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:

The SUPER data type in Amazon Redshift allows you to natively store semi-structured data such as JSON. By using the SUPER data type, you can store JSON data in a Redshift table without needing to flatten or transform it beforehand. This approach requires the least effort because it allows you to directly load the JSON data and query it using Redshift’s JSONPath and PARTITION BY capabilities without additional ETL processing.

QUESTION 195
A company wants to analyze sales records that the company stores in a MySQL database. The company wants to correlate the records with sales opportunities identified by Salesforce.

The company receives 2 GB of sales records every day. The company has 100 GB of identified sales opportunities. A data engineer needs to develop a process that will analyze and correlate sales records and sales opportunities. The process must run once each night.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to fetch both datasets. Use AWS Lambda functions to correlate the datasets. Use AWS Step Functions to orchestrate the process.
B. Use Amazon AppFlow to fetch sales opportunities from Salesforce. Use AWS Glue to fetch sales records from the MySQL database. Correlate the sales records with the sales opportunities. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to orchestrate the process.
C. Use Amazon AppFlow to fetch sales opportunities from Salesforce. Use AWS Glue to fetch sales records from the MySQL database. Correlate the sales records with sales opportunities. Use AWS Step Functions to orchestrate the process.
D. Use Amazon AppFlow to fetch sales opportunities from Salesforce. Use Amazon Kinesis Data Streams to fetch sales records from the MySQL database. Use Amazon Managed Service for Apache Flink to correlate the datasets. Use AWS Step Functions to orchestrate the process.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
This solution meets the requirements with the **least operational overhead** by utilizing managed AWS services that simplify the data ingestion, transformation, and orchestration processes:

Amazon AppFlow is a fully managed integration service that allows data ingestion from Salesforce without custom connectors or manual ETL processes.

AWS Glue provides serverless data integration, making it suitable for extracting data from the MySQL database and transforming it as needed.

AWS Step Functions can then be used to orchestrate and automate the nightly process, minimizing the need for complex management.

QUESTION 196
A company stores server logs in an Amazon S3 bucket. The company needs to keep the logs for 1 year. The logs are not required after 1 year.

A data engineer needs a solution to automatically delete logs that are older than 1 year.

Which solution will meet these requirements with the LEAST operational overhead?

A. Define an S3 Lifecycle configuration to delete the logs after 1 year.
B. Create an AWS Lambda function to delete the logs after 1 year.
C. Schedule a cron job on an Amazon EC2 instance to delete the logs after 1 year.
D. Configure an AWS Step Functions state machine to delete the logs after 1 year.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
An **S3 Lifecycle configuration** is the best choice for this requirement because it allows the company to define a rule that will automatically delete objects in an S3 bucket after a specified period, in this case, 1 year. This solution provides the **least operational overhead** because it is a built-in feature of Amazon S3, requires no additional infrastructure or management, and is designed specifically for managing object lifecycles.

QUESTION 197
A company is designing a serverless data processing workflow in AWS Step Functions that involves multiple steps. The processing workflow ingests data from an external API, transforms the data by using multiple AWS Lambda functions, and loads the transformed data into Amazon DynamoDB.

The company needs the workflow to perform specific steps based on the content of the incoming data.

Which Step Functions state type should the company use to meet this requirement?

A. Parallel
B. Choice
C. Task
D. Map

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
The **Choice** state in AWS Step Functions is designed to enable conditional branching within a workflow. It allows the workflow to evaluate data and make decisions based on specific conditions. This makes it ideal for cases where certain steps in the workflow need to be executed based on the content of the incoming data.

QUESTION 198
A data engineer created a table named cloudtrail_logs in Amazon Athena to query AWS CloudTrail logs and prepare data for audits. The data engineer needs to write a query to display errors with error codes that have occurred since the beginning of 2024. The query must return the 10 most recent errors.

Which query will meet these requirements?

A.
```
select count (*) as TotalEvents, eventname, errorcode, errormessage from
cloudtrail_logswhere errorcode is not nulland eventtime >= '2024-01-
01T00:00:00Z' group by eventname, errorcode, errormessageorder by
TotalEvents desclimit 10;
```
B.
```
select count (*) as TotalEvents, eventname, errorcode, errormessage from
cloudtrail_logs where eventtime >= '2024-01-01T00:00:00Z' group by
eventname, errorcode, errormessage order by TotalEvents desc limit 10;
```
C.
```
select count (*) as TotalEvents, eventname, errorcode, errormessage from
cloudtrail_logswhere eventtime >= '2024-01-01T00:00:00Z' group by eventname,
errorcode, errormessageorder by eventname asc
limit 10;
```
D.
```
select count (*) as TotalEvents, eventname, errorcode, errormessage from
cloudtrail_logs
where errorcode is not nulland eventtime >= '2024-01-01T00:00:00Z' group by
eventname, errorcode, errormessagelimit 10;
```

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:

```
select count(*) as TotalEvents, eventname, errorcode, errormessage
from cloudtrail_logs
where errorcode is not null
and eventtime >= '2024-01-01T00:00:00Z'
group by eventname, errorcode, errormessage
order by TotalEvents desc
limit 10;
```

This query meets the requirements by:

1. Filtering results where `errorcode` is not null, so only error events are included.
2. Filtering by `eventtime` to include events occurring since the beginning of 2024.
3. Grouping by `eventname`, `errorcode`, and `errormessage` to summarize the error events.
4. Sorting by `TotalEvents` in descending order to show the most recent errors.
5. Limiting the results to the 10 most recent errors.

QUESTION 199
An online retailer uses multiple delivery partners to deliver products to customers. The delivery partners send order summaries to the retailer. The retailer stores the order summaries in Amazon S3.

Some of the order summaries contain personally identifiable information (PII) about customers. A data engineer needs to detect PII in the order summaries so the company can redact the PII.

Which solution will meet these requirements with the LEAST operational overhead?

A. Amazon Textract
B. Amazon S3 Storage Lens
C. Amazon Macie
D. Amazon SageMaker Data Wrangler

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon Macie is a fully managed data security and privacy service that uses machine learning and pattern matching to discover and protect sensitive data, such as Personally Identifiable Information (PII), stored in Amazon S3. Macie can automatically scan the order summaries for PII, enabling the company to detect and then redact PII as required.

QUESTION 200
A company has an Amazon Redshift data warehouse that users access by using a variety of IAM roles. More than 100 users access the data warehouse every day.

The company wants to control user access to the objects based on each user's job role, permissions, and how sensitive the data is.

Which solution will meet these requirements?

A. Use the role-based access control (RBAC) feature of Amazon Redshift.
B. Use the row-level security (RLS) feature of Amazon Redshift.
C. Use the column-level security (CLS) feature of Amazon Redshift.
D. Use dynamic data masking policies in Amazon Redshift.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Role-based access control (RBAC) in Amazon Redshift enables administrators to define roles based on job functions and then assign these roles to users. RBAC allows for granular permissions management based on job roles, aligning well with the need to control access according to each user's role, permissions, and the sensitivity of data. This approach simplifies managing access for a large number of users by using predefined roles instead of individual permissions.

QUESTION 201
A company uses Amazon DataZone as a data governance and business catalog solution. The company stores data in an Amazon S3 data lake. The company uses AWS Glue with an AWS Glue Data Catalog.

A data engineer needs to publish AWS Glue Data Quality scores to the Amazon DataZone portal.

Which solution will meet this requirement?

A. Create a data quality ruleset with Data Quality Definition language (DQDL) rules that apply to a specific AWS Glue table. Schedule the ruleset to run daily. Configure the Amazon DataZone project to have an Amazon Redshift data source. Enable the data quality configuration for the data source.
B. Configure AWS Glue ETL jobs to use an Evaluate Data Quality transform. Define a data quality ruleset inside the jobs. Configure the Amazon DataZone project to have an AWS Glue data source. Enable the data quality configuration for the data source.
C. Create a data quality ruleset with Data Quality Definition language (DQDL) rules that apply to a specific AWS Glue table. Schedule the ruleset to run daily. Configure the Amazon DataZone project to have an AWS Glue data source. Enable the data quality configuration for the data source.
D. Configure AWS Glue ETL jobs to use an Evaluate Data Quality transform. Define a data quality ruleset inside the jobs. Configure the Amazon DataZone project to have an Amazon Redshift data source. Enable the data quality configuration for the data source.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Data Quality Ruleset: Creating a ruleset with Data Quality Definition Language (DQDL) rules allows for defining and evaluating data quality on specific AWS Glue tables, enabling automated checks on data quality.

Scheduled Execution: Running the ruleset daily ensures that data quality scores are regularly updated.

AWS Glue Data Source in Amazon DataZone: Configuring Amazon DataZone with an AWS Glue data source enables seamless integration, allowing data quality scores from AWS Glue Data Quality to be published to the Amazon DataZone portal.

QUESTION 202
A company has a data warehouse in Amazon Redshift. To comply with security regulations, the company needs to log and store all user activities and connection activities for the data warehouse.

Which solution will meet these requirements?

A. Create an Amazon S3 bucket. Enable logging for the Amazon Redshift cluster. Specify the S3 bucket in the logging configuration to store the logs.
B. Create an Amazon Elastic File System (Amazon EFS) file system. Enable logging for the Amazon Redshift cluster. Write logs to the EFS file system.
C. Create an Amazon Aurora MySQL database. Enable logging for the Amazon Redshift cluster. Write the logs to a table in the Aurora MySQL database.
D. Create an Amazon Elastic Block Store (Amazon EBS) volume. Enable logging for the Amazon Redshift cluster. Write the logs to the EBS volume.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift provides an option to log and store all user activities and connection activities for compliance and auditing purposes. Configuring logging to an **Amazon S3 bucket** is the recommended approach because:

Amazon S3 is cost-effective, scalable, and durable for storing large volumes of logs over time.

Amazon Redshift natively supports logging to S3, allowing user and connection logs to be easily enabled and stored directly in the specified S3 bucket.

Storing logs in S3 meets compliance needs and provides an efficient way to manage, archive, and retrieve logs for security reviews or audits.

QUESTION 203
A company wants to migrate a data warehouse from Teradata to Amazon Redshift.

Which solution will meet this requirement with the LEAST operational effort?

A. Use AWS Database Migration Service (AWS DMS) Schema Conversion to migrate the schema. Use AWS DMS to migrate the data.
B. Use the AWS Schema Conversion Tool (AWS SCT) to migrate the schema. Use AWS Database Migration Service (AWS DMS) to migrate the data.
C. Use AWS Database Migration Service (AWS DMS) to migrate the data. Use automatic schema conversion.
D. Manually export the schema definition from Teradata. Apply the schema to the Amazon Redshift database. Use AWS Database Migration Service (AWS DMS) to migrate the data.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
This solution provides the **least operational effort** for migrating a data warehouse from Teradata to Amazon Redshift by using AWS services that are specifically designed for schema and data migration:

AWS Schema Conversion Tool (SCT) is used to convert and migrate the schema from Teradata to Amazon Redshift, handling schema differences automatically and reducing manual work.

AWS Database Migration Service (DMS) is then used to transfer the data from Teradata to Redshift, allowing for efficient data migration.

QUESTION 204
A company uses a variety of AWS and third-party data stores. The company wants to consolidate all the data into a central data warehouse to perform analytics. Users need fast response times for analytics queries.

The company uses Amazon QuickSight in direct query mode to visualize the data. Users normally run queries during a few hours each day with unpredictable spikes.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Redshift Serverless to load all the data into Amazon Redshift managed storage (RMS).
B. Use Amazon Athena to load all the data into Amazon S3 in Apache Parquet format.
C. Use Amazon Redshift provisioned clusters to load all the data into Amazon Redshift managed storage (RMS).
D. Use Amazon Aurora PostgreSQL to load all the data into Aurora.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift Serverless is ideal for this scenario as it provides a fully managed, serverless data warehouse solution with high performance for analytics queries. Key reasons why it meets the requirements with the **least operational overhead**:

Serverless Configuration: Redshift Serverless automatically scales capacity based on query demand, making it well-suited for unpredictable spikes in usage without requiring manual adjustments.

Managed Storage (RMS): Loading data into Amazon Redshift managed storage provides optimized storage for analytics and integrates with Amazon QuickSight for fast query performance in direct query mode.

Cost-Effective and Low Management: Since the data warehouse is only actively used for a few hours daily, Redshift Serverless provides a cost-effective option by scaling resources dynamically, which eliminates the need to manage and pay for idle infrastructure.

QUESTION 205
A data engineer uses Amazon Kinesis Data Streams to ingest and process records that contain user behavior data from an application every day.

The data engineer notices that the data stream is experiencing throttling because hot shards receive much more data than other shards in the data stream.

How should the data engineer resolve the throttling issue?

A. Use a random partition key to distribute the ingested records.
B. Increase the number of shards in the data stream. Distribute the records across the shards.
C. Limit the number of records that are sent each second by the producer to match the capacity of the stream.
D. Decrease the size of the records that the producer sends to match the capacity of the stream.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon Kinesis Data Streams distribute data across multiple shards, with each shard having its own capacity for read and write operations. Throttling occurs when one or more shards, referred to as "hot shards," receive significantly more data than they can handle. To resolve this, increasing the number of shards in the data stream and redistributing the records across the shards is the appropriate solution. This approach ensures that the workload is spread more evenly, thereby preventing throttling on individual shards.

QUESTION 206
A company has a data processing pipeline that includes several dozen steps. The data processing pipeline needs to send alerts in real time when a step fails or succeeds. The data processing pipeline uses a combination of Amazon S3 buckets, AWS Lambda functions, and AWS Step Functions state machines.

A data engineer needs to create a solution to monitor the entire pipeline.

Which solution will meet these requirements?

A. Configure the Step Functions state machines to store notifications in an Amazon S3 bucket when the state machines finish running. Enable S3 event notifications on the S3 bucket.
B. figure the AWS Lambda functions to store notifications in an Amazon S3 bucket when the state machines finish running. Enable S3 event notifications on the S3 bucket.
C. Use AWS CloudTrail to send a message to an Amazon Simple Notification Service (Amazon SNS) topic that sends notifications when a state machine fails to run or succeeds to run.
D. Configure an Amazon EventBridge rule to react when the execution status of a state machine changes.

Configure the rule to send a message to an Amazon Simple Notification Service (Amazon SNS) topic that sends notifications.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Amazon EventBridge can monitor the execution status of AWS Step Functions state machines and trigger specific actions based on state changes, such as when a state machine succeeds or fails. By configuring an EventBridge rule to capture these execution status changes and forward the details to an Amazon SNS topic, real-time notifications can be sent to alert users. This solution ensures a scalable and event-driven approach to monitoring the entire data processing pipeline in real time.

QUESTION 207
A company has an application that uses an Amazon API Gateway REST API and an AWS Lambda function to retrieve data from an Amazon DynamoDB instance. Users recently reported intermittent high latency in the application's response times. A data engineer finds that the Lambda function experiences frequent throttling when the company's other Lambda functions experience increased invocations.

The company wants to ensure the API's Lambda function operate without being affected by other Lambda functions.

Which solution will meet this requirement MOST cost-effectively?

A. Increase the number of read capacity unit (RCU) in DynamoDB.
B. Configure provisioned concurrency for the Lambda function.
C. Configure reserved concurrency for the Lambda function.
D. Increase the Lambda function timeout and allocated memory.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Reserved concurrency ensures that a specific Lambda function has a set number of concurrent executions that cannot be throttled due to other Lambda functions consuming shared concurrency limits. By configuring reserved concurrency, the API's Lambda function will always have the necessary resources to execute without being affected by other functions, thereby preventing throttling and ensuring consistent performance.

QUESTION 208
A company has as JSON file that contains personally identifiable information (PII) data and non-PII data. The company needs to make the data available for querying and analysis.

The non-PII data must be available to everyone in the company. The PII data must be available only to a limited group of employees.

Which solution will meet these requirements with the LEAST operational overhead?

A. Store the JSON file in an Amazon S3 bucket. Configure AWS Glue to split the file into one file that contains the PII data and one file that contains the non-PII data. Store the output files in separate S3 buckets. Grant the required access to the buckets based on the type of user.
B. Store the JSON file in an Amazon S3 bucket. Use Amazon Macie to identify PII data and to grant access based on the type of user.
C. Store the JSON file in an Amazon S3 bucket. Catalog the file schema in AWS Lake Formation. Use Lake Formation permissions to provide access to the required data based on the type of user.
D. Create two Amazon RDS PostgreSQL databases. Load the PII data and the non-PII data into the separate databases. Grant access to the databases based on the type of user.

Correct Answer: C

Section: (none)

Explanation/Reference:
Explanation:
AWS Lake Formation lets you register your JSON file as a table in the data catalog and apply fine-grained permissions (down to columns or rows) without moving or transforming the underlying data. You can grant all employees access to the non-PII columns while restricting the PII columns to a small group, achieving your security requirements with minimal operational overhead.

QUESTION 209
A company uses AWS Key Management Service (AWS KMS) to encrypt an Amazon Redshift cluster. The company wants to configure a cross-Region snapshot of the Redshift cluster as part of disaster recovery (DR) strategy.

A data engineer needs to use the AWS CLI to create the cross-Region snapshot.

Which combination of steps will meet these requirements? (Choose two.)

A. Create a KMS key and configure a snapshot copy grant in the source AWS Region.
B. In the source AWS Region, enable snapshot copying. Specify the name of the snapshot copy grant that is created in the destination AWS Region.
C. In the source AWS Region, enable snapshot copying. Specify the name of the snapshot copy grant that is created in the source AWS Region.
D. Create a KMS key and configure a snapshot copy grant in the destination AWS Region.
E. Convert the cluster to a Multi-AZ deployment.

Correct Answer: BD
Section: (none)

Explanation/Reference:
Explanation:
You must create a customer managed KMS key in the destination Region and then create a snapshot copy grant against that key so Redshift can re-encrypt the incoming snapshot with the destination CMK.

In the source Region you then enable cross-Region snapshot copying and reference the grant name you created in the destination Region. This tells Redshift which CMK (via the grant) to use when copying the encrypted snapshot into the DR Region.

QUESTION 210
A company is using Amazon S3 to build a data lake. The company needs to replicate records from multiple source databases into Apache Parquet format.

Most of the source databases are hosted on Amazon RDS. However, one source database is an on-premises Microsoft SQL Server Enterprise instance. The company needs to implement a solution to replicate existing data from all source databases and all future changes to the target S3 data lake.

Which solution will meet these requirements MOST cost-effectively?

A. Use one AWS Glue job to replicate existing data. Use a second AWS Glue job to replicate future changes.
B. Use AWS Database Migration Service (AWS DMS) to replicate existing data. Use AWS Glue jobs to replicate future changes.
C. Use AWS Database Migration Service (AWS DMS) to replicate existing data and future changes.
D. Use AWS Glue jobs to replicate existing data. Use Amazon Kinesis Data Streams to replicate future changes.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:

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Amazon DMS can perform a one-time full load of each source (including the on-prem SQL Server) and then capture ongoing changes (CDC), delivering them directly to S3 in Apache Parquet format. Using a single DMS replication task per source minimizes infrastructure, operational overhead, and cost.

QUESTION 211
A data engineer needs to optimize the performance of a data pipeline that handles retail orders. Data about the orders is ingested daily into an Amazon S3 bucket.

The data engineer runs queries once each week to extract metrics from the orders data based the order date for multiple date ranges. The data engineer needs an optimization solution that ensures the query performance will not degrade when the volume of data increases.

Which solution will meet this requirement MOST cost-effectively?

A. Partition the data based on order date. Use Amazon Athena to query the data.
B. Partition the data based on order date. Use Amazon Redshift to query the data.
C. Partition the data based on load date. Use Amazon EMR to query the data.
D. Partition the data based on load date. Use Amazon Aurora to query the data.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Partitioning the S3 data by the order date lets Athena prune scans to only the relevant date folders, keeping query times stable as data grows. Because Athena is a serverless, pay-per-query service, you only pay for the data scanned, making it the most cost-effective way to run your weekly date-range metrics.

QUESTION 212
A data engineer has two datasets that contain sales information for multiple cities and states. One dataset is named reference, and the other dataset is named primary.

The data engineer needs a solution to determine whether a specific set of values in the city and state columns of the primary dataset exactly match the same specific values in the reference dataset. The data engineer wants to use Data Quality Definition Language (DQDL) rules in an AWS Glue Data Quality job.

Which rule will meet these requirements?

A. DatasetMatch "reference” “city->ref_city, state->ref_state” = 1.0
B. ReferentialIntegrity “city,state” “reference.{ref_city,ref_state}” = 1.0
C. DatasetMatch “reference” “city->ref_city, state->ref_state” = 100
D. Referentialintegrity “city,state” "reference.{ref_city,ref_state}” = 100

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
The ReferentialIntegrity rule checks that every (city, state) pair in the primary dataset has a matching (ref_city, ref_state) pair in the reference dataset, and setting the threshold to 1.0 enforces a 100% match rate. This directly validates exact correspondence of those columns without extra overhead.

QUESTION 213
A company has an on-premises PostgreSQL database that contains customer data. The company wants to migrate the customer data to an Amazon Redshift data warehouse. The company has established a VPN connection between the on-premises database and AWS.

The on-premises database is continuously updated. The company must ensure that the data in Amazon Redshift is updated as quickly as possible.

Which solution will meet these requirements?

A. Use the pg_dump utility to generate a backup of the PostgreSQL database. Use the AWS Schema Conversion Tool (AWS SCT) to upload the backup to Amazon Redshift. Set up a cron job to perform a backup. Upload the backup to Amazon Redshift every night.
B. Create an AWS Database Migration Service (AWS DMS) full-load task. Set Amazon Redshift as the target. Configure the task to use the change data capture (CDC) feature.
C. Use the pg_dump utility to generate a backup of the PostgreSQL database. Upload the backup to an Amazon S3 bucket. Use the COPY command to import the data into Amazon Redshift.
D. Create an AWS Database Migration Service (AWS DMS) full-load task. Set Amazon Redshift as the target. Configure the task to perform a full load of the database to Amazon Redshift every night.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS DMS can perform an initial full load of your on-premises PostgreSQL data into Redshift and then continuously capture and apply changes (CDC) over the VPN. This approach keeps your Redshift tables up to date with minimal latency and operational overhead.

QUESTION 214
A company has several new datasets in CSV and JSON formats. A data engineer needs to make the data available to a team of data analysts who will analyze the data by using SQL queries.

Which solution will meet these requirements in the MOST cost-effective way?

A. Create an Amazon RDS MySQL cluster. Use AWS Glue to transform and load the CSV and JSON files into database tables. Provide the data analysts access to the MySQL cluster.
B. Create an AWS Glue DataBrew project that contains the new data. Make the DataBrew project available to the data analysts.
C. Store the data in an Amazon S3 bucket. Use an AWS Glue crawler to catalog the S3 bucket as tables. Create an Amazon Athena workgroup that has a data usage threshold. Grant the data analysts access to the Athena workgroup.
D. Load the data into Super-fast, Parallel, In-memory Calculation Engine (SPICE) in Amazon QuickSight. Allow the data analysts to create analyses and dashboards in QuickSight.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
By storing the CSV and JSON files in Amazon S3 and running an AWS Glue crawler, you automatically catalog them as tables without upfront ETL. Amazon Athena then lets analysts run SQL queries directly against those tables on a pay-per-query basis. Using an Athena workgroup with a usage threshold keeps costs under control, making this the most cost-effective, low-operational-overhead solution.

QUESTION 215
A retail company stores order information in an Amazon Aurora table named Orders. The company needs to create operational reports from the Orders table with minimal latency. The Orders table contains billions of rows, and over 100,000 transactions can occur each second.

A marketing team needs to join the Orders data with an Amazon Redshift table named Campaigns in the marketing team's data warehouse. The operational Aurora database must not be affected.

Which solution will meet these requirements with the LEAST operational effort?

A. Use AWS Database Migration Service (AWS DMS) Serverless to replicate the Orders table to Amazon Redshift. Create a materialized view in Amazon Redshift to join with the Campaigns table.
B. Use the Aurora zero-ETL integration with Amazon Redshift to replicate the Orders table. Create a materialized view in Amazon Redshift to join with the Campaigns table.
C. Use AWS Glue to replicate the Orders table to Amazon Redshift. Create a materialized view in Amazon

Redshift to join with the Campaigns table.

D. Use federated queries to query the Orders table directly from Aurora. Create a materialized view in Amazon Redshift to join with the Campaigns table.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Aurora's zero-ETL integration with Redshift automatically and continuously streams data changes from the Orders table into a Redshift table with virtually no setup or maintenance. You can then define a materialized view in Redshift to join that replicated Orders table with Campaigns. This approach ensures minimal impact on the production Aurora workload and requires far less operational effort than building and managing your own ETL or replication pipelines.

QUESTION 216
A company is building a new application that ingests CSV files into Amazon Redshift. The company has developed the frontend for the application.

The files are stored in an Amazon S3 bucket. Files are no larger than 5 MB.

A data engineer is developing the extract, transform, and load (ETL) pipeline for the CSV files. The data engineer configured a Redshift cluster and an AWS Lambda function that copies the data out of the files into the Redshift cluster.

Which additional steps should the data engineer perform to meet these requirements?

A. Configure the bucket to send S3 event notifications to Amazon EventBridge. Configure an EventBridge rule that matches S3 new object created events. Set the Lambda function as the target.
B. Configure the $3 bucket to send S3 event notifications to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the Lambda function to process the queue.
C. Configure AWS Database Migration Service (AWS DMS) to stream new S3 objects to a data stream in Amazon Kinesis Data Streams. Set the Lambda function as the target of the data stream.
D. Configure an Amazon EventBridge rule that matches S3 new object created events. Set an Amazon Simple Queue Service (Amazon SQS) queue as the target of the rule. Configure the Lambda function to process the queue.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
By sending S3 "Object Created" events to EventBridge and matching those events with a rule that invokes your Lambda function, you trigger your ETL whenever a new CSV lands in S3, without extra polling or queue management. This direct, event-driven pattern keeps operational overhead to a minimum.

QUESTION 217
A company stores sensitive data in an Amazon Redshift table. The company needs to give specific users the ability to access the sensitive data. The company must not create duplication in the data.

Customer support users must be able to see the last four characters of the sensitive data. Audit users must be able to see the full value of the sensitive data. No other users can have the ability to access the sensitive information.

Which solution will meet these requirements?

A. Create a dynamic data masking policy to allow access based on each user role. Create IAM roles that have specific access permissions. Attach the masking policy to the column that contains sensitive data.
B. Enable metadata security on the Redshift cluster. Create IAM users and IAM roles for the customer support users and the audit users. Grant the IAM users and IAM roles permissions to view the metadata in the Redshift cluster.

C. Create a row-level security policy to allow access based on each user role. Create IAM roles that have specific access permissions. Attach the security policy to the table.
D. Create an AWS Glue job to redact the sensitive data and to load the data into a new Redshift table.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift's dynamic data masking lets you define masking policies on sensitive columns without duplicating data. You attach a masking policy that exposes only the last four characters to customer support IAM roles, while audit IAM roles see the full values. This meets the requirement for role-based column-level access with no data duplication.

QUESTION 218
A data engineer uses AWS Lake Formation to manage access to data that is stored in an Amazon S3 bucket. The data engineer configures an AWS Glue crawler to discover data at a specific file location in the bucket, s3://examplepath. The crawler execution fails with the following error: "The S3 location: s3://examplepath is not registered."

The data engineer needs to resolve the error.

Which solution will meet this requirement?

A. Attach an appropriate IAM policy to the IAM role of the AWS Glue crawler to grant the crawler permission to read the S3 location.
B. Register the S3 location in Lake Formation to allow the crawler to access the data.
C. Create a new AWS Glue database. Assign the correct permissions to the database for the crawler.
D. Configure the S3 bucket policy to allow cross-account access.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
In Lake Formation, you must explicitly register each S3 path as a data location before any Glue crawler or user can access it. Registering s3://examplepath in Lake Formation grants the crawler the metadata permissions it needs to crawl that location.

QUESTION 219
A company built a data lake and a data warehouse on AWS. The company wants to implement a data catalog to enhance the current data storage solutions. The company wants to have the capability to add business metadata and glossary information to the data catalog for every asset.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Glue Catalog. Create a user table for the business glossary. Use the AWS Glue API to change table properties to add business metadata. Create a web application to access the metadata.
B. Use an Apache Hive metastore. Create a user table for the business glossary. Use the ALTER TABLE command to change table properties to add business metadata. Create a web application to access the metadata.
C. Use Amazon DataZone. Create the business glossaries. Create metadata forms. Use the Amazon DataZone data portal to access the metadata.
D. Use Amazon OpenSearch Service. Create an index for the business glossary. Create a second index for the business metadata. Use the OpenSearch Service dashboard to access the metadata.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:

Amazon DataZone provides a fully managed data catalog and governance service with built-in support for business glossaries, custom metadata forms, and a user-friendly data portal. You can onboard your existing data assets, define glossaries, and capture business metadata without building or maintaining custom tables, APIs, or applications, minimizing operational overhead.

QUESTION 220
A data engineer is using an AWS Glue ETL job to remove outdated customer records from a table that contains customer account information. The data engineer is using the following SQL command to remove customers that exist in a table named monthly_accounts_update table from the customer accounts table:

```
MERGE INTO accounts t USING monthly_accounts_update s
ON t.customer = s.customer
WHEN MATCHED
THEN DELETE
```

What will happen when the data engineer runs the SQL command?

A. All customer records that exist in both the customer accounts table and the monthly_accounts_update table will be deleted from the accounts table.
B. Only customer records that are present in both tables will be retained in the customer accounts table.
C. The monthly_accounts_update table will be deleted.
D. No records will be deleted because the command syntax is not valid in AWS Glue.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
That MERGE statement deletes every row in accounts where the customer value matches a row in monthly_accounts_update. All overlapping customer records will be removed from the accounts table.

QUESTION 221
A company receives marketing campaign data from a vendor. The company ingests the data into an Amazon S3 bucket every 40 to 60 minutes. The data is in CSV format. File sizes are between 100 KB and 300 KB.

A data engineer needs to set-up an extract, transform, and load (ETL) pipeline to upload the content of each file to Amazon Redshift.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an AWS Lambda function that connects to Amazon Redshift and runs a COPY command. Use Amazon EventBridge to invoke the Lambda function based on an Amazon S3 upload trigger.
B. Create an Amazon Data Firehose stream. Configure the stream to use an AWS Lambda function as a source to pull data from the S3 bucket. Set Amazon Redshift as the destination.
C. Use Amazon Redshift Spectrum to query the S3 bucket. Configure an AWS Glue Crawler for the S3 bucket to update metadata in an AWS Glue Data Catalog.
D. Creates an AWS Database Migration Service (AWS DMS) task. Specify an appropriate data schema to migrate. Specify the appropriate type of migration to use.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Configuring your S3 bucket to emit “Object Created” events into EventBridge and using a rule to invoke a lightweight Lambda function lets you automatically run a Redshift COPY for each new CSV. This serverless, event-driven pattern requires no continuously running infrastructure and has minimal operational overhead.

QUESTION 222

A company wants to build a dimension table in an Amazon S3 bucket. The bucket contains historical data that includes 10 million records. The historical data is 1 TB in size.

A data engineer needs a solution to update changes for up to 10,000 records in the base table every day.

Which solution will meet this requirement with the LOWEST runtime?

A. Develop an Apache Spark job in Amazon EMR to read the historical data and the new changes into two Spark DataFrames. Use the Spark update method to update the base table.
B. Develop an AWS Glue Python job to read the historical data and new changes into two Pandas DataFrames. Use the Pandas update method to update the base table.
C. Develop an AWS Glue Apache Spark job to read the historical data and new changes into two Spark DataFrames. Use the Spark update method to update the base table.
D. Develop an Amazon EMR job to read new changes into Apache Spark DataFrames. Use the Apache Hudi framework to create the base table in Amazon S3. Use the Spark update method to update the base table.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
By using Apache Hudi on EMR you get native upsert support against your S3 “base table.” Hudi’s indexing and metadata management lets it rewrite only the small set of changed files (your ~10,000 daily records) rather than scanning and rewriting the full 1 TB. This minimizes processing and achieves the lowest runtime for daily updates.

QUESTION 223
A data engineer develops an AWS Glue Apache Spark ETL job to perform transformations on a dataset. When the data engineer runs the job, the job returns an error that reads, “No space left on device.”

The data engineer needs to identify the source of the error and provide a solution.

Which combinations of steps will meet this requirement MOST cost-effectively? (Choose two.)

A. Scale out the workers vertically to address data skewness.
B. Use the Spark UI and AWS Glue metrics to monitor data skew in the Spark executors.
C. Scale out the number of workers horizontally to address data skewness.
D. Enable the --write-shuffie-files-to-s3 job parameter. Use the salting technique.
E. Use error logs in Amazon CloudWatch to monitor data skew.

Correct Answer: BD
Section: (none)

Explanation/Reference:
Explanation:
Use the Spark UI and AWS Glue–exposed Spark metrics to pinpoint where partitions are disproportionately large (data skew) and where spill files are filling executor disk.

Enable the --write-shuffle-files-to-s3 job parameter so shuffle spills go to S3 instead of running out of local disk, and apply a salting technique on skewed keys to spread data more evenly across partitions.

QUESTION 224
A company has a data pipeline that uses an Amazon RDS instance, AWS Glue jobs, and an Amazon S3 bucket. The RDS instance and AWS Glue jobs run in a private subnet of a VPC and in the same security group.

A user made a change to the security group that prevents the AWS Glue jobs from connecting to the RDS instance. After the change, the security group contains a single rule that allows inbound SSH traffic from a specific IP address.

The company must resolve the connectivity issue.

Which solution will meet this requirement?

A. Add an inbound rule that allows all TCP traffic on all TCP ports. Set the security group as the source.
B. Add an inbound rule that allows all TCP traffic on all UDP ports. Set the private IP address of the RDS instance as the source.
C. Add an inbound rule that allows all TCP traffic on all TCP ports. Set the DNS name of the RDS instance as the source.
D. Replace the source of the existing SSH rule with the private IP address of the RDS instance. Create an outbound rule with the same source, destination, and protocol as the inbound SSH rule.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Since both the RDS instance and the Glue jobs share the same security group, allowing that security group as the source for inbound TCP traffic (on all ports) immediately restores connectivity from the Glue-attached ENIs to the database. This single rule change requires no additional infrastructure and incurs no extra maintenance.

QUESTION 225
A company builds a new data pipeline to process data for business intelligence reports. Users have noticed that data is missing from the reports.

A data engineer needs to add a data quality check for columns that contain null values and for referential integrity at a stage before the data is added to storage.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon SageMaker Data Wrangler to create a Data Quality and Insights report.
B. Use AWS Glue ETL jobs to perform a data quality evaluation transform on the data. Use an IsComplete rule on the requested columns. Use a ReferentialItegrity rule for each join.
C. Use AWS Glue ETL jobs to perform a SQL transform on the data to determine whether requested column contain null values. Use a second SQL transform to check referential integrity.
D. Use Amazon SageMaker Data Wrangler and a custom Python transform to create custom rules to check for null values and referential integrity.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue's built-in data quality evaluation transform lets you declaratively apply DQDL rules, like IsComplete for null checks and ReferentialIntegrity for joins, directly in your ETL job. This requires minimal custom code and no separate reporting or profiling infrastructure, giving you pre-load validations with the least operational overhead.

QUESTION 226
A company is setting up a data pipeline in AWS. The pipeline extracts client data from Amazon S3 buckets, performs quality checks, and transforms the data. The pipeline stores the processed data in a relational database. The company will use the processed data for future queries.

Which solution will meet these requirements MOST cost-effectively?

A. Use AWS Glue ETL to extract the data from the S3 buckets and perform the transformations. Use AWS Glue Data Quality to enforce suggested quality rules. Load the data and the quality check results into an Amazon RDS for MySQL instance.
B. Use AWS Glue Studio to extract the data from the S3 buckets. Use AWS Glue DataBrew to perform the transformations and quality checks. Load the processed data into an Amazon RDS for MySQL

instance. Load the quality check results into a new S3 bucket.

C. Use AWS Glue ETL to extract the data from the S3 buckets and perform the transformations. Use AWS Glue DataBrew to perform quality checks. Load the processed data and the quality check results into a new S3 bucket.
D. Use AWS Glue Studio to extract the data from the S3 buckets. Use AWS Glue DataBrew to perform the transformations and quality checks. Load the processed data and quality check results into an Amazon RDS for MySQL instance.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Using a single AWS Glue ETL job to both transform the data and invoke Glue Data Quality checks lets you declaratively enforce recommended rules without spinning up separate tools. You can then write the cleansed data and, if desired, the quality metrics, directly into your Amazon RDS for MySQL instance. This serverless, end-to-end approach minimizes service sprawl and only incurs Glue and RDS costs, making it the most cost-effective with the least operational overhead.

QUESTION 227
A company uses Amazon Redshift as a data warehouse solution. One of the datasets that the company stores in Amazon Redshift contains data for a vendor.

Recently, the vendor asked the company to transfer the vendor's data into the vendor's Amazon S3 bucket once each week.

Which solution will meet this requirement?

A. Create an AWS Lambda function to connect to the Redshift data warehouse. Configure the Lambda function to use the Redshift COPY command to copy the required data to the vendor's S3 bucket on a schedule.
B. Create an AWS Glue job to connect to the Redshift data warehouse. Configure the AWS Glue job to use the Redshift UNLOAD command to load the required data to the vendor's S3 bucket on a schedule.
C. Use the Amazon Redshift data sharing feature. Set the vendor's S3 bucket as the destination. Configure the source to be as a custom SQL query that selects the required data.
D. Configure Amazon Redshift Spectrum to use the vendor's S3 bucket a destination, Enable data querying in both directions.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
An AWS Glue job can connect to your Amazon Redshift cluster and execute the UNLOAD command to export the specified vendor data directly into the vendor's S3 bucket. Scheduling the Glue job to run weekly requires minimal operational effort, and UNLOAD is optimized for exporting large result sets efficiently.

QUESTION 228
A company uses an Amazon Redshift cluster as a data warehouse that is shared across two departments. To comply with a security policy, each department must have unique access permissions.

Department A must have access to tables and views for Department A. Department B must have access to tables and views for Department B.

The company often runs SQL queries that use objects from both departments in one query.

Which solution will meet these requirements with the LEAST operational overhead?

A. Group tables and views for each department into dedicated schemas. Manage permissions at the schema level.

B. Group tables and views for each department into dedicated databases. Manage permissions at the database level.
C. Update the names of the tables and views to follow a naming convention that contains the department names. Manage permissions based on the new naming convention.
D. Create an IAM user group for each department. Use identity-based IAM policies to grant table and view permissions based on the IAM user group.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
By organizing each department's tables and views into its own schema (for example, dept_a and dept_b), you can grant usage and object privileges at the schema level. Department A's role gets USAGE on schema dept_a and the necessary SELECT rights there (and no rights on dept_b), and vice versa for Department B. Because both schemas live in the same database, analysts can still run cross-department queries by schema-qualifying objects, and you avoid the extra complexity of multiple databases or intricate IAM policies.

QUESTION 229
A company wants to ingest streaming data into an Amazon Redshift data warehouse from an Amazon Managed Streaming for Apache Kafka (Amazon MSK) cluster. A data engineer needs to develop a solution that provides low data access time and that optimizes storage costs.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an external schema that maps to the MSK cluster. Create a materialized view that references the external schema to consume the streaming data from the MSK topic.
B. Develop an AWS Glue streaming extract, transform, and load (ETL) job to process the incoming data from Amazon MSK. Load the data into Amazon S3. Use Amazon Redshift Spectrum to read the data from Amazon S3.
C. Create an external schema that maps to the streaming data source. Create a new Amazon Redshift table that references the external schema.
D. Create an Amazon S3 bucket. Ingest the data from Amazon MSK. Create an event-driven AWS Lambda function to load the data from the S3 bucket to a new Amazon Redshift table.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
By using a serverless Glue streaming job to continuously pull your MSK records, transform them as needed, and land them in Parquet (or another columnar) files in S3, you:
1. Optimize storage costs – your data sits in S3, where you pay pennies per GB-month and can tier it further with lifecycle rules.
2. Get low-latency access – Redshift Spectrum lets you query S3-backed tables with millisecond planning time, so freshly landed data becomes queryable almost immediately.
3. Minimize ops overhead – you don't have to stand up or manage any EC2-based brokers, Lambda polling loops, or custom connector infrastructure. Glue's managed streaming runtime handles checkpointing, autoscaling, and fault tolerance for you.

Once the data lands in S3, you simply define an external schema in Redshift that points at the Glue Data Catalog database where your streaming job writes tables. Analysts can then query the "live" dataset via Spectrum as if it were inside Redshift, meeting both your performance and cost goals with minimal operational effort.

QUESTION 230
A sales company uses AWS Glue ETL to collect, process, and ingest data into an Amazon S3 bucket. The AWS Glue pipeline creates a new file in the S3 bucket every hour. File sizes vary from 200 KB to 300 KB. The company wants to build a sales prediction model by using data from the previous 5 years. The historic

data includes 44,000 files.

The company builds a second AWS Glue ETL pipeline by using the smallest worker type. The second pipeline retrieves the historic files from the S3 bucket and processes the files for downstream analysis. The company notices significant performance issues with the second ETL pipeline.

The company needs to improve the performance of the second pipeline.

Which solution will meet this requirement MOST cost-effectively?

A. Use a larger worker type.
B. Increase the number of workers in the AWS Glue ETL jobs.
C. Use the AWS Glue DynamicFrame grouping option.
D. Enable AWS Glue auto scaling.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Using the AWS Glue DynamicFrame grouping option (for example, groupFiles=True with an appropriate groupSize) combines many small input files into larger partitions at read time. This reduces the per-file overhead of task initialization and metadata operations, yielding much faster ETL runs without the added cost of more or bigger workers.

QUESTION 231
A company wants to combine data from multiple software as a service (SaaS) applications for analysis.

A data engineering team needs to use Amazon QuickSight to perform the analysis and build dashboards. A data engineer needs to extract the data from the SaaS applications and make the data available for QuickSight queries.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Create AWS Lambda functions that call the required APIs to extract the data from the applications. Store the data in an Amazon S3 bucket. Use AWS Glue to catalog the data in the S3 bucket. Create a data source and a dataset in QuickSight.
B. Use AWS Lambda functions as Amazon Athena data source connectors to run federated queries against the SaaS applications. Create an Athena data source and a dataset in QuickSight.
C. Use Amazon AppFlow to create a flow for each SaaS application. Set an Amazon S3 bucket as the destination. Schedule the flows to extract the data to the bucket. Use AWS Glue to catalog the data in the S3 bucket. Create a data source and a dataset in QuickSight.
D. Export data the from the SaaS applications as Microsoft Excel files. Create a data source and a dataset in QuickSight by uploading the Excel files.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon AppFlow provides fully managed, no-code connectors to a broad range of SaaS applications. By scheduling flows that land data in S3, you avoid building and maintaining custom API-calling code. Glue crawlers can catalog the files and make them available to QuickSight, giving you an end-to-end pipeline with minimal operational effort.

QUESTION 232
A company runs multiple applications on AWS. The company configured each application to output logs. The company wants to query and visualize the application logs in near real time.

Which solution will meet these requirements?

A. Configure the applications to output logs to Amazon CloudWatch Logs log groups. Create an Amazon S3 bucket. Create an AWS Lambda function that runs on a schedule to export the required log groups to the S3 bucket. Use Amazon Athena to query the log data in the S3 bucket.
B. Create an Amazon OpenSearch Service domain. Configure the applications to output logs to Amazon CloudWatch Logs log groups. Create an OpenSearch Service subscription filter for each log group to stream the data to OpenSearch. Create the required queries and dashboards in OpenSearch Service to analyze and visualize the data.
C. Configure the applications to output logs to Amazon CloudWatch Logs log groups. Use CloudWatch log anomaly detection to query and visualize the log data.
D. Update the application code to send the log data to Amazon QuickSight by using Super-fast, Parallel, In-memory Calculation Engine (SPICE). Create the required analyses and dashboards in QuickSight.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
By streaming your CloudWatch Logs log groups directly into an Amazon OpenSearch Service domain via subscription filters, you get near-real-time indexing and can build Kibana dashboards for querying and visualization with virtually no batching delays. This fully managed pipeline minimizes custom glue code and meets your requirements with low operational overhead.

QUESTION 233
An ecommerce company processes millions of orders each day. The company uses AWS Glue ETL to collect data from multiple sources, clean the data, and store the data in an Amazon S3 bucket in CSV format by using the S3 Standard storage class. The company uses the stored data to conduct daily analysis.

The company wants to optimize costs for data storage and retrieval.

Which solution will meet this requirement?

A. Transition the data to Amazon S3 Glacier Flexible Retrieval.
B. Transition the data from Amazon S3 to an Amazon Aurora cluster.
C. Configure AWS Glue ETL to transform the incoming data to Apache Parquet format.
D. Configure AWS Glue ETL to use Amazon EMR to process incoming data in parallel.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Converting the CSV files into Apache Parquet during your Glue ETL jobs dramatically reduces both storage size (because Parquet is a compressed, columnar format) and query cost (because analytics engines only scan the columns you need). This change requires no new infrastructure and pays off immediately for your daily analysis workloads.

QUESTION 234
A data engineer is optimizing query performance in Amazon Athena notebooks that use Apache Spark to analyze large datasets that are stored in Amazon S3. The data is partitioned.
An AWS Glue crawler updates the partitions.

The data engineer wants to minimize the amount of data that is scanned to improve efficiency of Athena queries.

Which solution will meet these requirements?

A. Apply partition filters in the queries.
B. Increase the frequency of AWS Glue crawler invocations to update the data catalog more often.
C. Organize the data that is in Amazon S3 by using a nested directory structure.

D. Configure Spark to use in-memory caching for frequently accessed data.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
By including predicates on your partition columns in each Athena query (for example, WHERE year = '2025' AND month = '05'), Athena prunes partitions at the optimizer level and reads only the matching S3 folders. This directly minimizes the data scanned and improves query efficiency with no additional infrastructure changes.

QUESTION 235
A company manages an Amazon Redshift data warehouse. The data warehouse is in a public subnet inside a custom VPC. A security group allows only traffic from within itself. An ACL is open to all traffic.

The company wants to generate several visualizations in Amazon QuickSight for an upcoming sales event. The company will run QuickSight Enterprise edition in a second AWS account inside a public subnet within a second custom VPC. The new public subnet has a security group that allows outbound traffic to the existing Redshift cluster.

A data engineer needs to establish connections between Amazon Redshift and QuickSight. QuickSight must refresh dashboards by querying the Redshift cluster.

Which solution will meet these requirements?

A. Configure the Redshift security group to allow inbound traffic on the Redshift port from the QuickSight security group.
B. Assign Elastic IP addresses to the QuickSight visualizations. Configure the QuickSight security group to allow inbound traffic on the Redshift port from the Elastic IP addresses.
C. Confirm that the CIDR ranges of the Redshift VPC and the QuickSight VPC are the same. If CIDR ranges are different, reconfigure one CIDR range to match the other. Establish network peering between the VPCs.
D. Create a QuickSight gateway endpoint in the Redshift VPC. Attach an endpoint policy to the gateway endpoint to ensure only specific QuickSight accounts can use the endpoint.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Because your Redshift cluster is publicly accessible in its VPC and QuickSight runs in a separate VPC and account, you simply need to allow QuickSight's outbound connections through the Redshift security group. By adding an inbound rule on the Redshift SG that permits traffic on port 5439 (or your cluster's port) from the QuickSight security group, QuickSight can refresh dashboards without any additional infrastructure changes.

QUESTION 236
A data engineer is building a data pipeline. A large data file is uploaded to an Amazon S3 bucket once each day at unpredictable times. An AWS Glue workflow uses hundreds of workers to process the file and load the data into Amazon Redshift. The company wants to process the file as quickly as possible.

Which solution will meet these requirements?

A. Create an on-demand AWS Glue trigger to start the workflow. Create an AWS Lambda function that runs every 15 minutes to check the S3 bucket for the daily file. Configure the function to start the AWS Glue workflow if the file is present.
B. Create an event-based AWS Glue trigger to start the workflow. Configure Amazon S3 to log events to AWS CloudTrail. Create a rule in Amazon EventBridge to forward PutObject events to the AWS Glue trigger.
C. Create a scheduled AWS Glue trigger to start the workflow. Create a cron job that runs the AWS Glue

job every 15 minutes. Set up the AWS Glue job to check the S3 bucket for the daily file. Configure the job to stop if the file is not present.
D. Create an on-demand AWS Glue trigger to start the workflow. Create an AWS Database Migration Service (AWS DMS) migration task. Set the DMS source as the S3 bucket. Set the target endpoint as the AWS Glue workflow.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
An event-based trigger driven by Amazon EventBridge provides the lowest-latency, lowest-overhead way to kick off your Glue workflow as soon as the file arrives. By enabling S3 data events in CloudTrail and writing a simple EventBridge rule that matches the PutObject event for your bucket, you can target your Glue workflow's trigger directly, eliminating continuous polling or extra Lambda or DMS infrastructure.

QUESTION 237
A data engineer needs to run a data transformation job whenever a user adds a file to an Amazon S3 bucket. The job will run for less than 1 minute. The job must send the output through an email message to the data engineer. The data engineer expects users to add one file every hour of the day.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Create a small Amazon EC2 instance that polls the S3 bucket for new files. Run transformation code on a schedule to generate the output. Use operating system commands to send email messages.
B. Run an Amazon Elastic Container Service (Amazon ECS) task to poll the S3 bucket for new files. Run transformation code on a schedule to generate the output. Use operating system commands to send email messages.
C. Create an AWS Lambda function to transform the data. Use Amazon S3 Event Notifications to invoke the Lambda function when a new object is created. Publish the output to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe the data engineer's email account to the topic.
D. Deploy an Amazon EMR cluster. Use EMR File System (EMRFS) to access the files in the S3 bucket. Run transformation code on a schedule to generate the output to a second S3 bucket. Create an Amazon Simple Notification Service (Amazon SNS) topic. Configure Amazon S3 Event Notifications to notify the topic when a new object is created.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
An S3 event notification on ObjectCreated can directly invoke a Lambda function as soon as a new file arrives. Lambda can perform your <1 minute transformation and then publish the result to an SNS topic. With the data engineer subscribed to that topic by email, the output is delivered automatically - no servers to manage, no polling, and minimal ongoing operations.

QUESTION 238
A company uses Amazon S3 and AWS Glue Data Catalog to manage a data lake that contains contact information for customers. The company uses PySpark and AWS Glue jobs with a DynamicFrame to run a workflow that processes data within the data lake.

A data engineer notices that the workflow is generating errors as a result of how customer postal codes are stored in the data lake. Some postal codes include unnecessary numbers or invalid characters.

The data engineer needs a solution to address the errors and correct the postal codes in the data lake.

A. Create a schema definition for PySpark that matches the format the processing workflow requires for postal codes. Pass the schema to the DynamicFrame during processing.
B. Use AWS Glue workflow properties to allow job state sharing. Configure the AWS Glue jobs to read values from the postal code column by using the properties from a previously successful run of the jobs.

C. Configure the column.push_down_predicate setting and the catalogPartitionPredicate settings for the postal code column in the DynamicFrame.
D. Set the DynamicFrame additional_options parameter ‘useS3ListImplementation’ to True.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
By defining and enforcing a strict schema in PySpark that specifies the exact format you expect for postal codes (for example, a string of five digits), Spark will automatically flag or coerce any values that don’t match. Passing that schema into your Glue DynamicFrame ensures that malformed postal codes are identified and can be corrected or filtered out as part of your ETL logic, eliminating the downstream errors.

QUESTION 239
A data engineer is troubleshooting an AWS Glue workflow that occasionally fails. The engineer determines that the failures are a result of data quality issues. A business reporting team needs to receive an email notification any time the workflow fails in the future.

Which solution will meet this requirement?

A. Create an Amazon Simple Notification Service (Amazon SNS) FIFO topic. Subscribe the team’s email account to the SNS topic. Create an AWS Lambda function that initiates when the AWS Glue job state changes to FAILED. Set the SNS topic as the target.
B. Create an Amazon Simple Notification Service (Amazon SNS) standard topic. Subscribe the team’s email account to the SNS topic. Create an Amazon EventBridge rule that triggers when the AWS Glue job state changes to FAILED. Set the SNS topic as the target.
C. Create an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Subscribe the team’s email account to the SQS queue. Create an AWS Config rule that triggers when the AWS Glue job state changes to FAILED. Set the SQS queue as the target.
D. Create an Amazon Simple Queue Service (Amazon SQS) standard queue. Subscribe the team’s email account to the SQS queue. Create an Amazon EventBridge rule that triggers when the AWS Glue job state changes to FAILED. Set the SQS queue as the target.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon EventBridge can detect AWS Glue job state change events natively. You configure a rule that matches when a job’s state transitions to FAILED and set a standard Amazon SNS topic as the target. By subscribing the business reporting team’s email addresses to that SNS topic, they’ll receive automatic email notifications whenever the workflow fails, with no custom polling or Lambda required.

QUESTION 240
A company uses AWS Glue jobs to implement several data pipelines. The pipelines are critical to the company.

The company needs to implement a monitoring mechanism that will alert stakeholders if the pipelines fail.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an Amazon EventBridge rule to match AWS Glue job failure events. Configure the rule to target an AWS Lambda function to process events. Configure the function to send notifications to an Amazon Simple Notification Service (Amazon SNS) topic.
B. Configure an Amazon CloudWatch Logs log group for the AWS Glue jobs. Create an Amazon EventBridge rule to match new log creation events in the log group. Configure the rule to target an AWS Lambda function that reads the logs and sends notifications to an Amazon Simple Notification Service (Amazon SNS) topic if AWS Glue job failure logs are present.
C. Create an Amazon EventBridge rule to match AWS Glue job failure events. Define an Amazon CloudWatch metric based on the EventBridge rule. Set up a CloudWatch alarm based on the metric to

send notifications to an Amazon Simple Notification Service (Amazon SNS) topic.

D. Configure an Amazon CloudWatch Logs log group for the AWS Glue jobs. Create an Amazon EventBridge rule to match new log creation events in the log group. Configure the rule to send notifications to an Amazon Simple Notification Service (Amazon SNS) topic.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
By creating an EventBridge rule that matches Glue job state-change events where the state is FAILED, you can emit a custom CloudWatch metric without writing any code. Then you define a CloudWatch alarm on that metric to notify an SNS topic upon breach. This approach requires no Lambda functions or log parsing (just the rule, the metric, and the alarm) minimizing operational overhead.

QUESTION 241
A company uses AWS Glue Apache Spark jobs to handle extract, transform, and load (ETL) workloads. The company has enabled logging and monitoring for all AWS Glue jobs.

One of the AWS Glue jobs begins to fail. A data engineer investigates the error and wants to examine metrics for all individual stages within the job.

How can the data engineer access the stage metrics?

A. Examine the AWS Glue job and stage details in the Spark UI.
B. Examine the AWS Glue job and stage metrics in Amazon CloudWatch.
C. Examine the AWS Glue job and stage logs in AWS CloudTrail logs.
D. Examine the AWS Glue job and stage details by using the run insights feature on the job.

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Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue provides a built-in Spark UI (accessible from the Glue console’s job run details) that exposes per-stage metrics, such as task counts, shuffle read/write sizes, and execution times, for each Spark stage. This is the most direct way to drill down into the individual stages when troubleshooting a failing Glue Spark job.

QUESTION 242
A data engineer notices slow query performance on a highly partitioned table that is in Amazon Athena. The table contains daily data for the previous 5 years, partitioned by date.

The data engineer wants to improve query performance and to automate partition management.

Which solution will meet these requirements?

A. Use an AWS Lambda function that runs daily. Configure the function to manually create new partitions in AWS Glue for each day’s data.
B. Use partition projection in Athena. Configure the table properties by using a date range from 5 years ago to the present.
C. Reduce the number of partitions by changing the partitioning schema from daily to monthly granularity.
D. Increase the processing capacity of Athena queries by allocating more compute resources.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Partition projection eliminates the need for frequent metadata lookups or explicit partition registration in the AWS Glue Data Catalog. By defining a date range in your table’s properties (for example,

projection.enabled=true, projection.order_date.type=DATE, projection.order_date.range=2020/01/01,NOW, and the appropriate format), Athena will calculate partitions on-the-fly. This both speeds up queries (avoiding catalog calls for thousands of partitions) and removes the need for any external partition-management jobs.

QUESTION 243

A data engineer is using an Apache Iceberg framework to build a data lake that contains 100 TB of data. The data engineer wants to run AWS Glue Apache Spark jobs that use the Iceberg framework.

What combination of steps will meet these requirements? (Choose two.)

A. Create a key named --conf for an AWS Glue job. Set Iceberg as a value for the --datalake-formats job parameter.

B. Specify the path to a specific version of Iceberg by using the -extra-jars job parameter. Set Iceberg as a value for the datalake-formats job parameter.

C. Set Iceberg as a value for the --datalake-formats job parameter.

D. Set the --enable-auto-scaling parameter to true.

E. Add the --job-bookmark-option: job-bookmark-enable parameter to an AWS Glue job.

Correct Answer: BC
Section: (none)

Explanation/Reference:
Explanation:
To run AWS Glue Apache Spark jobs with the Apache Iceberg framework, you must:
- Set --datalake-formats to iceberg so Glue knows to use the Iceberg format.
- Specify the Iceberg library by using the --extra-jars parameter to include the appropriate Iceberg JAR files for Glue to access the Iceberg table format.

QUESTION 244

A data engineer is configuring an AWS Glue Apache Spark extract, transform, and load (ETL) job. The job contains a sort-merge join of two large and equally sized DataFrames.

The job is failing with the following error: No space left on device.

Which solution will resolve the error?

A. Use the AWS Glue Spark shuffle manager.

B. Deploy are Amazon Elastic Block Store (Amazon EBS) volume for the job to use.

C. Convert the sort-merge join in the job to be a broadcast join.

D. Convert the DataFrames to DynamicFrames, and perform a DynamicFrame join in the job.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
The error "No space left on device" during a sort-merge join typically results from insufficient disk space during the shuffle phase of a Spark job. Using the AWS Glue Spark shuffle manager (optimized shuffle manager) helps reduce disk usage and improves shuffle performance by optimizing how intermediate data is written and read. This directly addresses the issue without requiring job logic changes or manual resource management.

QUESTION 245
A company needs to implement a workflow to process transactions. Each transaction goes through multiple levels of validation. Each validation level depends on the preceding validation level.

The workflow must either process or reject each transaction within 24-hours. The workflow must run for less than 24 hours total.

Which solution will meet these requirements with the LEAST operational cost?

A. Create a standard workflow in AWS Step Functions. Implement a Wait for Callback pattern to wait for the validation steps to finish.
B. Create an express workflow in AWS Step Functions. Implement a Wait for Callback pattern to wait for the validation steps to finish.
C. Use AWS Lambda functions to implement the workflow. Use Amazon EventBridge to invoke the validation steps.
D. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to implement the workflow.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Standard Step Functions support long-running, sequential workflows and the Wait for Callback pattern without charging for idle wait time, minimizing cost while ensuring completion within 24 hours. Express would bill for long waits, Lambda+EventBridge adds complexity and limits, and MWAA has higher operational overhead and cost.

QUESTION 246
A data engineer configures a large number of AWS Glue jobs that all start up around the same time. All the jobs run for less than 1 hour in the same subnet of the same VPC. All the AWS Glue jobs run on a G1.X worker type.

Some of the jobs occasionally fail with the following error: "The specified subnet does not have enough free addresses to satisfy the request".

What is the likely root cause of the error?

A. There are not enough IP addresses in the subnet.
B. The G1.X worker type cannot access the subnet.
C. AWS Glue does not have the correct IAM permissions to add additional IP addresses to the subnet.
D. There are not enough IP addresses in the VPC.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Each AWS Glue worker requires an elastic network interface (ENI) and consumes an IP address in the subnet. When a large number of Glue jobs start simultaneously, the subnet can run out of available IP addresses. This causes the error “The specified subnet does not have enough free addresses.” The root cause is insufficient IPs in the subnet, not the VPC overall.

QUESTION 247
A company runs an Apache Spark application every night in an Amazon EMR cluster. The company uses Amazon EC2 instances to supply compute capacity for the EMR cluster. The company deployed the Spark application in cluster mode.

An error occurs in the Spark application. A log for the error is stored in the application's Spark driver standard error logs. A data engineer needs to investigate the error.

Where can the data engineer find this error log?

A. The engineer can connect to the web UI on the live cluster to see the YARN ResourceManager logs.
B. The engineer can connect to the persistent application UI to see the first YARN container log in the Spark UI.
C. The engineer can connect to the Amazon EMR console to see the Amazon EMR step logs that are archived in Amazon S3.
D. The engineer can connect to the primary node of the cluster by using SSH to see the Spark history server logs.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
In EMR cluster mode, the Spark driver runs inside the cluster and its standard error logs are captured as step logs. These logs are automatically archived to Amazon S3 and accessible from the Amazon EMR console under step logs. This is the correct location for investigating Spark driver errors.

QUESTION 248
A company processes 500 GB of audience and advertising data daily, storing CSV files in Amazon S3 with schemas registered in AWS Glue Data Catalog. They need to convert these files to Apache Parquet format and store them in an S3 bucket.

The solution requires a long-running workflow with 15 GiB memory capacity to process the data concurrently, followed by a correlation process that begins only after the first two processes complete.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to orchestrate the workflow by using AWS Glue. Configure AWS Glue to begin the third process after the first two processes have finished.
B. Use Amazon EMR to run each process in the workflow. Create an Amazon Simple Queue Service (Amazon SQS) queue to handle messages that indicate the completion of the first two processes. Configure an AWS Lambda function to process the SQS queue by running the third process.
C. Use AWS Glue workflows to run the first two processes in parallel. Ensure that the third process starts after the first two processes have finished.
D. Use AWS Step Functions to orchestrate a workflow that uses multiple AWS Lambda functions. Ensure that the third process starts after the first two processes have finished.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue workflows natively orchestrate multiple Glue jobs in parallel and support dependency triggers so the third job starts only after the first two finish. Glue jobs provide the required ~15 GiB memory capacity and easily convert CSV to Parquet using the Glue Data Catalog, delivering a fully managed, low-operations solution.

QUESTION 249
A company uses an organization in AWS Organizations to manage multiple AWS accounts. The company uses an enhanced fanout data stream in Amazon Kinesis Data Streams to receive streaming data from multiple producers. The company runs the data stream in an account named Account A. The company wants to use an AWS Lambda function in an account named Account B to process the data from the data stream. The company creates a Lambda execution role in Account B that has permissions to access data from the data stream in Account A.

What additional step must the company take to meet this requirement?

A. Create a service control policy (SCP) to grant the data stream read access to the cross-account Lambda execution role. Attach the SCP to Account A.
B. Add a resource-based policy to the data stream to allow read access for the cross-account Lambda execution role.
C. Create a service control policy (SCP) to grant the data stream read access to the cross-account Lambda execution role. Attach the SCP to Account B.
D. Add a resource-based policy to the cross-account Lambda function to grant the data stream read access to the function.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
To enable cross-account Lambda processing of Kinesis Data Streams, the stream in Account A must explicitly allow the Lambda execution role from Account B. This is done by adding a resource-based policy on the Kinesis data stream to grant kinesis:SubscribeToShard and related read permissions to the cross-account role. Without this resource-based policy, the Lambda in Account B cannot consume the data.

QUESTION 250
A company needs to use Amazon Athena to analyze data that is in an Amazon S3 bucket. A data engineer needs to configure AWS Glue table partitions for year, month, and day. The data engineer needs to create the partitions every day to adjust to schema changes in the data.

Which solution will meet these requirements?

A. Use AWS Glue DataBrew to create the partitions for the AWS Glue table.
B. Use an AWS Lambda function to create the partitions for the AWS Glue table.
C. Set partition projection properties for the AWS Glue table.
D. Configure an AWS Glue crawler to run on a set schedule.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Partition projection in AWS Glue lets Athena automatically calculate partitions based on table properties without explicitly creating them each day. This eliminates the need for crawlers or Lambda jobs, reduces operational overhead, and adapts to schema changes while supporting year, month, and day partitions efficiently.

QUESTION 251
A hotel management company receives daily data files from each of its hotels. The company wants to upload its data to AWS. The company plans to use Amazon Athena to access the files. The company needs to protect the files from accidental deletion. The company will develop an application on its on-premises servers to automatically forward the files to a fully managed AWS ingestion service.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS DataSync to replicate data from the on-premises servers to Amazon Elastic File System (Amazon EFS). Configure automatic backups in AWS Backup.
B. Use the Amazon Kinesis Agent on the on-premises servers to send data to Amazon Data Firehose. Store the data in an Amazon S3 bucket that has versioning enabled.
C. Use AWS Glue jobs to ingest data from the on-premises servers into Amazon RDS. Enable automated backups for data protection.
D. Use a self-managed Apache Kafka agent on the on-premises servers to stream data to Amazon Managed Streaming for Apache Kafka (Amazon MSK). Store the data in an Amazon S3 bucket with

versioning enabled.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Kinesis Agent on-prem sends files to fully managed Kinesis Data Firehose, which delivers them to Amazon S3. Enabling S3 versioning protects against accidental deletions, and Athena can query directly from S3 - meeting requirements with minimal operational overhead.

QUESTION 252
A company wants to use Apache Spark jobs that run on an Amazon EMR cluster to process streaming data. The Spark jobs will transform and store the data in an Amazon S3 bucket. The company will use Amazon Athena to perform analysis.

The company needs to optimize the data format for analytical queries.

Which solutions will meet these requirements with the SHORTEST query times? (Choose two.)

A. Use Avro format. Use AWS Glue Data Catalog to track schema changes.
B. Use ORC format. Use AWS Glue Data Catalog to track schema changes.
C. Use Apache Parquet format. Use an external Amazon DynamoDB table to track schema changes.
D. Use Apache Parquet format. Use AWS Glue Data Catalog to track schema changes.
E. Use ORC format. Store schema definitions in separate files in Amazon S3.

Correct Answer: BD
Section: (none)

Explanation/Reference:
Explanation:
ORC is a columnar format optimized for Athena and provides high compression and predicate pushdown for faster queries; Glue Data Catalog manages schema evolution.

Parquet is a columnar format optimized for Athena with efficient compression and predicate pushdown; Glue Data Catalog tracks schema changes.

QUESTION 253
A company adds new data to a large CSV file in an Amazon S3 bucket every day. The file contains company sales data from the previous 5 years. The file currently includes more than 5,000 rows. The CSV file structure is shown below with sample data:

ID	SALE_DATE	ITEM_SOLD	SALE_PRICE	SALES_REP	STORE_NAME
1	01-Jan-2024	TV	1000	Terry	Alaska
2	02-Jan-2024	DVD player	100	Diego	Boston

The company needs to use Amazon Athena to run queries on the CSV file to fetch data from a specific time period.

Which solution will meet this requirement MOST cost-effectively?

A. Write an Apache Spark script to convert the CSV data to JSON format. Create an AWS Glue job to run the script every day. Catalog the JSON data in AWS Glue. Run the Athena queries on the JSON data.
B. Use prefixes to partition the data in the S3 bucket. Use the SALE_DATE column to create a partition for each day. Catalog the data in AWS Glue and ensure that the partitions are added. Update the Athena queries to use the new partitions.
C. Launch an Amazon EMR cluster. Specify AWS Glue Data Catalog as the default Apache Hive metastore. Use Presto in Trino to run queries on the data.
D. Create an Amazon RDS database. Create a table named SALES that matches the schema of the CSV

file. Create an index on the SALE_DATE column. Create an AWS Lambda function to load the CSV data into the RDS database. Use S3 Event Notifications to invoke the Lambda function.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Partitioning the S3 data by SALE_DATE (daily prefixes) lets Athena prune non-matching partitions so it scans only the requested time range, minimizing bytes scanned and cost while working directly on the existing CSVs. Cataloging the partitions in AWS Glue enables efficient querying with minimal operational overhead.

QUESTION 254
A data engineer must implement a data cataloging solution to track schema changes in an Amazon Redshift table.

Which solution will meet these requirements?

A. Schedule an AWS Glue crawler to run every day on the table by using the Java Database Connectivity (JDBC) driver. Configure the crawler to update an AWS Glue Data Catalog.
B. Use AWS DataSync to log the table metadata to an AWS Glue Data Catalog. Use an AWS Glue crawler to update the Data Catalog every day.
C. Use the AWS Schema Conversion Tool (AWS SCT) to log the table metadata to an Apache Hive metastore. Use Amazon EventBridge Scheduler to update the metastore every day.
D. Schedule an AWS Glue crawler to run every day on the table. Configure the crawler to update an Apache Hive metastore.

Correct Answer: A
Section: (none)

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Explanation/Reference:
Explanation:
An AWS Glue crawler can connect to Amazon Redshift via JDBC, detect schema changes, and automatically update the AWS Glue Data Catalog on a schedule, providing the required ongoing schema tracking.

QUESTION 255
A data engineer is building a solution to detect sensitive information that is stored in a data lake across multiple Amazon S3 buckets. The solution must detect personally identifiable information (PII) that is in a proprietary data format.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use the AWS Glue Detect PII transform with specific patterns.
B. Use Amazon Made with managed data identifiers.
C. Use an AWS Lambda function with custom regular expressions.
D. Use Amazon Athena with a SQL query to match the custom formats.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
The AWS Glue Detect PII transform is a built-in feature that can automatically identify personally identifiable information (PII) using predefined patterns or custom regular expressions. It works directly within AWS Glue jobs on proprietary data formats and integrates with the Glue Data Catalog, delivering a managed, low-overhead solution compared to building and maintaining custom Lambda or SQL-based detection.

QUESTION 256

A data engineer is building a new data pipeline that stores metadata in an Amazon DynamoDB table. The data engineer must ensure that all items that are older than a specified age are removed from the DynamoDB table daily.

Which solution will meet this requirement with the LEAST configuration effort?

A. Enable DynamoDB TTL on the DynamoDB table. Adjust the application source code to set the TTL attribute appropriately.
B. Create an Amazon EventBridge rule that uses a daily cron expression to trigger an AWS Lambda function to delete items that are older than the specified age.
C. Add a lifecycle configuration to the DynamoDB table that deletes items that are older than the specified age.
D. Create a DynamoDB stream that has an AWS Lambda function that reacts to data modifications. Configure the Lambda function to delete items that are older than the specified age.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
DynamoDB Time to Live (TTL) is a fully managed feature that automatically expires and deletes items based on a timestamp attribute you define. By enabling TTL and setting the TTL attribute in the application code, items older than the specified age are removed without the need for custom scheduling or Lambda functions, offering the least configuration effort.

QUESTION 257
A ride-sharing company stores records for all rides in an Amazon DynamoDB table. The table includes the following columns and types of values:

RideID	RiderID	DriverID	RideStatus	TripStartTime	TripEndTime
XA1231	AXEF1	BN123	Active	2025-02-11 12:23:34:00	NULL
XA1232	AXEF2	BN124	Completed	2025-02-11 08:36:12:00	2025-02-11 08:55:02:00

The table currently contains billions of items. The table is partitioned by RideID and uses TripStartTime as the sort key. The company wants to use the data to build a personal interface to give drivers the ability to view the rides that each driver has completed, based on RideStatus. The solution must access the necessary data without scanning the entire table.

Which solution will meet these requirements?

A. Create a local secondary index (LSI) on DriverID.
B. Create a global secondary index (GSI) that uses RiderID as the partition key and RideStatus as the sort key.
C. Create a global secondary index (GSI) that uses DriverID as the partition key and RideStatus as the sort key.
D. Create a filter expression that uses RiderID and RideStatus.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
To let drivers efficiently query only their completed rides, you need a global secondary index (GSI) with DriverID as the partition key (so queries can be targeted per driver) and RideStatus as the sort key (so you can query for “Completed” rides without scanning the full table). This avoids costly scans and supports fast, targeted lookups at scale.

QUESTION 258
A company stores information about its subscribers in an Amazon S3 bucket. The company runs an analysis every time a subscriber ends their subscription. The company uses AWS Lambda functions to

respond to events from the S3 bucket by performing analyses.

The Lambda functions clean data from the S3 bucket and initiate an AWS Glue workflow. The Lambda functions have 128 MB of memory and 512 MB of ephemeral storage. The Lambda functions have a timeout of 15 seconds.

All three functions successfully finish running. However, CPU usage is often near 100%, which causes slow performance. The company wants to improve the performance of the functions and reduce the total runtime of the pipeline.

Which solution will meet these requirements?

A. Increase the memory of the Lambda functions to 512 MB.
B. Increase the number of retries by using the Maximum Retry Attempts setting.
C. Configure the Lambda functions to run in the company's VPC.
D. Increase the timeout value for the Lambda functions from 15 seconds to 30 seconds.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
In AWS Lambda, increasing memory also proportionally increases the allocated CPU. By raising the Lambda functions' memory from 128 MB to 512 MB, the functions receive more CPU power, which reduces execution time and improves performance without changing logic or timeouts.

QUESTION 259
A company uses a data stream in Amazon Kinesis Data Streams to collect transactional data from multiple sources. The company uses an AWS Glue extract, transform, and load (ETL) pipeline to look for outliers in the data from the stream. When the workflow detects an outlier, it sends a notification to an Amazon Simple Notification Service (Amazon SNS) topic. The SNS topic initiates a second workflow to retrieve logs for the outliers and stores the logs in an Amazon S3 bucket.

The company experiences delays in the notifications to the SNS topic during periods when the data stream is processing a high volume of data. When the company examines Amazon CloudWatch logs, the company notices a high value for the glue.driver.BlockManager.disk.diskSpaceUsed_MB metric when the traffic is high. The company must resolve this issue.

Which solution will meet this requirement with the LEAST operational effort?

A. Increase the number of data processing units (DPUs) in AWS Glue ETL jobs.
B. Use Amazon EMR to manage the ETL pipeline instead of AWS Glue.
C. Use AWS Step Functions to orchestrate a parallel workflow state.
D. Enable auto scaling for the AWS Glue ETL jobs.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
The high BlockManager disk usage indicates memory pressure and spilling due to under-provisioned Spark executors. Enabling AWS Glue auto scaling lets the job automatically add workers during traffic spikes, alleviating memory/disk pressure and reducing delays with minimal operational effort.

QUESTION 260
A company has a data processing pipeline that runs multiple SQL queries in sequence against an Amazon Redshift cluster. The company merges with a second company. The original company modifies a query that aggregates sales revenue data to join sales tables from both companies. The sales table for the first company is named Table S1. The sales table for the second company is named Table S2. Table S1 contains 10 billion records. Table S2 contains 900 million records.
The query becomes slow after the modification. A data engineer must improve the query performance.

Which solutions will meet these requirements? (Choose two.)

A. Use the KEY distribution style for both sales tables. Select a low cardinality column to use for the join.
B. Use the KEY distribution style for both sales tables. Select a high cardinality column to use for the join.
C. Use the EVEN distribution style for Table S1. Use the ALL distribution style for Table S2.
D. Use the Amazon Redshift query optimizer to review and select optimizations to implement.
E. Use Amazon Redshift Advisor to review and select optimizations to implement.

Correct Answer: BE
Section: (none)

Explanation/Reference:
Explanation:
Choosing KEY distribution on both tables with a high-cardinality join column colocates matching rows across nodes and avoids data skew, improving join performance.

Redshift Advisor provides automated, actionable recommendations (e.g., distribution and sort keys, stats) to further optimize the slow query with minimal effort.

QUESTION 261
A gaming company uses AWS Glue to perform read and write operations on Apache Iceberg tables for real-time streaming data. The data in the Iceberg tables is in Apache Parquet format. The company is experiencing slow query performance.

Which solutions will improve query performance? (Choose two.)

A. Use AWS Glue Data Catalog to generate column-level statistics for the Iceberg tables on a schedule.
B. Use AWS Glue Data Catalog to automatically compact the Iceberg tables.
C. Use AWS Glue Data Catalog to automatically optimize indexes for the Iceberg tables.
D. Use AWS Glue Data Catalog to enable copy-on-write for the Iceberg tables.
E. Use AWS Glue Data Catalog to generate views for the Iceberg tables.

Correct Answer: BD
Section: (none)

Explanation/Reference:
Explanation:
Compaction reduces many small Parquet files into larger ones, lowering file-listing and open/scan overhead for faster reads.

Copy-on-write favors read performance by materializing updates into rewritten data files, resulting in more contiguous, scan-efficient files for queries.

QUESTION 262
A data engineer at a company is optimizing extract, transform, and load (ETL) workflows. The current architecture uses Amazon EMR and Apache Spark for large-scale transformations and AWS Glue for other ETL tasks. The workflows load processed data into an Amazon S3 based data lake.

The company wants to move to a fully managed serverless solution that can orchestrate multiple ETL jobs and automate execution. The new solution must continue to use Spark to process data. The company needs to orchestrate and automate the ETL workflows with minimal manual intervention.

Which solution will meet these requirements?

A. Migrate all ETL jobs to AWS Glue. Use AWS Glue workflows to orchestrate the pipeline.
B. Configure AWS Step Functions and Amazon EventBridge to orchestrate and invoke ETL workflows in AWS Glue and Amazon EMR.
C. Configure AWS Lambda functions to process Amazon S3 event notifications for data transformation tasks when new data is uploaded.

D. Use Amazon Managed Workflows for Apache Airflow automatic scheduling to orchestrate the Spark-based ETL jobs.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue provides a fully managed serverless ETL service with native Spark support. By migrating all ETL jobs to Glue, the company eliminates the need to manage Amazon EMR clusters. Glue workflows can orchestrate and automate multiple ETL jobs, ensuring dependencies are respected and executions are automated with minimal manual intervention. This satisfies the requirement for a serverless, Spark-based, fully managed solution.

QUESTION 263
A company that operates globally must follow regulations that require data from an AWS Region to be accessible only within that Region.

A data engineer is creating a data pipeline that will create resources in the Region where the data engineer works. The data pipeline should have access to data only from the Region where the data engineer works. The pipeline uses Active Directory as an identity and authentication system. The pipeline uses a custom identity broker application to verify that employees are signed in to Active Directory and to obtain temporary credentials by using the AssumeRole API operation.

Which solution will meet the locality requirements with the LEAST administrative effort?

A. Create an IAM role that has permissions to create resources. Create a policy for each Region that ensures users can create resources only in that Region. Pass the policy as the session policy when employees obtain the temporary credentials.
B. Create an IAM role for data engineers in each Region separately. Instruct each data engineer to obtain temporary credentials by assuming the appropriate Region specific IAM role.
C. Create an IAM group for each Region. Include the required IAM policies for each IAM group. Add users to each IAM group so that when users log in by obtaining the temporary credentials, the users will receive the appropriate access based on the IAM group.
D. Create individual IAM policies that allow users to create resources in a specific Region. Assign the policies to each data engineer. Allow users to assume the individually assigned role when the users log in to AWS.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Using a single IAM role with region-scoped session policies (using the aws:RequestedRegion condition) applied at AssumeRole time lets the identity broker grant temporary credentials limited to the engineer's Region. This enforces Regional access while avoiding per-Region roles or per-user policies, minimizing administration.

QUESTION 264
A company needs a solution to automatically ingest data from an Amazon S3 bucket to an Amazon Redshift table every day. The company must obfuscate sensitive data in Amazon Redshift. Users must not be able to view the sensitive data as plaintext.

Which solution will meet these requirements with the LEAST operational overhead?

A. Enable Amazon Macie for the company's AWS account. Configure a sensitive data discovery job for the S3 bucket. Configure an AWS Glue job to load the data into Amazon Redshift.
B. Enable Amazon Macie for the company's AWS account. Configure an AWS Glue job to flag and obfuscate data columns based on the Macie analysis. Load the data into Amazon Redshift.
C. Configure an AWS Glue job to read the data from the S3 bucket, detect and obfuscate sensitive data, and load the data into Amazon Redshift.

D. Configure an AWS Glue job to load the data into Amazon Redshift. Use Amazon Redshift to detect the sensitive data, create masking policies, and apply dynamic masking.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Schedule an AWS Glue job to load S3 data into Amazon Redshift, then use Redshift's built-in sensitive data discovery and dynamic data masking to identify sensitive columns and enforce masking at query time. This meets the daily ingest and non-plaintext visibility requirements with the least operational overhead, since masking is managed natively in Redshift without custom obfuscation pipelines or Macie-based preprocessing.

QUESTION 265
A company needs to aggregate and filter a large amount of streaming data in real-time with low latency. The company needs to store the data in Amazon S3 for analysis.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Use Amazon Kinesis Data Streams with provisioned capacity and AWS Lambda functions to perform custom transformations and to integrate with Amazon S3.
B. Use Amazon Data Firehose with built-in data transformations. Deliver the data directly to Amazon S3.
C. Use Amazon Kinesis Data Streams and Amazon Managed Service for Apache Flink to perform complex processing and to integrate with Amazon S3.
D. Use Amazon Data Firehose and AWS Lambda functions to perform custom transformations and to deliver the data to Amazon S3.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon Kinesis Data Streams with Amazon Managed Service for Apache Flink performs stateful, low-latency streaming aggregations and filtering at scale, then writes results to Amazon S3 - meeting real-time requirements while remaining fully managed and operationally efficient.

QUESTION 266
A company is developing machine learning (ML) models. A data engineer needs to apply data quality rules to training data. The company stores the training data in an Amazon S3 bucket.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an AWS Lambda function to check data quality and to raise exceptions in the code. Run the function when data is added to the S3 bucket. Create an Amazon CloudWatch alarm for exceptions in the code.
B. Create an AWS Glue DataBrew project for the data in the S3 bucket. Create a ruleset for the data quality rules. Create a profile job to run the data quality rules. Use Amazon EventBridge to run the profile job when data is added to the S3 bucket.
C. Create an Amazon EMR provisioned cluster. Add a Python open source data quality package to the EMR cluster. Use the Python package to write code for data quality rules and to copy the data from the S3 bucket to the EMR cluster. Copy the data from the S3 bucket to the EMR cluster. Run the data quality rules.
D. Create AWS Lambda functions to evaluate data quality rules. Use AWS Step Functions to orchestrate a workflow that publishes notifications when the data fails to meet data quality rules.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:

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AWS Glue DataBrew is a fully managed, serverless service with built-in data quality rulesets and profiling jobs that run directly on S3 data. Triggering the profile job via EventBridge on object creation applies rules with minimal code and management overhead.

QUESTION 267
A data engineer needs to analyze time-sensitive sales data. The company stores the data in an Amazon S3 bucket. The data engineer uses AWS Glue Data Catalog to access the data.

When the data engineer performs the analysis, the data engineer notices that some records are missing or out of date.

What is the likely cause of these issues?

A. AWS Glue Data Catalog is not up to date with the latest S3 partition changes.
B. Incorrect IAM roles are assigned to the AWS Glue jobs.
C. Versioning is not enabled on the S3 bucket.
D. The AWS Glue job schedules overlap with one another.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Missing or outdated records typically occur when the AWS Glue Data Catalog has not been updated to reflect the latest partitions in the Amazon S3 bucket. The catalog must be crawled or updated so queries and analyses can see the most recent data.

QUESTION 268
A retail company stores point-of-sale transaction data in an Amazon RDS for MySQL database. The company maintains historical sales analytics in Amazon Redshift. The company needs to create daily reports that combine the current day's transactions with historical sales patterns for trend analysis. The company requires a solution that provides near real-time insights while minimizing data transfer costs and maintenance overhead.

Which solution will meet these requirements?

A. Configure AWS Database Migration Service (AWS DMS) to continuously replicate data from RDS for MySQL to Amazon Redshift. Use Redshift queries to create consolidated reports.
B. Implement Amazon Redshift federated queries to directly access RDS for MySQL data and join it with existing Redshift tables in a single query.
C. Use AWS Glue to create an extract, transform, and load (ETL) pipeline that runs every hour to copy incremental data from RDS for MySQL to Amazon Redshift. Generate reports.
D. Export RDS for MySQL data to an Amazon S3 bucket on a regular schedule. Use the COPY command to load the data into Amazon Redshift staging tables. Join the data with historical data.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS DMS with ongoing replication (CDC) continuously streams new RDS for MySQL transactions into Amazon Redshift, providing near real-time freshness with minimal maintenance and transferring only changes instead of full batches. Reports can then run in Redshift joining current and historical data without hourly ETL jobs or S3 staging.

QUESTION 269
A company is creating a new data pipeline to populate a data lake. A data analyst needs to prepare and standardize the data before a data engineering team can perform advanced data transformations. The data analyst needs a solution to process the data that does not require writing new code.

Which solution will meet these requirements with the LEAST operational effort?

A. Use Python and Pandas in an AWS Glue Studio notebook. Ensure that the data engineers add additional transformations to complete the pipeline.
B. Use Amazon SageMaker Canvas and SageMaker Data Wrangler to write to a new dataset. Ensure that the data engineers add additional transformations to complete the pipeline by using AWS Glue.
C. Use AWS Glue Studio with data preparation recipe transformations. Ensure that the data engineers add additional transformations to complete the pipeline.
D. Create a document that includes the data preparation rules. Ensure that the data engineers implement the rules in AWS Glue.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue Studio lets analysts build no-code/low-code visual ETL with built-in preparation transformations (recipe-style), producing Glue jobs that engineers can extend - minimizing coding and operational overhead.

QUESTION 270
A company uses an Amazon RDS for Oracle database to store data for an ecommerce application. The company needs a solution to make the data in all tables available for future analytics. The solution must process data from a table named customer_complaints in near real time by using a REST API.

Which solution will meet these requirements MOST cost-effectively?

A. Use AWS Database Migration Service (AWS DMS) to replicate the Oracle database to an Amazon RDS for PostgreSQL database instance. Create an AWS Lambda function to process data from the PostgreSQL customer_complaints table and to call the REST API.
B. Create a task in AWS Database Migration Service (AWS DMS) to replicate the Oracle database to Amazon Redshift. Set the Oracle database as the source. Configure an AWS Lambda function to use the Amazon Redshift Data API to query the data from the customer_complaints table and to call the REST API. Unload the data from Amazon Redshift to Amazon S3.
C. Set up one AWS Database Migration Service (AWS DMS) task to stream the customer_complaints table data to Amazon Kinesis Data Streams. Set up a second AWS DMS task to send data to an Amazon S3 bucket. Configure an AWS Lambda function to process Kinesis data and to call the REST API.
D. Create a read replica of the Amazon RDS for Oracle database. Configure an AWS Lambda function to process data from the read replica customer_complaints table and to call the REST API.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS DMS can stream change data from the Oracle customer_complaints table to Kinesis Data Streams for near real-time processing by a Lambda that calls the REST API, while a separate DMS task delivers all tables to Amazon S3 for future analytics. This achieves both streaming and batch needs with minimal infrastructure and cost.

QUESTION 271
A company maintains a central Amazon Redshift data warehouse that aggregates daily transactional data from Amazon RDS for PostgreSQL and Amazon Aurora MySQL. A data engineer notices that some complex transformation queries take hours to finish. The data engineer wants to optimize query performance to reduce query execution time as much as possible.

Which solution will meet this requirement?

A. Increase the concurrency scaling quota for the Redshift cluster.
B. Export the tables to an Amazon S3 bucket. Use Amazon Athena to query the data in the bucket.
C. Use Amazon Redshift Spectrum to create external tables based on the Redshift tables.

D. Use materialized views in Amazon Redshift for frequently queried data patterns.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift materialized views precompute and store results of complex queries, allowing subsequent queries to run much faster by reading from the precomputed data. This significantly reduces execution time for frequently used transformation patterns.

QUESTION 272
A company uses an Amazon S3 bucket to integrate multiple data sources into a central data lake. The company needs to perform multiple transformations and data cleaning processes on the data to make the data accessible to business partners.

The company needs a solution that will give multiple business partners the ability to run SQL queries on the central data lake during normal business hours.

Which solution will meet these requirements MOST cost-effectively?

A. Use a provisioned Amazon EMR cluster after normal business hours to process the previous day's data, apply all necessary transformations, and load the prepared data into Amazon Redshift Serverless.
B. Use an AWS Glue Flex Job after normal business hours to process the previous day's data, apply all necessary transformations. and load the prepared data into Amazon Redshift Serverless.
C. Use an AWS Lambda function after normal business hours to process the previous day's data, apply all necessary transformations, and load the prepared data into an Amazon Redshift provisioned cluster.
D. Use an AWS Glue Flex job after normal business hours to process the previous day's data, apply all necessary transformations, and load the prepared data into an Amazon Redshift provisioned cluster.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue Flex Jobs provide discounted, off-hours batch ETL to transform S3 data, and Amazon Redshift Serverless offers on-demand SQL access during business hours without paying for idle clusters minimizing cost and operations.

QUESTION 273
A company needs to build an extract, transform, and load (ETL) pipeline that has separate stages for batch data ingestion, transformation, and storage. The pipeline must store the transformed data in an Amazon S3 bucket. Each stage must automatically retry failures. The pipeline must provide visibility into the success or failure of individual stages.

Which solution will meet these requirements with the LEAST operational overhead?

A. Chain AWS Glue jobs that perform each stage together by using job triggers. Set the MaxRetries field to 0.
B. Deploy AWS Step Functions workflows to orchestrate AWS Lambda functions that ingest data. Use AWS Glue jobs to transform the data and store the data in the S3 bucket.
C. Build an Amazon EventBridge based pipeline that invokes AWS Lambda functions to perform each stage.
D. Schedule Apache Airflow directed acyclic graphs (DAGs) on Amazon Manages Workflows tor Apache Airflow (Amazon MWAA) to orchestrate pipeline steps. Use Amazon Simple Queue Service (Amazon SQS) to ingest data. Use AWS Glue jobs to transform data and store the data in the S3 bucket.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Step Functions provides managed orchestration with per-step retries, error handling, and execution history. Orchestrating Lambda for ingestion and AWS Glue jobs for transformation to S3 delivers a serverless, low-maintenance pipeline with built-in visibility into each stage’s success or failure.

QUESTION 274
A company needs to implement a data mesh architecture in which domains for trading, risk, and compliance teams each have own their data. The teams need to share specific views with one another. The teams have over 1,000 tables across 50 databases in AWS Glue Data Catalog. All three teams use Amazon Athena to perform on-demand analysis. The teams use Amazon Redshift to generate complex reports. The compliance team must audit all data access. Access to personally identifiable information (PII) data must be restricted.

The company requires a scalable solution to meet the team requirements. The solution must provide the ability to perform analysis across team domains.

Which solution will meet these requirements?

A. Create views in Athena for on-demand analysis. Use the Athena views in Amazon Redshift to perform cross-domain analytics. Use AWS CloudTrail to audit data access. Use AWS Lake Formation to establish fine-grained access control.
B. Use AWS Glue Data Catalog views to perform analysis. Use AWS CloudTrail logs to audit data access. Use AWS Lake Formation to manage access permissions. Use security definer views to mask PII.
C. Use AWS Lake Formation to set up cross-domain access to tables. Set up fine-grained access controls.
D. Create materialized views and enable Amazon Redshift datashares for each domain. Configure cross-domain access policies.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Lake Formation provides centralized, fine-grained governance (LF-Tags, column/row-level permissions) across domains, integrates with Athena and Redshift for cross-domain analytics, and supports comprehensive auditing via Lake Formation access logs and CloudTrail meeting scalability, PII restriction, and audit requirements.

QUESTION 275
A company stores data in Amazon S3 and Amazon RDS. The company needs to apply technical controls to meet security requirements. The company must generate backups of the data and retain the backups for specified retention periods. The company must protect the backups from accidental or malicious deletion during the retention periods.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create S3 buckets with versioning and MFA Delete enabled. Set a backup retention period and backup window for each RDS DB instance. Use the automated backup feature of Amazon RDS, and take additional snapshots as required.
B. Create S3 buckets with versioning and Same-Region Replication enabled. Set a backup retention period and backup window for each RDS DB instance. Use the automated backup feature of Amazon RDS. Take additional snapshots as required.
C. Create a backup plan in AWS Backup. Specify the backup schedule, set window and lifecycle values, and select a backup vault. Assign the Amazon S3 and Amazon RDS resources to the backup plan. Create a vault lock in compliance mode.
D. Create S3 buckets with versioning and Cross-Region Replication enabled. Set a backup retention period and backup window for each RDS DB instance. Use Amazon RDS automated backups. Take additional snapshots as required.

Correct Answer: C

Section: (none)

Explanation/Reference:
Explanation:
AWS Backup centrally automates backups for S3 and RDS with scheduled plans and lifecycle policies, and Vault Lock (compliance mode) enforces WORM immutability to prevent modification or deletion during retention, meeting security and retention requirements with minimal administration.

QUESTION 276
A data engineer uploads confidential documents to an Amazon S3 bucket every day. The data engineer requires a solution to independently verify the integrity of all uploaded data to confirm that there was no corruption during the transfer process.

Which solution will meet this requirement?

A. Download a subset of the data after the data is uploaded to the S3 bucket. Manually validate the objects for integrity.
B. Change the default encryption on the S3 bucket to server-side encryption with customer-provided keys (SSE-C). Turn on S3 bucket keys to validate data integrity.
C. Calculate the SHA-256 checksum for the objects before uploading the objects. Pass the calculated value to the AWS SDK in each upload request.
D. Download the complete data after the data is uploaded to the S3 bucket. Programmatically validate the objects for integrity.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
By calculating a checksum (e.g., SHA-256) before upload and providing it in the upload request, Amazon S3 verifies the checksum on receipt and rejects the upload if the data is corrupted. This provides automatic, independent validation of data integrity without requiring full data downloads.

QUESTION 277
A company needs to optimize storage for an Amazon S3 bucket. Objects older than 1 year must be accessible within 5 hours. All versions of the objects must be retained and immutable for 7 years. All versions of the objects must use the write-once-read-many (WORM) model.

Which solution will meet these requirements?

A. Configure S3 Versioning on the bucket and use the S3 Intelligent-Tiering storage class. Configure a lifecycle policy for the bucket to transition objects that are older than 1 year to S3 Glacier Flexible Retrieval. Configure the policy to delete objects that are older than 7 years.
B. Configure S3 Object Lock on the bucket and use the S3 Intelligent-Tiering storage class. Configure a lifecycle policy for the bucket to transition objects that are older than 1 year to S3 Glacier Deep Archive. Configure the policy to delete objects that are older than 7 years.
C. Configure S3 Object Lock on the bucket and use the S3 Intelligent-Tiering storage class. Configure a lifecycle policy for the bucket to transition objects that are older than 1 year to S3 Glacier Flexible Retrieval. Configure the policy to delete objects that are older than 7 years.
D. Configure S3 Versioning on the bucket and use the S3 Intelligent-Tiering storage class. Configure a lifecycle policy for the bucket to transition objects that are older than 1 year to S3 Glacier Deep Archive. Configure the policy to deletes objects that are older than 7 years.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
S3 Object Lock enforces the write-once-read-many model and ensures that all object versions remain immutable for the required 7-year retention period. Transitioning objects older than 1 year to S3 Glacier

Flexible Retrieval satisfies the requirement that data be accessible within 5 hours, as this storage class supports retrieval within minutes to a few hours. The lifecycle policy can expire objects after 7 years, aligning with the retention requirement once the Object Lock retention period has ended.

QUESTION 278
A data engineer needs to deploy a complex pipeline. The stages of the pipeline must be able to run a script. The data engineer must use only fully managed and serverless services in the pipeline.

Which solution will meet these requirements?

A. Deploy AWS Glue jobs and workflows. Use AWS Glue to run the jobs and workflows on a schedule.
B. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to build and schedule the pipeline.
C. Deploy the script to Amazon EC2 instances. Use Amazon EventBridge to run the script on a schedule.
D. Use Aws Glue DataBrew to build the pipeline. Use Amazon EventBridge to run the pipeline on a schedule.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon Managed Workflows for Apache Airflow is a fully managed, serverless service designed for orchestrating complex pipelines. It supports defining multiple stages as tasks that can run scripts, manage dependencies, and handle scheduling, retries, and monitoring without requiring infrastructure management, fully meeting the requirement for a complex, script-based pipeline using only managed and serverless services.

QUESTION 279
A company needs to optimize storage costs for an Amazon S3 bucket. The S3 bucket receives 10 million objects every day. The objects range in size from 2 KB to 5 MB. The objects need to be immediately accessible for the first 60 days. Users access objects infrequently from 61 to 180 days. The objects must be accessible within an hour from 181 to 365 days. The company can delete the objects after 365 days.

Which solution will meet these requirements?

A. Use S3 Intelligent-Tiering to automatically transition objects. Select the Archive Access tier for Intelligent-Tiering. Configure an S3 bucket policy to expire objects that are older than 365 days.
B. Create an S3 Lifecycle policy to move objects. Configure the policy to move objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) after 60 days. Move the objects to S3 Glacier Flexible Retrieval after 180 days. Expire objects after 365 days.
C. Enable S3 Inventory. Use a daily inventory report to configure an S3 Batch Operations job that moves objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) after 60 days. Move objects to S3 Glacier Flexible Retrieval after 180 days. Expire objects after 365 days.
D. Enable S3 Inventory. Run an AWS Lambda function each day to fetch an inventory report and move objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) after 60 days. Move objects to S3 Glacier Flexible Retrieval after 180 days. Expire objects after 365 days.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
An S3 Lifecycle policy provides native, automatic tiering by object age without operational overhead, which is critical at 10 million new objects per day. Keeping objects in S3 Standard for the first 60 days satisfies immediate access. Transitioning to S3 Standard-IA from days 61–180 matches infrequent access with millisecond retrieval. Transitioning to S3 Glacier Flexible Retrieval after 180 days meets the “accessible within an hour” requirement. Expiring objects after 365 days satisfies the deletion requirement.

QUESTION 280
A company needs to use an Aws Glue PySpark job to read specific data from an Amazon DynamoDB

table. The company knows the partition key values for the required records. The existing processing logic of the AWS Glue PySpark job requires the data to be in DynamicFrame format. The company needs a solution to ensure that the job reads only the specified data.

Which solution will meet this requirement with the MINIMUM number of read capacity units (RCUs)?

A. Use the AWS Glue DynamoDB ETL connector to read the DynamoDB table. Use the filter option to read the required partition key.
B. Perform a query on the DynamoDB table in the AWS Glue job by using only the sort key in the key condition expression. Load the data into a DynamicFrame.
C. Perform a scan on the DynamoDB table in the Aws Glue job. Put the data into a DynamicFrame. Filter the DynamicFrame on the partition key.
D. Perform a query on the DynamoDB table in the AWS Glue job. Use the partition key in the key condition expression. Put the data into a DynamicFrame.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
A DynamoDB Query that uses the partition key in the key condition expression reads only the items in the specified partition(s), which minimizes consumed RCUs compared with any Scan-based approach. The job can then convert the query results into a DynamicFrame to satisfy the existing processing logic.

QUESTION 281
A company runs a multi-tenant Amazon EMR cluster on Amazon EC2 instances. Multiple teams perform interactive query analyses and data transformations on the data in the EMR cluster. The teams can access the cluster only through EMR Studio workspaces and EMR steps.

The teams need to use EMR steps to run Apache Spark jobs to fetch data from an Amazon DynamoDB table. The DynamoDB table contains confidential data that must be accessible to only one specific team. The company needs to ensure that only the appropriate team can access the confidential data in the EMR cluster.

Which solution will meet these requirements?

A. Set up runtime roles for EMR steps.
B. Set up AWS Lake Formation permissions.
C. Set up IAM roles for EMR File System (EMRFS) requests.
D. Set up a DynamoDB resource-based policy.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Runtime roles for EMR steps allow each submitted step (such as a Spark job) to assume a distinct IAM role at execution time. By granting DynamoDB read permissions only to the role used by the authorized team's steps, the confidential table data becomes accessible only to that team's Spark jobs, while other teams' steps run with roles that lack access.

QUESTION 282
A company stores Apache Parquet files in an Amazon S3 data lake. The data lake receives thousands of files from multiple sources every hour. The files range in size from 50 KB to 100 KB.

The company is evaluating the implementation of Apache Iceberg tables for the data lake. The company is using AWS Glue Data Catalog as part of the evaluation. The company needs a solution to optimize query performance in Iceberg. The solution must ensure that Iceberg table performance does not degrade when more files are added over time.

Which solution will meet these requirements?

A. Use an AWS Glue job to compact the files into a standard size of 512 MB at the end of each day. Run an AWS Glue crawler to update the Data Catalog.
B. Configure the Data Catalog to automatically compact the files every minute.
C. Configure Iceberg table properties to enable automatic compaction based on thresholds for file size and the number of files.
D. Implement a partition strategy in Amazon S3. Run an AWS Glue crawler to update the Data Catalog every 5 minutes.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Apache Iceberg is designed to manage table performance over time by controlling file layout through compaction. Enabling Iceberg's automatic compaction via table properties allows the table to rewrite many small files into fewer larger files based on configured thresholds, preventing the "small files problem" and keeping query planning and scan performance from degrading as new files continue to arrive.

QUESTION 283
A data engineer must maintain and monitor a data pipeline on AWS that processes streaming data from Internet of Things (IoT) devices. The pipeline uses Amazon Kinesis Data Streams to ingest data and Amazon Data Firehose to deliver data to an Amazon S3 bucket. The data engineer needs to monitor the health of the pipeline.

Which solution will meet these requirements with the LEAST operational effort?

A. Use Amazon CloudWatch Logs to manually review logs that are generated by Kinesis Data Streams and Firehose.
B. Configure Amazon CloudWatch alarms to monitor key metrics such as IncomingBytes, OutgoingBytes, and DeliveryToS3.Success for Kinesis Data Streams and Firehose
C. Use an AWS Lambda function to run daily checks on the status of the Kinesis Data Streams and Firehose. Configure the Lambda function to use Amazon Simple Notification Service (Amazon SNS) to send notifications.
D. Use Amazon Managed Service for Apache Flink to perform near real-time anomaly detection on the streaming data and to invoke alerts if unusual patterns are detected.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
CloudWatch metrics for Kinesis Data Streams and Firehose are available automatically and provide direct visibility into ingestion, throughput, and delivery success. Configuring CloudWatch alarms on key metrics (such as stream input/output and Firehose delivery success) enables proactive, automated health monitoring and alerting with minimal setup and no custom code or additional services to operate.

QUESTION 284
A company needs a solution to store and query product data that has variable attributes. The solution must support unpredictable and high-volume queries with single-digit millisecond latency, even during sudden traffic spikes. The solution must retrieve items by a primary identifier named Product ID. The solution must allow flexible queries by secondary attributes named Category and Brand.

Which solution will meet these requirements?

A. Use an Amazon DynamoDB table with on-demand capacity to store product data. Store products by primary key. Use global secondary indexes (GSIs) to store secondary attributes.
B. Use Amazon Aurora with a Multi-AZ deployment to store product data. Use read replicas. Create indexes for primary and secondary attributes.
C. Use an Amazon OpenSearch Serverless cluster with dynamic scaling to store product data. Index product data by primary and secondary attributes.

D. Use Amazon ElastiCache (Redis OSS) and Amazon S3 to store product data. Use Amazon Athena to run flexible secondary attribute queries.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon DynamoDB with on-demand capacity is designed to handle unpredictable, high-volume request patterns and sudden traffic spikes while maintaining single-digit millisecond latency. Using Product ID as the table's primary key enables efficient item retrieval by the primary identifier, and global secondary indexes on Category and Brand provide flexible, low-latency queries on those secondary attributes.

QUESTION 285
A company is uploading log files from on-premises servers to an Amazon S3 bucket. The company needs to validate that the logs from the on-premises server are the same as the logs that are stored in the S3 bucket.

Which solution will meet this requirement?

A. Use the AWS SDK to automatically compute CRC32 checksums during the upload. Store the checksums in S3 object metadata.
B. Create an AWS Lambda function to calculate SHA-256 checksums. Store results in a separate metadata table. Validate the logs after the upload.
C. Enable S3 Object Lock in compliance mode on the S3 bucket. Upload the objects to the bucket.
D. After uploading the objects to the S3 bucket, enable S3 Object Lock in governance mode on the S3 objects.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Computing a checksum during upload and storing it with the object provides an integrity value that can be compared later to confirm the uploaded object matches the original log file. Using an SDK-managed CRC32 checksum gives a straightforward, automated validation approach without additional processing infrastructure.

QUESTION 286
An ecommerce company uses AWS Glue ETL to process and analyze orders. The company wants to build an extract, transform, and load (ETL) pipeline that processes placed, shipped, delivered, and canceled orders differently.

The company integrates the order processing system with Amazon EventBridge. The company configures EventBridge Scheduler rules for each order status to invoke different AWS Glue workflows. When the company examines Amazon CloudWatch metrics for the workflow, the company notices that the FailedInvocations metric shows a high value for canceled orders.

The company must determine the cause of the failed invocations.

Which solution will meet this requirement?

A. Configure a dead-letter queue in EventBridge Scheduler to store failed events. Analyze the failed order events.
B. Use the archive and replay features in EventBridge Scheduler to investigate the issue.
C. Change the retry policy in EventBridge Scheduler to reduce the value for maximum retries.
D. Change the retry policy in EventBridge Scheduler to increase the value for maximum age of event.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Configuring a dead-letter queue for EventBridge Scheduler captures the events that could not be delivered to the target along with the failure context. Reviewing the failed canceled-order events in the DLQ provides the most direct way to identify why the invocations are failing (for example, payload issues, permission errors, or target configuration problems) without changing retry behavior.

QUESTION 287
A company needs to build a data pipeline to process a 1-TB file from an Amazon S3 bucket. The pipeline needs to create three DataFrames based on business logic. The pipeline must save all three DataFrames to a second S3 bucket in parallel. The company needs to set the pipeline to be the target of an Amazon EventBridge rule that matches file uploads to the source S3 bucket.

Which solution will meet these requirements with the LEAST maintenance overhead?

A. Configure an Apache Spark Streaming application on Amazon EMR to process data from the S3 source bucket in batches, create DataFrames, and save the output to the destination S3 bucket.
B. Configure three AWS Lambda functions to process the business logic and to save the DataFrames to the destination S3 bucket in parallel.
C. Configure an AWS Glue workflow to run three AWS Glue jobs in parallel to process the file.
D. Configure an AWS Step Functions state machine to initiate an AWS Glue workflow to run three AWS Glue jobs in parallel to process the file.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
An AWS Glue workflow can be triggered by an EventBridge rule when a file is uploaded to the source S3 bucket, and it can orchestrate three AWS Glue jobs to run in parallel. This provides a fully managed, serverless approach for Spark-based DataFrame processing at the 1-TB scale while keeping orchestration and operational management minimal.

QUESTION 288
A company needs to transform IoT sensor data in near real time before the company stores the data in an Amazon S3 bucket. The data is available from a data stream in Amazon Kinesis Data Streams.

The company needs to apply complex and stateful transformations to the data before the company stores the data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Schedule AWS Glue ETL jobs to process the data stream.
B. Configure an application in Amazon Managed Service for Apache Flink to process the data stream.
C. Configure an AWS Lambda function to process the data stream.
D. Schedule Apache Spark jobs on an Amazon EMR cluster to process the data stream.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon Managed Service for Apache Flink is purpose-built for near-real-time stream processing on Kinesis Data Streams and supports complex, stateful transformations with managed scaling, state management, and checkpointing, resulting in the lowest operational overhead for this use case.

QUESTION 289
A company stores time-series data that is collected from streaming services in an Amazon S3 bucket. The company must ensure that only workloads that are deployed within the company's VPC can access the data.

Which solution will meet this requirement?

A. Create an S3 bucket policy that uses a condition to allow access only to traffic that originates from the company's VPC.
B. Apply a security group to the S3 bucket that allows connections only from the company's VPC CIDR block.
C. Define an IAM policy that denies access to all users unless the request originates from within the company's VPC.
D. Use a network ACL on the VPC subnets to allow only specific resources to access the S3 bucket.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
An S3 bucket policy can restrict access to requests that come through a specific VPC endpoint by using a condition on the VPC endpoint (or VPC) context, ensuring only workloads running inside the company's VPC (using that endpoint) can access the bucket.

QUESTION 290
A company runs a data platform on AWS. The data platform uses AWS Glue to provide a data catalog and to perform processing. The company notices quality issues in the data.

The company needs to implement data quality validations. The validations must include rules for known issues. The validations must have the ability to automatically detect unexpected data quality issues.

Which solution will meet these requirements with the LEAST operation overhead?

A. Use AWS Glue jobs to implement AWS Glue Data Quality validations that include anomaly detection.
B. Use AWS Glue jobs to implement data quality rules that use open source data quality frameworks.
C. Use AWS Glue DataBrew to profile the data. Configure data quality rules based on the data quality results from the profiling.
D. Use AWS Glue jobs to implement data quality validations that use SQL statements.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue Data Quality provides native, managed data quality rules with built-in anomaly detection, allowing validation of known issues while automatically identifying unexpected data quality problems with minimal operational effort.

QUESTION 291
A company uses an Amazon S3 Standard bucket to maintain a self-managed transactional data lake that uses Apache Iceberg tables. The data lake ingests data both in real time and in batches.

Users report slow performance for real-time tables. A data engineer reviews the real-time tables and notices that the tables are made up of many small data files The data engineer must improve the performance of the real-time tables.

Which solution will meet this requirement?

A. Expire historic snapshots.
B. Archive historic snapshots.
C. Delete S3 objects that are not linked from the Iceberg table.
D. Apply compaction.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Compaction merges many small data files into fewer, larger files, which reduces file count and metadata overhead in Apache Iceberg tables, directly improving query performance for real-time workloads.

QUESTION 292
An ecommerce company collects daily customer transaction logs in CSV format and stores the logs in Amazon S3. The company uses Amazon Athena to scan a subset of attributes from the logs on the same day the company receives each log.

Query times are increasing because of increasing transaction volume. The company wants to improve query performance.

Which solution will meet these requirements with the SHORTEST query times?

A. Convert the CSV logs into multiple ORC files for better parallelism in Athena. Partition by date in Amazon S3. Use columnar pushdown filters.
B. Convert the CSV logs to JSON. Partition by date in Amazon S3. Use Athena with dynamic filtering to reduce data scans.
C. Convert the CSV logs to Avro. Partition by date in Amazon S3. Use Athena with projection-based partitioning.
D. Convert the CSV logs to a single Apache Parquet file for each day Partition the data by date in Amazon S3. Use Athena with predicate pushdown filters.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Partitioning by date limits each query to only the relevant day's data, and Parquet enables efficient columnar reads and predicate pushdown for scanning only the needed columns and row groups. Writing a single Parquet file per day minimizes file listing and small-file overhead, typically delivering the shortest Athena query times for same-day, subset-column analytics.

QUESTION 293
A company stores customer data in an Amazon S3 bucket. The company must permanently delete all customer data that is older than 7 years.

Which solution will meet this requirement?

A. Configure an S3 Lifecycle policy to permanently delete objects that are older than 7 years.
B. Use Amazon Athena to query the S3 bucket for objects that are older than 7 years. Configure Athena to delete the results.
C. Configure an S3 Lifecycle policy to move objects that are older than 7 years to S3 Glacier Deep Archive.
D. Configure an S3 Lifecycle policy to enable S3 Object Lock on all objects that are older than 7 years.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
An S3 Lifecycle policy can automatically and permanently expire (delete) objects after they reach an age of 7 years, ensuring the data is removed without ongoing manual processes.

QUESTION 294
Two data engineering teams use separate AWS accounts. Both teams request access to the same datashare in an Amazon Redshift cluster that is in a third AWS account. The datashare is named salesshare.

A data engineer must use the Amazon Redshift SQL interface to grant both data engineering teams' access to the datashare.

Which command or commands will meet this requirement?

A. GRANT USAGE ON DATASHARE salesshare TO ACCOUNTS ‘<account ID 1>’ AND ‘<account ID 2>’;
B. GRANT USAGE ON DATASHARE salesshare TO NAMESPACES ‘<account ID 1>’ AND ‘<account ID 2>’;
C. GRANT USAGE ON DATASHARE salesshare TO ACCOUNT ‘<account ID 1>’;
GRANT USAGE ON DATASHARE salesshare TO ACCOUNT ‘<account ID 2>’;
D. GRANT USAGE ON DATASHARE salesshare TO NAMESPACE ‘<account ID 1>’;
GRANT USAGE ON DATASHARE salesshare TO NAMESPACE ‘<account ID 2>’;

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
In Amazon Redshift data sharing, cross-account permissions for a datashare are granted to consumer cluster/serverless namespaces. Granting USAGE on the datashare to each consumer namespace enables both teams’ Redshift environments in their respective accounts to create and query databases from the shared data.

QUESTION 295
A company regularly loads data into an Amazon DynamoDB table. The company must retain the data in DynamoDB for 1 year. After 1 year, the company must archive the data for 5 years. After 5 years, the company must delete the data.

Which solution will meet these requirements in the MOST operationally efficient way?

A. Include a timestamp with the data that is loaded into DynamoDB. Create an AWS Lambda function to read the timestamps and move items that are 1 year old to Amazon S3 Glacier Flexible Retrieval. Create an S3 Lifecycle policy to delete the items after 5 years.
B. Enable Amazon DynamoDB Streams. Configure a TTL rule to delete items from the DynamoDB table after 1 year. Use an AWS Lambda function to consume the stream and send items to Amazon S3 Glacier Flexible Retrieval. Delete the items from S3 Glacier Flexible Retrieval after 5 years.
C. Include a timestamp with the data that is loaded into DynamoDB. Use an AWS Lambda function to read timestamps and move items that are 1 year old directly to Amazon S3 Glacier Deep Archive. Configure the function to delete items after 5 years.
D. Enable Amazon DynamoDB Streams. Configure a TTL rule to delete items from the DynamoDB table after 1 year. Use an AWS Lambda function to consume the stream and send items directly to Amazon S3 Glacier Deep Archive. Create an S3 Lifecycle policy to delete the items after 5 years.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
DynamoDB TTL provides a fully managed way to remove items after 1 year with minimal operational work. DynamoDB Streams can capture the item changes so a Lambda function can archive the expiring data to Amazon S3 before it is removed from the table. Using S3 Glacier Deep Archive minimizes archive storage cost for long retention, and an S3 Lifecycle policy can automatically delete the archived objects after 5 years, avoiding custom deletion logic.

QUESTION 296
A company must retain specific data for 1 year. A data engineer observes that one of the company’s Amazon S3 buckets contains millions of objects that are older than 3 years. Versioning is enabled on the bucket.

To reduce costs, the data engineer implements an S3 Lifecycle rule to expire objects after 365 days. The new S3 Lifecycle rule causes the object count to double instead of decrease.

Which additional step must the data engineer take to permanently delete the old objects?

A. Disable versioning on the S3 bucket.
B. Use an AWS Lambda function to run a Python job to identify and delete objects that are older than 365 days.
C. Suspend versioning on the S3 bucket.
D. Add an additional S3 Lifecycle rule to delete the current and expired versions of objects that are older than 365 days.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
With S3 versioning enabled, lifecycle expiration of an object typically adds a delete marker instead of removing prior versions, which can increase the total number of object versions. To permanently remove the older data, the lifecycle configuration must also delete noncurrent (previous) versions and expired delete markers based on age so that both current and prior versions are actually removed.

QUESTION 297
A manufacturing company uses AWS Glue jobs to process IoT sensor data to generate predictive maintenance models. A data engineer needs to implement automated data quality checks to identify temperature readings that are outside the expected range of -50°C to 150°C. The data quality checks must also identify records that are missing timestamp values.

The data engineer needs a solution that requires minimal coding and can automatically flag the specified issues.

Which solution will meet these requirements?

A. Create an AWS Glue DataBrew project to profile the sensor data Define completeness rules for timestamps. Set up numeric range validation for temperature values.
B. Use AWS Glue’s Data Quality rules and machine learning (ML)-based anomaly detection to identify missing timestamps and to detect temperature anomalies.
C. Create an AWS Lambda function to scan the sensor data files to validate temperature ranges. Use AWS Glue Data Catalog tables to check timestamp completeness.
D. Create an AWS Glue DynamicFrame that uses a custom data quality operator to profile the sensor data. Use Amazon SageMaker Data Wrangler transforms to validate timestamps and temperature ranges.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue DataBrew provides a low-code way to implement automated data quality checks, including completeness rules to flag missing timestamp values and numeric range validation to flag temperature readings outside -50°C to 150°C.

QUESTION 298
A media company wants to improve a system that recommends media content to customers based on user behavior and preferences. To improve the recommendation system, the company needs to incorporate insights from third-party datasets into the company’s existing analytics platform.

The company wants to minimize the effort and time required to incorporate third-party datasets.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use API calls to access and integrate third-party datasets from AWS Data Exchange.

B. Use API calls to access and integrate third-party datasets from AWS DataSync.

C. Use Amazon Kinesis Data Streams to access and integrate third-party datasets from a Git repository.

D. Use Amazon Kinesis Data Streams to access and integrate third-party datasets from Amazon Elastic Container Registry (Amazon ECR).

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Data Exchange is designed to quickly subscribe to and consume third-party datasets through a managed service interface and APIs, minimizing the integration effort and operational overhead compared to building custom ingestion from other sources.

QUESTION 299
A company generates yearly financial statements for customers and stores the statements in an Amazon S3 bucket. Customers rarely access the documents after 1 week. The company must retain the statements for 7 years. The statements must remain readily accessible for customers.

Which solution will meet these requirements in the MOST cost-effective way?

A. Create an S3 Lifecycle rule to transition objects to S3 Glacier Deep Archive after 7 days. Expire the objects after 7 years.
B. Set the S3 bucket to use S3 Intelligent-Tiering when new objects are uploaded. Set objects to expire after 7 years.
C. Create an S3 Lifecycle rule to transition objects to S3 Glacier Instant Retrieval after 7 days. Expire the objects after 7 years.
D. Set the S3 bucket to use S3 Glacier Instant Retrieval when new objects are uploaded. Create an AWS Lambda function that runs daily to delete any objects that are older than 7 years.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
S3 Glacier Instant Retrieval is the most cost-effective option that still keeps the statements readily accessible because it is designed for archival data that is rarely accessed but must be retrieved immediately, with millisecond access. A lifecycle rule can keep the files in a standard S3 class for the first 7 days, then transition them to Glacier Instant Retrieval for the remainder of the 7-year retention period, and expire them automatically at the end. S3 Glacier Deep Archive is cheaper but does not provide immediate access, and S3 Intelligent-Tiering is less cost-effective here because the access pattern is already known.

Reference: https://aws.amazon.com/ru/s3/storage-classes/glacier/instant-retrieval/
https://aws.amazon.com/ru/s3/storage-classes/glacier/instant-retrieval/

QUESTION 300
A company runs an extract, transform, and load (ETL) job in AWS Glue. The job processes personally identifiable information (PII) data and writes logs to an Amazon CloudWatch Logs log group. A data engineer needs to mask PII data in the CloudWatch logs group.

Which solution will meet these requirements?

A. Attach an AWS Glue security configuration to the ETL job.
B. Configure a data protection policy. Attach the policy to the CloudWatch log group.
C. Run an Amazon Macie sensitive data discovery job.
D. Call AWS Glue sensitive data detection APIs in the ETL job.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
A CloudWatch Logs data protection policy is specifically designed to detect, audit, and mask sensitive data such as PII in log events that are ingested into a log group. Attaching the policy directly to the CloudWatch log group meets the requirement to mask PII in the logs without changing the ETL logic itself.

Reference: https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/mask-sensitive-log-data.html
https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/cloudwatch-logs-data-protection-policies.html

QUESTION 301
A company creates a new non-production application that runs on an Amazon EC2 instance. The application needs to communicate with an Amazon RDS database instance using Java Database Connectivity (JDBC). The EC2 instances and the RDS database instance are in the same subnet.

Which solution will meet this requirement?

A. Modify the IAM role that is assigned to the database instance to allow connections from the EC2 instances.
B. Modify the ec2_authorized_hosts parameter in the RDS parameter group to include the EC2 instances. Restart the database instance.
C. Update the database security group to allow connections from the EC2 instances.
D. Enable the Amazon RDS Data API and specify the Amazon Resource Name (ARN) of the database instance in the JDBC connection string.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Connectivity from Amazon EC2 to an Amazon RDS DB instance in the same subnet or VPC is controlled by VPC security groups. Updating the database security group to allow inbound connections from the EC2 instances enables the application to connect to the database by using JDBC. IAM roles do not control direct JDBC network access, and the RDS Data API is not used through a JDBC connection string.

Reference: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Overview.RDSSecurityGroups.html
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_VPC.Scenarios.html

QUESTION 302
A retail company wants to implement real-time analytics for an ecommerce platform. The company needs to collect clickstream data from the company's website and mobile apps. The company needs to store the data for analytics.

Which solution will meet these requirements with the LEAST ongoing maintenance?

A. Use Amazon Data Firehose to ingest the streaming data. Deliver the processed data directly to a provisioned Amazon Redshift cluster.
B. Deploy agents on Amazon EC2 instances to collect the streaming data. Use the AWS CLI to periodically batch upload the data to an Amazon S3 bucket.
C. Use Amazon Kinesis Data Streams to collect the streaming data. Use Amazon Data Firehose to deliver the data to an Amazon S3 bucket.
D. Use an Amazon Managed Streaming for Apache Kafka (Amazon MSK) broker to collect the data. Store the data in an Amazon RDS DB instance.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon Kinesis Data Streams is well suited to collect high-volume real-time clickstream data from

websites and mobile apps, and Amazon Data Firehose can then deliver that streaming data to Amazon S3 with very little operational effort. This approach is highly managed and avoids the ongoing administration required for EC2-based collectors, self-managed Kafka patterns, or maintaining a provisioned Redshift cluster for raw landing storage.

Reference: https://docs.aws.amazon.com/firehose/latest/dev/what-is-this-service.html
https://docs.aws.amazon.com/firehose/latest/dev/basic-deliver.html

QUESTION 303
A company uses AWS Glue ETL pipelines to process data. The company uses Amazon Athena to analyze data in an Amazon S3 bucket.

To better understand shipping timelines, the company decides to collect and store shipping and delivery dates in addition to order data. The company adds a data quality check to ensure that shipping date is greater than order date and that delivery date is greater than shipping date. Orders that fail the quality check must be stored in a second S3 bucket.

Which solution will meet these requirements MOST cost-effectively?

A. Use the AWS Glue DataBrew DATEDIFF function to create two additional columns. Check the new columns.
B. Use Athena to query all three date columns, and compare the columns.
C. Use AWS Glue Data Quality to create a custom rule that uses the three date columns.
D. Use an AWS Glue crawler to populate an AWS Glue Data Catalog. Use the three date columns to create a filter.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue Data Quality is the most cost-effective choice because it is built for ETL data validation and supports custom rules that compare multiple columns, such as ensuring shipping date is after order date and delivery date is after shipping date. It can also identify the specific records that fail validation so those failed orders can be written to a separate Amazon S3 bucket as part of the ETL flow.

Reference: https://docs.aws.amazon.com/glue/latest/dg/glue-data-quality.html
https://docs.aws.amazon.com/glue/latest/dg/tutorial-data-quality.html
https://docs.aws.amazon.com/glue/latest/dg/dqdl-rule-types-CustomSql.html

QUESTION 304
A data engineer is designing a log table for an application that requires continuous ingestion. The application must provide dependable API-based access to specific records from other applications. The application must handle more than 4,000 concurrent write operations and 6,500 read operations every second.

Which solution will meet these requirements?

A. Create an Amazon Redshift table with the KEY distribution style. Use the Amazon Redshift Data API to perform all read and write operations.
B. Store the log files in an Amazon S3 Standard bucket. Register the schema in AWS Glue Data Catalog. Create an external Redshift table that points to the AWS Glue schema. Use the table to perform Amazon Redshift Spectrum read operations.
C. Create an Amazon Redshift table with the EVEN distribution style. Use the Amazon Redshift Java Database Connectivity (JDBC) connector to establish a database connection. Use the database connection to perform all read and write operations.
D. Create an Amazon DynamoDB table that has provisioned capacity to meet the application's capacity needs. Use the DynamoDB table to perform all read and write operations by using DynamoDB APIs.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Amazon DynamoDB is the best fit because it is designed for dependable API-based access with very high request rates and can be provisioned to support the required thousands of concurrent write and read operations per second. It also supports direct item retrieval and query operations through DynamoDB APIs, which matches the requirement for other applications to access specific records reliably. Amazon Redshift is designed for analytics workloads rather than high-throughput transactional API reads and writes.

Reference: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/read-write-operations.html
https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Query.html

QUESTION 305
A company needs to store and analyze a large amount of IoT sensor data. The company needs to retain the data indefinitely. The company analyzes the data in an Amazon Redshift cluster.

Which solution will meet these requirements MOST cost-effectively?

A. Store the data in an Amazon S3 bucket in JSON format. Configure auto-copy data ingestion from the S3 bucket to the Redshift cluster.
B. Store the data in an Amazon S3 bucket in Apache Parquet format. Configure query access through Amazon Redshift Spectrum.
C. Store the data in an Amazon S3 bucket in JSON format. Configure query access through Amazon Redshift Spectrum.
D. Store the data in an Amazon S3 bucket in Apache Parquet format. Configure auto-copy data ingestion from the S3 bucket to the Redshift cluster.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Storing the IoT data in Amazon S3 in Apache Parquet format and querying it through Amazon Redshift Spectrum is the most cost-effective approach because the data can be retained indefinitely in low-cost S3 storage while still being analyzed from Redshift without loading all of it into the cluster. Parquet further reduces cost because it is a columnar format that lets Redshift Spectrum scan only the needed columns, which lowers data scanned and improves query efficiency.

Reference: https://docs.aws.amazon.com/redshift/latest/dg/c-spectrum-external-tables.html
https://docs.aws.amazon.com/redshift/latest/dg/c-spectrum-external-performance.html

QUESTION 306
A company has an Amazon S3 based data lake. The data lake contains datasets that belong to multiple departments. The data lake ingests millions of customer records each day.

A data engineer needs to design an access and storage solution that allows departments to access only the subset of the company's dataset that each department requires. The solution must follow the principle of least privilege.

Which solution will meet these requirements with the LEAST operational effort?

A. Define IAM policies and IAM roles for each department. Specify the S3 access paths from the data lake that each team can access.
B. Set up Amazon Redshift and Amazon Redshift Spectrum as the primary entry points for the data lake. Define an IAM role that Amazon Redshift can assume. Configure the IAM role to grant access to the data that is in Amazon S3.
C. Set up AWS Lake Formation. Assign LF-Tags to AWS Glue Data Catalog resources. Enable Lake Formation tag-based access control (LF-TBAC).
D. Deploy an Amazon RDS for PostgreSQL database that has the aws_s3 extension installed. Configure

AWS Step Functions events to invoke an AWS Lambda function to sync the data lake with the database.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Lake Formation with LF-Tags and LF-TBAC is designed for fine-grained, scalable access control across shared data lake resources. It lets the company grant each department access only to the datasets it needs while following least privilege, and it reduces ongoing administrative work compared with managing separate IAM policies and S3 paths for every team.

QUESTION 307
A healthcare company stores patient records in an on-premises MySQL database. The company creates an application to access the MySQL database. The company must enforce security protocols to protect the patient records. The company currently rotates database credentials every 30 days to minimize the risk of unauthorized access.

The company wants a solution that does require the company to modify the application code for each credential rotation.

Which solution will meet this requirement with the LEAST operational overhead?

A. Assign an IAM role access permissions to the database. Configure the application to obtain temporary credentials through the IAM role.
B. Use AWS Key Management Service (AWS KMS) to generate encryption keys. Configure automatic key rotation. Store the encrypted credentials in an Amazon DynamoDB table.
C. Use AWS Secrets Manager to automatically rotate credentials. Allow the application to retrieve the credentials by using API calls.
D. Store credentials in an encrypted Amazon S3 bucket. Rotate the credentials every month by using an S3 Lifecycle policy. Use bucket policies to control access.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Secrets Manager is purpose-built to store sensitive credentials and rotate them automatically without requiring the application code to be updated for each password change. The application retrieves the current secret dynamically through API calls, so credential rotation happens transparently while minimizing operational overhead.

QUESTION 308
A retail company needs to implement a solution to capture data updates from multiple Amazon Aurora MySQL databases. The company needs to make the updates available for analytics in near real time. The solution must be serverless and require minimal maintenance.

Which solution will meet these requirements with the LEAST operational overhead?

A. Set up AWS Database Migration Service (AWS DMS) tasks that perform schema conversions for each database. Load the changes into Amazon Redshift Serverless.
B. Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) Connect with Debezium connectors to load data into Amazon Redshift Serverless.
C. Use AWS Database Migration Service (AWS DMS) to set up binary log replication to Amazon Kinesis Data Streams. Load the data into Amazon Redshift Serverless after schema conversion.
D. Use Aurora zero-ETL integrations with Amazon Redshift Serverless for each database to load Aurora MySQL changes in Amazon Redshift Serverless.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Aurora zero-ETL integration with Amazon Redshift Serverless is the most direct fit because it delivers near real-time data availability for analytics with a fully managed approach and minimal maintenance. It avoids building and operating custom CDC pipelines, streaming infrastructure, or connector frameworks, which gives it the least operational overhead for multiple Aurora MySQL databases.

QUESTION 309
A data engineer is writing a query to join two tables in Amazon Athena. The data engineer needs to choose the correct join order for the tables to optimize query performance.

Which solution will meet these requirements?

A. Specify the smaller table on the left side of the join and the larger table on the right side of the join.
B. Specify the larger table on the left side of the join and the smaller table on the right side of the join.
C. Use AWS Glue to pre-process the tables before performing the join.
D. Use table statistics to automatically determine the join order.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Athena (based on Presto/Trino) performs joins more efficiently when the smaller table is placed on the left side because it is used as the build side of the join and can be loaded into memory. This minimizes data shuffling and improves performance when joining with a larger table on the right side.

QUESTION 310
A company generates reports from 30 tables in an Amazon Redshift data warehouse. The data source is an operational Amazon Aurora MySQL database that contains 100 tables. Currently, the company refreshes all data from Aurora to Amazon Redshift every hour, which causes delays in report generation.

Which combination of steps will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Use AWS Database Migration Service (AWS DMS) to create a replication task. Select only the required tables.
B. Create a database in Amazon Redshift that uses the integration.
C. Create a zero-ETL integration in Amazon Aurora. Select only the required tables.
D. Use query editor v2 in Amazon Redshift to access the data in Aurora.
E. Create an AWS Glue job to transfer each required table. Run an AWS Glue workflow to initiate the jobs every 5 minutes.

Correct Answer: BC
Section: (none)

Explanation/Reference:
Explanation:
Aurora zero-ETL integration replicates operational data from Aurora into Amazon Redshift with low latency and minimal management overhead, and AWS supports filtering so only the required tables are replicated instead of all 100 tables. After the integration is created, Amazon Redshift requires a destination database to be created from that integration so the replicated data can be queried for reporting.

Reference: https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/zero-etl.filtering.html
https://docs.aws.amazon.com/redshift/latest/mgmt/zero-etl-using.creating-db.html
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/zero-etl.html

QUESTION 311
A data engineer needs a fully automated solution to check for new data in multiple databases and process data that the solution finds. The solution must run every hour. The solution must be compatible with

Amazon RDS, Amazon DynamoDB, and Amazon OpenSearch Service. The solution must be able to process up to 10 MB of data at one time. The solution must be optimized for costs and operational overhead. The solution must have robust error handling capabilities.

Which solution will meet these requirements?

A. Use Amazon EventBridge to invoke AWS Step Functions every hour to deploy an AWS Lambda function to check for data. Configure Step Functions steps to process data that the Lambda function finds. Implement error handling in each state.
B. Use Amazon EventBridge to invoke an AWS Lambda function every hour to check for data. Configure the function to send a message to an Amazon Simple Queue Service (Amazon SQS) queue when the function finds new data. Use a second Lambda function to read the queue and perform the processing.
C. Configure an Apache Spark application to run on Amazon EMR to check for data. Implement error handling in the application. Use Amazon EventBridge to invoke the application every hour.
D. Use Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to create a workflow that runs a directed acyclic graph (DAG) every hour to check for data. Configure the DAG to process identified data. Implement error handling in a Python operator.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon EventBridge can invoke AWS Step Functions on an hourly schedule, and Step Functions is a serverless orchestration service that is well suited for coordinating checks across multiple data sources while providing built-in retry and catch mechanisms for robust error handling. AWS Lambda can connect to Amazon RDS, Amazon DynamoDB, and Amazon OpenSearch Service through AWS SDKs, and Lambda can handle workloads well beyond 10 MB in memory, so this design meets the processing-size requirement with low cost and minimal operational overhead compared with EMR or MWAA.

Reference: https://docs.aws.amazon.com/step-functions/latest/dg/tutorial-handling-error-conditions.html
https://docs.aws.amazon.com/opensearch-service/latest/developerguide/configuration-samples.html
https://docs.aws.amazon.com/lambda/latest/dg/configuration-memory.html
https://docs.aws.amazon.com/step-functions/latest/dg/workflow-studio-process-error.html

QUESTION 312
A company uses an Amazon Redshift cluster to manage data, including vendor sales data. The company wants to store a copy of the vendor data in an Amazon S3 bucket.

A data engineer sets up an AWS Glue job to upload the data to the S3 bucket data on a schedule. The data engineer set up a network connection to allow private traffic between Amazon Redshift and Amazon S3.

What is the next step required to meet this requirement?

A. Create an IAM role that has permission to write to the S3 bucket. Associate the IAM role with the Amazon Redshift cluster.
B. Add the S3 bucket to an AWS Glue Data Catalog. Configure Amazon Redshift Spectrum to access the Data Catalog.
C. Enable the Amazon Redshift data sharing feature. Set the S3 bucket as a target bucket for data sharing.
D. Store login credentials for Amazon Redshift in AWS Secrets Manager. Add a reference to the secret to the Glue job configuration.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift needs an IAM role that grants permission to write to the target Amazon S3 bucket. After the private network path is in place, associating that role with the Redshift cluster enables the export

operation to store the vendor data in S3.

QUESTION 313
A company stores a 100 MB dataset in an Amazon S3 bucket as an Apache Parquet file. A data engineer needs to profile the data before performing data preparation steps on the data.

Which solution will meet this requirement in the MOST operationally efficient way?

A. Create a profile job on the dataset in AWS Glue DataBrew. Review the profile job results.
B. Stream the data into Amazon Managed Service for Apache Flink for SQL queries. Use the Apache Flink dashboard to profile the data.
C. Ingest the data into Amazon Redshift Spectrum. Use SQL queries to profile the data.
D. Load the data into an Amazon QuickSight dataset. Build a topic to profile the data with questions.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue DataBrew provides built-in data profiling specifically for exploring dataset structure, data types, distributions, missing values, and anomalies before preparation. For a 100 MB Parquet file in Amazon S3, creating a profile job is the most operationally efficient approach because it requires minimal setup and no custom infrastructure or query framework.

QUESTION 314
A company needs to implement real-time analytics for a retail shopping platform. The company wants to capture clickstream data, process the data, and load the data into Amazon Redshift for analysis. The solution must handle hundreds of megabytes of data every second.

Which solution will meet these requirements with the LEAST query latency for analytics?

A. Use Amazon Data Firehose to capture the data. Store the data in an Amazon S3 bucket. Use the COPY command to load data into Amazon Redshift.
B. Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) to capture the data. Use Amazon EMR to process the data. Use federated queries to access data in Amazon Redshift.
C. Use Amazon Kinesis Data Streams to capture the data. Use Amazon Redshift streaming ingestion to load data directly into materialized views.
D. Use Amazon DynamoDB Streams to capture the data. Use AWS Glue to process the data. Use a zero-ETL integration to load the data into Amazon Redshift.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon Kinesis Data Streams with Amazon Redshift streaming ingestion provides the lowest analytics latency because Amazon Redshift can ingest streaming data directly into materialized views without first staging the data in Amazon S3. AWS documentation describes Redshift streaming ingestion from Kinesis as low-latency and high-speed, which makes it the best fit for near real-time clickstream analytics at high throughput.

Reference: https://docs.aws.amazon.com/redshift/latest/dg/materialized-view-streaming-ingestion.html
https://docs.aws.amazon.com/streams/latest/dev/using-other-services-redshift.html

QUESTION 315
A company has a data pipeline that processes transaction data in real time. The company needs a notification system that alerts different teams based on the type of processing error without any delay. For security-related errors, the system must immediately notify the security team. For data validation errors, the system must notify the data quality team. For system errors, the system must notify the operations team.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an Amazon Simple Notification Service (Amazon SNS) topic with an AWS Lambda function subscriber that evaluates the error type and forwards the error to the appropriate email addresses.
B. Configure Amazon EventBridge rules with distinct event patterns for each error type. Route each error type to a dedicated Amazon Simple Notification Service (Amazon SNS) topic for team-specific alerts.
C. Use Amazon Simple Queue Service (Amazon SQS) with message attributes to categorize errors. Allow each team to poll their respective SQS queue for relevant errors.
D. Set up Amazon CloudWatch alarms with different metrics for each error type. Invoke a different Amazon Simple Notification Service (Amazon SNS) notification each time a metrics threshold is crossed.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon EventBridge is designed to route events based on matching rules and event patterns, so the company can classify each processing error type and immediately send it to the correct destination without custom routing code. By using separate EventBridge rules for security, data validation, and system errors and sending each to its own Amazon SNS topic, the solution stays fully managed, low latency, and operationally simpler than introducing Lambda logic, queue polling, or metric-based alarms.

Reference: https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-event-patterns.html
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-what-is.html
https://docs.aws.amazon.com/sns/latest/dg/welcome.html

QUESTION 316
A global ecommerce company processes customer transactions, inventory updates, and user activity logs across multiple AWS services. The company needs a scalable, fully managed, and event-driven orchestration solution to coordinate complex extract, transform, and load (ETL) workflows. The solution must use AWS Glue and Amazon EMR to process data. The data will be stored in Amazon Redshift and Amazon S3. The solution must support dependency management, automated retries, and data pipeline monitoring.

Which solution will meet these requirements?

A. Use AWS Step Functions to define an express workflow that invokes the data transformation and loading tasks across Amazon EMR and AWS Glue.
B. Create AWS Lambda functions for each step of the workflow Configure Amazon EventBridge to invoke AWS Glue jobs. Configure the Lambda functions to process and move data through the pipeline.
C. Use Apache Airflow on Amazon Managed Workflows for Apache Airflow (Amazon MWAA) to create Directed Acyclic Graphs (DAGs) to manage ETL workflows.
D. Create an AWS Lambda function that runs each step of the workflow. Create an Amazon EventBridge scheduled rule to invoke the function every day.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon Managed Workflows for Apache Airflow is a fully managed orchestration service built specifically for coordinating complex ETL workflows with dependencies, retries, scheduling, and monitoring. Airflow DAGs can natively orchestrate AWS Glue and Amazon EMR jobs and manage data movement into Amazon Redshift and Amazon S3, making it the best fit for a scalable, event-driven ETL orchestration requirement.

QUESTION 317
A global company currently uses Amazon Redshift to store data and Amazon Quick Suite (previously known as Amazon QuickSight) to generate reports.

A team of business analysts have varying levels of technical expertise. Some analysts lack SQL knowledge. All the analysts need to create new reports frequently. The company wants to use natural program language queries to create dashboards and reports more efficiently.

Which solution will meet these requirements with the LEAST operational effort?

A. Use Quick Suite dashboards that have zero-ETL access to Amazon Redshift.
B. Enable Amazon Q in Quick Suite. Generate Quick Suite dashboards and reports.
C. Integrate Tableau with Amazon Redshift to give Tableau direct access to the data.
D. Use Quick Suite dashboards that have federated query access to Amazon Redshift.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon Q in QuickSight enables users to create dashboards and reports using natural language queries, removing the need for SQL knowledge. It is fully integrated and managed within QuickSight, providing the most operationally efficient way for analysts of varying skill levels to generate insights quickly without additional tooling or infrastructure.

QUESTION 318
A company is setting up a new Amazon SageMaker Unified Studio domain. Each of the company's business units needs isolated control over its own assets, projects, and metadata. Specific datasets must be shareable with other business units upon approval. The company also requires centralized user authentication and identity mapping.

Which solution will meet these requirements?

A. Configure each business unit as a domain unit with delegated ownership and fine-grained permissions policies. Give users the ability to share assets across domain units with explicit access control. Assign API keys to users for authentication to access the domain portal.
B. Configure business units as separate domain units with owner permissions. Restrict projects exclusively to owners to prevent data sharing between domains. Configure AWS IAM Identity Center for centralized authentication. Map user profiles to their respective domain units.
C. Configure business units to be represented as separate domains. Establish isolated environments with no shared administrative policies. Configure AWS IAM Identity Center for centralized authentication. Delegate administration at the domain level.
D. Configure each business unit as a separate domain unit to manage permissions on assets, projects, and metadata. Configure AWS IAM Identity Center for centralized authentication. Map user profiles to their respective domain units. Enable cross-business unit sharing through access requests. Instruct domain unit owners to approve or deny the requests.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
SageMaker Unified Studio supports domain units within a single domain to provide isolation of assets, projects, and metadata while still allowing controlled sharing between units. Using AWS IAM Identity Center provides centralized authentication and identity mapping. Cross-unit sharing with approval workflows ensures datasets can be shared securely while maintaining isolation and least privilege.

QUESTION 319
A data engineer at a large company needs to create centralized datasets that are optimized for Amazon Redshift performance. The company has multiple downstream teams that use their own AWS accounts and dedicated Amazon Redshift clusters with RA3 nodes. All downstream teams need access to the centralized datasets.

Which solution will provide immediate access to the datasets and maintain the current Amazon Redshift performance?

A. Copy the datasets to an Amazon S3 bucket by using the UNLOAD command. Register the table definitions in a dedicated AWS Glue Data Catalog schema. Share the schema with the other AWS accounts by using AWS Lake Formation. Use Amazon Redshift Spectrum to access the data.
B. Create a daily extract, transform, and load (ETL) job to unload the data to an Amazon S3 staging area. Instruct the teams to copy the data into their Amazon Redshift clusters.
C. Set up Amazon Redshift data sharing between the Amazon Redshift producer clusters and the consumer clusters to provide access to the centralized datasets.
D. Set up an AWS DataSync job that automatically syncs the data between the Amazon Redshift producer clusters and the consumer clusters.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift data sharing allows multiple Redshift clusters (including across AWS accounts) to access centralized datasets in real time without copying or moving data. This provides immediate access while maintaining performance because queries run directly on the producer cluster's managed storage, eliminating duplication and ETL overhead.

QUESTION 320
A global finance company needs to implement near real-time cross-Region synchronization of trading data between trading centers in the us-east-1 Region, the eu-west-2 Region, and the ap-northeast-1 Region. The company must ensure that data is encrypted in transit. The solution must ensure data ordering and consistency and must support cross-Region disaster recovery. The solution must provide data latency of less than 500 milliseconds.

Which solution will meet these requirements with the LEAST operational effort?

A. Deploy Apache Kafka Connect in each AWS Region. Use custom-developed connectors to set up cross-Region data replication. Configure the SSL security protocol.
B. Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) Replicator to establish fully interconnected replication relationships between MSK clusters in the three AWS Regions. Enable TLS encryption and IAM authentication. Set up cross-Region backup configurations.
C. Deploy Apache Kafka Mirror Maker 2.0 in each AWS Region. Set up custom replication policies to handle cross-Region data synchronization. Configure the SSL security protocol.
D. Use Amazon Kinesis Data Streams to receive trading data from each AWS Region. Use Amazon Data Firehose to replicate data between Amazon Managed Streaming for Apache Kafka (Amazon MSK) clusters in each Region. Configure AWS Key Management Service (AWS KMS) encryption and IAM roles to manage access.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon MSK Replicator is a fully managed service for replicating data across MSK clusters in different AWS Regions, which makes it the lowest-effort option for near real-time cross-Region synchronization and disaster recovery. It supports secure encrypted connections, preserves Kafka replication behavior needed for ordered streaming data, and is designed for multi-Region resilience without requiring the company to operate custom replication infrastructure.

Reference: https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator.html
https://docs.aws.amazon.com/msk/latest/developerguide/msk-replicator-how-it-works.html

QUESTION 321
A company's application needs to search and analyze data in near real time. The application must handle up to 1,000 requests each second with low query latency. The company wants a solution that individual data teams can own and configure to meet each team's cost and performance optimization requirements.

Which solution will meet these requirements?

A. Use Amazon S3 buckets to store the data. Use Amazon Athena to query and analyze the data. Assign each data team a separate S3 bucket prefix to optimize queries.
B. Use streams in Amazon Kinesis Data Streams and Amazon Managed Service for Apache Flink to query and analyze the data. Assign each data team a separate stream to manage and consume.
C. Use Amazon OpenSearch Service clusters with indexing to query the data. Assign each data team a separate cluster to configure for storage and queries.
D. Use Amazon Aurora clusters that run on Aurora I/O-Optimized instances. Assign each data team a separate Aurora cluster to configure for storage and queries.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Amazon OpenSearch Service is built for near real-time search and analytics with low query latency, making it a strong fit for applications that must handle about 1,000 requests per second. Giving each data team its own OpenSearch cluster lets each team independently tune storage, indexing, scaling, and cost/performance settings to match its own workload requirements.

Reference: https://docs.aws.amazon.com/opensearch-service/latest/developerguide/bp.html
https://docs.aws.amazon.com/opensearch-service/latest/developerguide/serverless-overview.html

QUESTION 322
A company uses Amazon Redshift for its data warehouse. A data engineer must query a table named orders.complete_orders_history, which contains 100 columns. The query must return all columns except columns named companyId and unique_system_id.

Which Amazon Redshift SQL statement will meet this requirement?

A.
```
SELECT * EXCLUDE company_id, unique_system_id
FROM
orders.complete_orders_history;
```
B.
```
SELECT * NOT IN company_id, unique_system_id
FROM
orders.complete_orders_history;
```
C.
```
SELECT * EXCEPT company_id, unique_system_id
FROM
orders.complete_orders_history;
```
D.
```
SELECT * TRUNCATE company_id, unique_system_id
FROM
orders.complete_orders_history;
```

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Redshift supports the SELECT * EXCLUDE syntax, which allows returning all columns from a table except the specified ones. This is the most efficient and direct way to exclude specific columns like company_id and unique_system_id without listing all remaining columns.

QUESTION 323
A company stores sensitive transaction data in an Amazon S3 bucket. A data engineer must implement controls to prevent accidental deletions.

Which solution will meet this requirement?

A. Enable versioning on the S3 bucket and configure MFA delete.
B. Configure an S3 bucket policy rule that denies the creation of S3 delete markers.
C. Create an S3 Lifecycle rule that moves deleted files to S3 Glacier Deep Archive.
D. Set up AWS Config remediation actions to prevent users from deleting S3 objects.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Enabling S3 versioning preserves all object versions so data can be recovered if deletion occurs, and MFA delete adds an additional layer of protection by requiring multi-factor authentication to permanently delete objects or change versioning settings, effectively preventing accidental deletions.

QUESTION 324
A company stores historical customer data in an Amazon Redshift table. A column named Email contains null entries and values that are not email addresses. The quality of the Email column is critical for multiple downstream processes. A data engineer must create an AWS Glue Data Quality rule that fails when the percentage of valid email addresses in the Email column is less than 90%.

Which component of an AWS Glue Data Quality rule will meet these requirements?

A. Uniqueness "Email" matches "[%@%.%]" with a threshold set to > 0.9
B. ColumnValues "Email” matches "[%@%.%]" with a threshold set to > 0.1
C. ColumnValues "Email" matches "[%@%.%]" with a threshold set to > 0.9
D. UniqueValueRatio "Email" matches "[%@%.%]" with a threshold set to > 0.1

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Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue Data Quality uses the ColumnValues rule with pattern matching to validate that values conform to a required format, such as an email pattern. Setting the threshold to greater than 0.9 ensures that at least 90% of the values in the Email column match the expected email format, causing the rule to fail if data quality drops below this level.

QUESTION 325
A company is building data processing pipelines by using AWS Glue. The pipelines access data stored in Amazon S3. The company has organized the data into folders with prefixes that represent different classification levels. The company needs to restrict AWS Glue jobs to access only specific prefixes based on the data classification. The company must also restrict access to business hours (9 AM to 5 PM).

Which elements must the company include in a custom IAM policy to meet these requirements?

A. A Resource element with S3 object Amazon Resource Name (ARN) patterns that use wildcards for each prefix and a Condition element that uses the $util.time variable with TimeGreaterThan and TimeLessThan operators
B. A Resource element with S3 object Amazon Resource Name (ARN) patterns that use wildcards for each prefix and a Condition element that uses the aws:CurrentTime condition key with DateGreaterThan and DateLessThan operators
C. A Condition element that uses the s3:prefix condition key to restrict folder access and aws:CurrentTime with DateGreaterThanEquals and DateLessThanEquals to restrict hours of operation

D. A Condition element that uses the s3:ResourceAccount condition key to restrict bucket access and a Deny statement that applies outside of business hours

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
IAM policies can restrict access to specific S3 prefixes by defining object ARNs with wildcard patterns in the Resource element. Time-based access control is implemented using the aws:CurrentTime condition key with DateGreaterThan and DateLessThan operators to enforce business-hour access.

QUESTION 326
A company needs to collect logs for an Amazon RDS for MySQL database and make the logs available for audits. The logs must track each user that modifies data in the database or makes changes to the database instance.

Which solution will meet these requirements?

A. Enable Amazon CloudWatch Logs. Create metric filters to monitor database changes and instance-level changes. Configure automated notification systems to send near real-time alerts for suspicious database operations.
B. Configure an Amazon EventBridge rule to monitor database activity. Create an AWS Lambda function to process EventBridge events and store them in Amazon OpenSearch Service.
C. Configure AWS CloudTrail to log API calls. Use Amazon CloudWatch Logs for basic monitoring. Use IAM policies to control access to the logs. Set up scheduled reporting for log audits.
D. Enable and configure native Amazon RDS database audit logging. Enable Amazon CloudWatch Logs. Configure metric filters and alarms. Configure AWS CloudTrail audit logging.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Native Amazon RDS database audit logging captures user-level database activity such as data modifications, while AWS CloudTrail logs API-level changes to the database instance. Publishing logs to Amazon CloudWatch Logs enables centralized storage and monitoring, ensuring both database and instance changes are tracked for audit requirements.

QUESTION 327
A company processes a CSV file that contains millions of transaction records every day. The file is stored in Amazon S3. Each transaction must be validated before updating a database. The company needs a solution that will process the data in parallel. The solution must use error handling that stops the entire process if more than 15% of the records fail validation.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create an AWS Batch job that processes chunks of the file in parallel with a custom error tracking mechanism.
B. Use AWS Step Functions Distributed Map state with the ToleratedFailurePercentage field set to 15%.
C. Deploy an Amazon EMR cluster with Spark to process the file Configure a custom failure threshold to 15%.
D. Use AWS Lambda with S3 Batch Operations to process the file and track validation failures to be less than 15%.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
AWS Step Functions Distributed Map is designed to process large datasets from Amazon S3 in parallel

with minimal infrastructure management. The ToleratedFailurePercentage setting directly supports the requirement to stop the overall workflow when more than 15% of records fail validation, which makes it the lowest-overhead solution for this use case.

QUESTION 328
A company needs to generate a one-time performance report by joining data that is stored in Amazon DynamoDB. Amazon RDS. Amazon Redshift. and Amazon S3. The company wants to avoid unnecessary data movement and to minimize query execution time.

Which solution will meet these requirements?

A. Capture data from DynamoDB by using DynamoDB Streams. Migrate data from Amazon RDS by using AWS DMS. Export Amazon Redshift data. Store all data in Amazon S3. Use Redshift Spectrum to run queries.
B. Set up an AWS Glue ETL pipeline to extract, transform, and centralize data in Amazon S3. Use Amazon Athena to run analytical queries.
C. Deploy an Amazon EMR cluster powered by Apache Spark to ingest, process, and merge datasets from multiple sources. Run analytical workloads on the merged data.
D. Use Amazon Athena Federated Query to perform one-time joins and analysis across DynamoDB, Amazon RDS, Amazon Redshift, and Amazon S3.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Amazon Athena Federated Query can run SQL across data in Amazon S3 and external sources by using data source connectors, and AWS provides connectors for Amazon DynamoDB, Amazon RDS, and Amazon Redshift. That lets the company perform a one-time cross-source join and analysis without first centralizing or copying the data, which minimizes unnecessary data movement and setup effort.

Reference: https://docs.aws.amazon.com/athena/latest/ug/federated-queries.html
https://docs.aws.amazon.com/athena/latest/ug/connectors-redshift.html
https://docs.aws.amazon.com/athena/latest/ug/connectors-dynamodb.html

QUESTION 329
A data engineer is building a serverless. multi-step extract, transform, and load (ETL) pipeline. The pipeline extracts data from an Amazon S3 data lake and transforms the data by using AWS Glue ETL jobs. The pipeline then loads the results into an Amazon Redshift database. The data engineer needs to orchestrate the serverless ETL workflow.

Which solutions will meet these requirements? (Choose two.)

A. Implement the workflow by using AWS Step Functions. Configure Step Functions to coordinate the AWS Glue ETL jobs and handle error conditions with automatic retries.
B. Use AWS Glue workflows to create a graph of the ETL tasks that visually represents the dependencies between jobs and the job triggers.
C. Provision an always on Amazon EC2 instance. Create a cron job that invokes the AWS Glue ETL jobs in sequence based on a predefined scheduleю
D. Use Amazon EventBridge rules to invoke the AWS Glue ETL jobs based on S3 object creation events. Configure the rules to chain the AWS Glue ETL jobs in sequence and handle complex job dependencies.
E. Build an orchestration solution by using AWS CodePipeline to coordinate the ETL pipeline and infrastructure changes based on the dependencies.

Correct Answer: AB
Section: (none)

Explanation/Reference:
Explanation:
AWS Step Functions is a serverless orchestration service that can coordinate multi-step ETL workflows,

invoke AWS Glue jobs, manage dependencies, and provide built-in retry and error-handling logic. This makes it a strong fit for orchestrating a serverless ETL pipeline end to end.

AWS Glue workflows are specifically designed to orchestrate AWS Glue jobs, crawlers, and triggers in a dependency-based ETL graph. They provide a native way to represent and manage task order and workflow execution for serverless Glue-based pipelines.

QUESTION 330
A media company uploads large video files to Amazon S3 for processing. After processing, the company needs to keep the original files for 90 days in case the files require reprocessing. After 90 days, the company can delete the files to reduce storage costs. The company stores the processed videos in a different S3 bucket.

Which S3 Lifecycle configuration will meet these requirements for the original files MOST cost-effectively?

A. Store the files in S3 Standard for 90 days. Transition the files to S3 Glacier Flexible Retrieval for long-term storage. Then expire the files.
B. Store the files in S3 Standard for 90 days. Enable versioning. Enable Object Lock on the files for 90 days. Then expire the files.
C. Store the files in S3 Standard for 90 days. Implement S3 Lifecycle management to expire the files.
D. Store the files in S3 Intelligent-Tiering for 90 days. Enable versioning. Add S3 Lifecycle management to expire the files.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Keeping the files in S3 Standard for 90 days and using an S3 Lifecycle expiration rule is the most cost-effective option among the choices because the objects only need to be retained for 90 days and then deleted. Adding Glacier Flexible Retrieval would introduce archival and restore behavior intended for longer-term storage, and AWS notes that S3 Glacier Flexible Retrieval is a long-term archival class with a 90-day minimum storage duration. S3 Intelligent-Tiering also adds monitoring and automation charges, and Object Lock/versioning add unnecessary cost and controls beyond the stated requirement.

Reference: https://docs.aws.amazon.com/AmazonS3/latest/userguide/glacier-storage-classes.html
https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-expire-general-considerations.html
https://docs.aws.amazon.com/AmazonS3/latest/userguide/intelligent-tiering.html

QUESTION 331
A university is developing an educational application that analyzes student essays. The application provides personalized feedback with accurate citations to the university's textbooks. The application needs to process essays in multiple languages. Application responses must include direct references to specific sections in the course materials and must be in the student's selected language.

Which solution will meet these requirements with the LEAST operational overhead?

A. Build a custom vector database by using Amazon OpenSearch Serverless. Store textbook content as multilingual embeddings. Create an AWS Lambda function that queues the database when generating responses with Amazon Bedrock.
B. Create a knowledge base in Amazon Bedrock Knowledge Bases with the university's textbooks. Configure a multilingual model to generate responses with source citations.
C. Use Amazon Comprehend to detect the language and key topics in the essays. Use Amazon Kendra to search for relevant textbook passages. Create an AWS Lambda function that formats the textbook passages into feedback.
D. Use Amazon SageMaker to host a custom-trained large language model (LLM) that has been fine-tuned on the university's textbooks to generate personalized feedback with citations.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
Amazon Bedrock Knowledge Bases is the lowest-overhead solution because it provides a managed retrieval-augmented generation workflow over the university's textbooks and can return generated responses with citations to the source material. The RetrieveAndGenerate capability is designed to generate answers from retrieved textbook content and include source citations, which directly matches the requirement for accurate references to specific course materials. By choosing a multilingual model in Bedrock for generation, the application can respond in the student's selected language without building and operating a custom retrieval stack.

Reference: https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base.html
https://docs.aws.amazon.com/bedrock/latest/userguide/kb-test-retrieve-generate.html
https://docs.aws.amazon.com/bedrock/latest/userguide/kb-how-retrieval.html

QUESTION 332
A company needs a solution that restricts access to Amazon S3 data and encrypts the data by using AWS managed keys. The solution must manage database credentials that an AWS Lambda function uses and must rotate the credentials automatically.

Which solution will meet these requirements?

A. Use S3 bucket policies to control access. Use server-side encryption with Amazon S3 managed keys (SSE-S3) to encrypt the data. Store the database credentials as Lambda environment variables.
B. Use IAM policies to control access. Use server-side encryption with AWS KMS keys (SSE-KMS) to encrypt the data. Configure AWS Secrets Manager to store and automatically rotate the credentials by using a Lambda function.
C. Use S3 ACLs to control access. Use server-side encryption with AWS KMS keys (SSE-KMS) to encrypt the data. Store the credentials in AWS Systems Manager Parameter Store and automatically rotate the credentials by using a Lambda function.
D. Use IAM policies to control access. Use server-side encryption with Amazon S3 managed keys (SSE-S3) to encrypt the data. Store the credentials in AWS Systems Manager Parameter Store. Configure a scheduled Lambda function to rotate the credentials.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:
IAM policies can be used to control access to Amazon S3 data, and server-side encryption with AWS KMS keys (SSE-KMS) supports encryption by using an AWS managed KMS key such as aws/s3. AWS Secrets Manager is the managed service designed to store database credentials and rotate them automatically, including rotation by using a Lambda function. This combination satisfies the requirements for S3 access control, encryption with AWS managed keys, and automatic credential rotation for the Lambda-based application.

Reference: https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingKMSEncryption.html
https://docs.aws.amazon.com/AmazonS3/latest/userguide/access-policy-language-overview.html
https://docs.aws.amazon.com/secretsmanager/latest/userguide/intro.html

QUESTION 333
A company needs to store semi-structured transactional data for an application in a database. The database must be serverless. The application writes the data infrequently, but it reads the data frequently. The application must retrieve the data within milliseconds.

Which solution will meet these requirements with the LEAST operational overhead?

A. Store the data in an Amazon S3 Standard bucket. Enable S3 Transfer Acceleration.
B. Store the data in an Amazon S3 Apache Iceberg table. Enable S3 Transfer Acceleration.
C. Store the data in an Amazon RDS for MySQL cluster. Configure RDS Optimized Reads for the cluster.
D. Store the data in an Amazon DynamoDB table. Configure a DynamoDB Accelerator cache.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Amazon DynamoDB is a fully managed serverless database that is designed for millisecond response times and works well for semi-structured application data. Because the application reads frequently and writes infrequently, adding DynamoDB Accelerator (DAX) provides an in-memory cache that further reduces read latency with minimal operational overhead.

QUESTION 334
A company runs concurrent analytical queries on Amazon Redshift tables multiple times each day. The queries require consistent data views three times each day. The company runs extract, transform, and load (ETL) operations that update dimension tables while the queries run.

The company has noticed that the queries cause table-level locks during the ETL operations. The company's current solution experiences query timeouts and deadlocks during peak processing hours, which affects analytical reporting and on-demand analysis.

Which solution will fix this issue?

A. Use Amazon Redshift materialized views for analytical queries. Schedule ETL operations during off-peak hours to minimize lock contention.
B. Configure Amazon Redshift federated queries to access source data directly. Use read replicas to isolate analytical workloads from ETL operations.
C. Use Amazon Redshift Spectrum to query data in Amazon S3 for analytical workloads. Maintain ETL operations on Amazon Redshift tables with transaction isolation.
D. Deploy separate Amazon Redshift clusters for ETL and analytics workloads. Use cross-database queries and data sharing to maintain data consistency.

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Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
Using separate Amazon Redshift clusters for ETL and analytics isolates write-heavy ETL activity from read-heavy analytical workloads, which eliminates the lock contention that is causing timeouts and deadlocks. Redshift data sharing provides near real-time, transactionally consistent access to the same data across clusters, so analysts can query a stable view of the data while ETL processes continue independently.

QUESTION 335
A company aggregates high-frequency sensor telemetry into an Amazon S3 data lake. Each sensor stream emits structured records every hour. The records include metadata such as sensor category, unit ID, operational state, event timestamp, and site location. The data scales up to millions of records each day. The company runs complex queries each day to uncover performance insights specific to sensor categories.

Which solution will meet these requirements with the FASTEST query execution time?

A. Persist the data in Apache ORC format. Partition the data by date. Sort the data by sensor category.
B. Persist the data in CSV format. Partition the data by date. Sort the data by operational status.
C. Persist the data in Parquet format. Partition the data by sensor category. Sort the data by date
D. Persist the data in CSV format. Partition the data by date. Sort the data by sensor category.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Parquet is a columnar storage format that significantly improves query performance for analytical

workloads by reading only the required columns and applying efficient compression. Partitioning by sensor category aligns directly with the primary query filter, which minimizes the amount of data scanned. Sorting by date further improves query efficiency for time-based filtering within each partition, resulting in the fastest query execution.

QUESTION 336
A company uses Amazon Redshift to store order transactions from the current day. The company has an orders table that contains the previous order data. The company also has a staging table that contains new or updated order records.

The company needs to remove stale records from the orders table and insert the most recent data in the orders table from the staging table. Several downstream applications need the orders table to display up-to-date information.

Which solution will meet these requirements?

A. Use Amazon Redshift Spectrum to delete stale records from the orders table and insert records from the staging table into the orders table.
B. Unload the orders table and the staging table to Amazon S3. Delete stale orders table data and insert new staging table data in Amazon S3 by using Amazon Athena. Copy the orders S3 table to the orders Amazon Redshift table.
C. Use Amazon Athena federated queries to read stale records from the orders table. Delete the stale records and insert the records from the staging table into the orders table.
D. Write an Amazon Redshift stored procedure that deletes the stale records from the orders table and inserts new records from the staging table.

Correct Answer: D
Section: (none)

Explanation/Reference:
Explanation:
An Amazon Redshift stored procedure can perform the delete-and-insert logic directly inside the warehouse in a single controlled operation. This is the most appropriate way to remove stale rows from the target table and load the latest records from the staging table while keeping the orders table current for downstream applications, without introducing unnecessary data movement or external processing.

QUESTION 337
A company is developing a log streaming pipeline that uses Amazon Data Firehose. The pipeline streams Amazon CloudWatch Logs data to an Amazon S3 bucket. The company's analytics team needs to use the data in audits. The pipeline must deliver only the relevant logs to the S3 bucket in a compatible format for the team's analysis.

Which solution will meet these requirements and maintain reliable performance?

A. Set the S3 bucket rules to allow logs from only specific timestamp ranges. Create an AWS Lambda function that converts the log files to the desired format. Use an S3 trigger to invoke the Lambda function.
B. Create a subscription filter in the CloudWatch Logs log group that uses the Firehose delivery stream as the destination. Create an AWS Lambda function that converts the log files to the desired format. Configure Firehose to invoke the Lambda function.
C. Create a subscription filter in the CloudWatch Logs log group. Configure the filter to monitor the Firehose stream. Create an AWS Lambda function to convert the log files to the desired format. Configure Firehose to invoke the Lambda function.
D. Tag the CloudWatch Logs log groups that the analytics team needs. Configure Firehose to ingest only the tagged log groups. Configure Firehose to write the output in the desired format.

Correct Answer: B
Section: (none)

Explanation/Reference:
Explanation:

A CloudWatch Logs subscription filter can send only the relevant log events from the log group to an Amazon Data Firehose delivery stream. Firehose can then invoke an AWS Lambda function to transform the records into the format the analytics team needs before delivering them to Amazon S3. This approach is managed, scalable, and reliable for continuous log delivery.

QUESTION 338
A media company wants to build a real-time analytics pipeline to process customer activity events across the company's website and mobile app. The company wants to build a solution to ingest millions of events with minimum latency. The solution must be scalable and durable enough so that no data is lost.

Which solution will meet these requirements in the MOST cost-effective way?

A. Set up an Amazon Kinesis Data Streams pipeline to ingest data, process the data by using AWS Lambda functions, and store the results in Amazon Redshift for analytics.
B. Schedule an AWS Glue job to fetch user interaction logs every 10 minutes from Amazon S3. Configure the AWS Glue job to transform and store the data in Amazon Redshift for analytics.
C. Configure Amazon S3 Event Notifications to invoke an AWS Lambda function to process every new interaction log file. Store the result in Amazon Redshift for analytics.
D. Deploy an Amazon Managed Streaming for Apache Kafka (Amazon MSK) cluster. Use self-managed consumers to process and distribute data in real-time. Integrate with Amazon Redshift for enhanced analytics.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
Amazon Kinesis Data Streams is built for durable, scalable, real-time ingestion of streaming events and can process data with very low latency at high volume. AWS documents that Kinesis Data Streams supports real-time stream processing and integrates directly with Amazon Redshift for streaming ingestion, which fits a pipeline handling millions of events without waiting for batch intervals. Using AWS Lambda for stream processing keeps the solution managed and cost-effective, while Amazon MSK would introduce higher cluster cost and operational complexity for this requirement.

Reference: https://docs.aws.amazon.com/streams/latest/dev/introduction.html

QUESTION 339
A company stores a large dataset in an Amazon S3 bucket. A data engineer frequently runs complex queries on the dataset by using Amazon Athena. The data engineer needs to optimize query performance and optimize costs for queries that are run multiple times with the same parameters.

Which solution will meet these requirements?

A. Convert the dataset to JSON format before running Athena queries.
B. Use Amazon EMR to pre-process the data before running Athena queries.
C. Configure query result reuse settings in the Athena workgroup.
D. Use Amazon Redshift Spectrum to query the data in Amazon S3.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
Athena query result reuse allows repeated queries with the same text and parameters to return previously computed results instead of scanning the source data again. This improves performance for repeated queries and reduces query costs by avoiding unnecessary data scans.

QUESTION 340
A company stores raw clickstream data in an Amazon S3 bucket. The company needs a solution to process the data every day by using complex PySpark transformations that rely on custom internal libraries. After the data is transformed, the company must store the data in Amazon Redshift for analytics.

The solution must be highly scalable to handle large data workloads.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Glue Studio to build and schedule PySpark jobs. Configure an AWS Glue data connection that includes the custom libraries.
B. Use Amazon EC2 Auto Scaling groups with a custom AMI that contains the custom libraries to run a PySpark application.
C. Use Amazon EMR to run PySpark jobs. Use bootstrap actions to install the custom libraries
D. Use Amazon SageMaker Processing jobs to run PySpark code that uses native SageMaker libraries.

Correct Answer: A
Section: (none)

Explanation/Reference:
Explanation:
AWS Glue is a fully managed, highly scalable serverless service for running PySpark ETL jobs, and it supports adding custom Python libraries to Glue jobs from Amazon S3 by using job parameters such as --additional-python-modules and --extra-py-files. AWS Glue can also write to Amazon Redshift, making it the best fit for daily large-scale PySpark transformations with the least operational overhead compared with managing EC2 instances or EMR clusters.

Reference: https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-python-libraries.html
https://docs.aws.amazon.com/glue/latest/dg/aws-glue-programming-etl-connect-redshift-home.html

QUESTION 341
An application uses an AWS Lambda function that is configured with managed runtimes. The Lambda function successfully writes logs to the default Amazon CloudWatch Logs log group. A data engineer wants to modify the logging behavior to show only ERROR level logs for application logs and WARN level logs for system logs.

Which solution will meet these requirements?

A. Add additional permissions to the Lambda execution role.
B. Set the log level to ERROR in the Lambda function code.
C. Configure the Lambda function to use the JSON log format.
D. Configure the Lambda function to send logs to a custom log group.

Correct Answer: C
Section: (none)

Explanation/Reference:
Explanation:
AWS Lambda supports configuring advanced logging controls for managed runtimes when using structured JSON log format. This allows separate log level settings for application logs and system logs, enabling ERROR level for application logs and WARN level for system logs without modifying permissions or log destinations.